UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, DC 20549

FORM 10-K

(Mark One)

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2023

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from_____to _____

Commission file number 001-32597

CF INDUSTRIES HOLDINGS, INC.

(Exact name of registrant as specified in its charter)

Delaware	**20-2697511**
(State or other jurisdiction of incorporation or organization)	(I.R.S. Employer Identification No.)
2375 Waterview Drive **Northbrook, Illinois**	**60062**
(Address of principal executive offices)	(Zip Code)

Registrant's telephone number, including area code: **(847) 405-2400**

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading symbol(s)	Name of each exchange on which registered
common stock, par value $0.01 per share	CF	New York Stock Exchange

Securities registered pursuant to section 12(g) of the Act: **None**

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer ☒ Accelerated filer ☐ Non-accelerated filer ☐ Smaller reporting company ☐ Emerging growth company [

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to § 240.10D-1(b) ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

The aggregate market value of the registrant's common stock held by non-affiliates as of June 30, 2023 (the last business day of the registrant's most recently completed second fiscal quarter), computed by reference to the closing sale price of the registrant's common stock, was $13,335,083,067.

188,337,896 shares of the registrant's common stock, par value $0.01 per share, were outstanding as of January 31, 2024.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the registrant's definitive proxy statement for its 2024 annual meeting of shareholders (Proxy Statement) are incorporated by reference into Part III of this Annual Report on Form 10-K. The Proxy Statement will be filed with the Securities and Exchange Commission, pursuant to Regulation 14A, not later than 120 days after the end of the 2023 fiscal year, or, if the registrant does not file the Proxy Statement within such 120-day period, the registrant will amend this Annual Report on Form 10-K to include the information required under Part III of Form 10-K not later than the end of such 120-day period.

CF INDUSTRIES HOLDINGS, INC.

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CF INDUSTRIES HOLDINGS, INC.

PART I

ITEM 1. BUSINESS.

Our Company

All references to "CF Holdings," "we," "us," "our" and "the Company," refer to CF Industries Holdings, Inc. and its subsidiaries, except where the context makes clear that the reference is to CF Industries Holdings, Inc. only and not its subsidiaries. All references to "CF Industries" refer to CF Industries, Inc., a 100% owned subsidiary of CF Industries Holdings, Inc. References to tons refer to short tons and references to tonnes refer to metric tons. Notes referenced throughout this document refer to consolidated financial statement note disclosures that are found in Item 8. Financial Statements and Supplementary Data—Notes to Consolidated Financial Statements.

Our mission is to provide clean energy to feed and fuel the world sustainably. With our employees focused on safe and reliable operations, environmental stewardship, and disciplined capital and corporate management, we are on a path to decarbonize our ammonia production network – the world's largest – to enable green and low-carbon hydrogen and nitrogen products for energy, fertilizer, emissions abatement and other industrial activities. Our nitrogen manufacturing complexes in the United States, Canada and the United Kingdom, an extensive storage, transportation and distribution network in North America, and logistics capabilities enabling a global reach underpin our strategy to leverage our unique capabilities to accelerate the world's transition to clean energy. Our principal customers are cooperatives, independent fertilizer distributors, traders, wholesalers and industrial users. Our core product is anhydrous ammonia (ammonia), which contains 82% nitrogen and 18% hydrogen. Our nitrogen products that are upgraded from ammonia are granular urea, urea ammonium nitrate solution (UAN) and ammonium nitrate (AN). Our other nitrogen products include diesel exhaust fluid (DEF), urea liquor, nitric acid and aqua ammonia, which are sold primarily to our industrial customers.

Our principal assets as of December 31, 2023 include:

- six U.S. nitrogen manufacturing facilities, located in Donaldsonville, Louisiana (the largest nitrogen complex in the world); Sergeant Bluff, Iowa (our Port Neal complex); Yazoo City, Mississippi; Claremore, Oklahoma (our Verdigris complex); Woodward, Oklahoma; and Waggaman, Louisiana. The Waggaman facility is wholly owned by us, and the other five U.S. nitrogen manufacturing facilities are wholly owned directly or indirectly by CF Industries Nitrogen, LLC (CFN), of which we own approximately 89% and CHS Inc. (CHS) owns the remainder;
- two Canadian nitrogen manufacturing facilities, located in Medicine Hat, Alberta (the largest nitrogen complex in Canada) and Courtright, Ontario;
- a United Kingdom nitrogen manufacturing facility located in Billingham;
- an extensive system of terminals and associated transportation equipment located primarily in the Midwestern United States; and
- a 50% interest in Point Lisas Nitrogen Limited (PLNL), an ammonia production joint venture located in the Republic of Trinidad and Tobago (Trinidad) that we account for under the equity method.

We have a strategic venture with CHS under which CHS owns an equity interest in CFN, a subsidiary of CF Holdings, which represents approximately 11% of the membership interests of CFN. We own the remaining membership interests. CHS also receives deliveries pursuant to a supply agreement under which CHS has the right to purchase annually from CFN up to approximately 1.1 million tons of granular urea and 580,000 tons of UAN at market prices. As a result of its minority equity interest in CFN, CHS is entitled to semi-annual cash distributions from CFN. We are also entitled to semi-annual cash distributions from CFN. See Note 19—Noncontrolling Interest for additional information on our strategic venture with CHS.

For the years ended December 31, 2023, 2022 and 2021, we sold 19.1 million, 18.3 million and 18.5 million product tons generating net sales of $6.63 billion, $11.19 billion and $6.54 billion, respectively.

Our principal executive offices are located outside of Chicago, Illinois, at 2375 Waterview Drive, Northbrook, Illinois 60062, and our telephone number is 847-405-2400. Our Internet website address is *www.cfindustries.com.* Information made available on our website does not constitute part of this Annual Report on Form 10-K.

We make available free of charge on or through our Internet website,*www.cfindustries.com*, all of our reports on Forms 10-K, 10-Q and 8-K and all amendments to those reports as soon as reasonably practicable after such material is filed electronically with, or furnished to, the Securities and Exchange Commission (SEC). Copies of our Corporate Governance Guidelines, Code of Corporate Conduct and charters for the Audit Committee, Compensation and Management Development

Committee, Corporate Governance and Nominating Committee, and Environmental Sustainability and Community Committee of our Board of Directors (the Board) are also available on our Internet website. We will provide electronic or paper copies of these documents free of charge upon request. The SEC also maintains a website at *www.sec.gov* that contains reports, proxy and information statements and other information regarding issuers that file electronically with the SEC.

Our Strategy

Our strategy is to leverage our unique capabilities to accelerate the world's transition to clean energy. We believe this strategy builds upon the Company's leadership in ammonia production to capture emerging opportunities available to ammonia produced with a lower carbon intensity than that of ammonia produced through traditional processes. These opportunities include traditional applications in agriculture to help reduce the carbon footprint of food production and the life cycle carbon intensity of ethanol production, enabling its use for sustainable aviation fuel, among other purposes. They also include new applications, such as power generation and marine shipping, that would use the hydrogen component of the ammonia molecule for clean energy given that ammonia does not contain or emit carbon when combusted.

We execute our strategy across four dimensions: decarbonizing our existing network to accelerate the availability of low-carbon ammonia; evaluating new low-carbon ammonia capacity growth; forging partnerships to accelerate our timeline and bridge gaps in areas where we do not have expertise; and collaborating to build understanding of ammonia's clean energy capability, safety track record and regulatory environment.

Decarbonization projects in our existing network include our green ammonia project at our Donaldsonville, Louisiana complex. Green ammonia refers to ammonia produced with hydrogen sourced through an electrolysis process that produces no carbon emissions. In April 2021, we signed an engineering and procurement contract with thyssenkrupp to supply a 20 MW alkaline water electrolysis plant to produce green hydrogen at our Donaldsonville complex. We will integrate the green hydrogen generated by the electrolysis plant into existing ammonia synthesis loops to enable the production of approximately 20,000 tons per year of green ammonia. The green hydrogen production facility is mechanically complete, and commissioning activities began in early 2024. We believe that the Donaldsonville green ammonia project will be the largest of its kind in North America at the time of its startup.

Decarbonization projects in our existing network also include the production of low-carbon ammonia. Low-carbon ammonia is ammonia produced by conventional processes but with approximately 60-98% of the process and flue gas CO_2 generated by ammonia production removed through carbon capture and sequestration (CCS). We are executing a project also at our Donaldsonville complex that will enable us to produce a significant volume of low-carbon ammonia. At an estimated cost of $200 million, we are constructing a CO_2 dehydration and compression facility to enable CCS at the facility. Engineering activities for the construction of the dehydration and compression unit continue to advance, all major equipment for the facility has been procured, fabrication of the CO_2 compressors is proceeding, and construction activities at the Donaldsonville site are underway. Once the dehydration and compression unit is in service and sequestration is initiated, we expect that the Donaldsonville complex will have the capacity to dehydrate and compress up to 2 million tons per year of process CO_2, thereby converting a portion of our existing ammonia production to low-carbon ammonia. In October 2022, we announced that we had entered into a definitive CO_2 offtake agreement with ExxonMobil to transport and permanently sequester the CO_2 from Donaldsonville. Start-up for the project is planned for 2025. Under current regulations, the project would be expected to qualify for tax credits under Section 45Q of the Internal Revenue Code, which provides a credit per tonne of CO_2 sequestered.

Alongside these projects, we are also evaluating the construction of greenfield low-carbon ammonia capacity in Louisiana. In the fourth quarter of 2023, we and Mitsui & Co., Ltd. (Mitsui) completed a front-end engineering and design (FEED) study on a greenfield steam methane reforming (SMR) ammonia facility with CCS technologies. The FEED study estimates the cost of a project with these attributes to be in the range of $3 billion, with approximately $2.5 billion allocated to the ammonia facility and CCS technologies and approximately $500 million allocated to scalable common infrastructure for the site, such as ammonia storage and vessel loading docks.

We and Mitsui are progressing two additional FEED studies focused on technologies with the potential to further reduce the carbon intensity of the proposed low-carbon ammonia facility, including a FEED study evaluating autothermal reforming (ATR) ammonia production technology and a FEED study assessing the cost and viability of adding flue gas capture to an SMR ammonia facility. We expect to complete both FEED studies in the second half of 2024.

We and Mitsui are targeting the second half of 2024 for the final investment decision on the proposed greenfield low-carbon ammonia facility. Should the companies agree to move forward, the ammonia facility would be constructed at our new Blue Point complex. We own the land for the complex, which is located on the west bank of the Mississippi river in Ascension Parish, Louisiana. Construction and commissioning of a new world-scale ammonia plant typically takes approximately four years from the time construction begins.

In addition to ongoing discussions with existing customers who have interest in forthcoming availability of low-carbon ammonia for traditional applications, we are engaged in advanced discussions regarding the supply of low-carbon ammonia for new applications. In the first quarter of 2023, we signed a memorandum of understanding (MOU) with JERA Co., Inc. (JERA), Japan's largest energy generator, regarding the long-term supply of up to 500,000 tonnes per year of clean ammonia beginning in 2027. The execution of the MOU was the result of a supplier comparison and evaluation process for the procurement of clean ammonia that JERA initiated in February 2022 for the world's first commercial scale ammonia co-firing operations that JERA is developing. The MOU establishes a framework for JERA and us to assess how we would best supply JERA with clean ammonia, which will be required to be produced with at least 60% lower carbon emissions than conventionally produced ammonia, under a long-term offtake agreement. We and JERA are evaluating a range of potential supply options, including JERA making an equity investment with us to develop a clean ammonia facility in Louisiana and a supplementary long-term offtake agreement.

We are also evaluating and in various stages of developing discussions and agreements with other companies for clean ammonia long-term offtake opportunities related to new applications of ammonia.

Company History

We were founded in 1946 as Central Farmers Fertilizer Company, and were owned by a group of regional agriculture cooperatives for the first 59 years of our existence. Central Farmers became CF Industries in 1970.

Originally established as a fertilizer brokerage company, we expanded owning and operating fertilizer manufacturing and distribution facilities in the early 1950s with a principal objective of assured supply for our owners. At various times in our history, we manufactured and/or distributed nitrogen, phosphate and potash fertilizers.

We operated as a traditional manufacturing and supply cooperative until 2002, when we adopted a new business model that established financial performance as our principal objective, rather than assured supply for our owners. A critical aspect of the new business model was to establish a more economically driven approach to the marketplace.

In August 2005, we completed the initial public offering (IPO) of our common stock, which is listed on the New York Stock Exchange. In connection with the IPO, we consummated a reorganization transaction whereby we ceased to be a cooperative and our pre-IPO owners' equity interests in CF Industries were cancelled in exchange for all of the proceeds of the offering and shares of our common stock. At the time of the IPO, our assets consisted of one wholly owned nitrogen manufacturing facility in Louisiana, United States; a joint venture nitrogen manufacturing facility in Alberta, Canada, of which we owned 66 percent; a phosphate mining and manufacturing operation in Florida, United States; and distribution facilities throughout North America.

In April 2010, we acquired Terra Industries Inc. (Terra), a leading North American producer and marketer of nitrogen fertilizer products for a purchase price of $4.6 billion, which was paid in cash and shares of our common stock. As a result of the Terra acquisition, we acquired five nitrogen fertilizer manufacturing facilities; an approximately 75.3% interest in Terra Nitrogen Company, L.P. (TNCLP), a publicly traded limited partnership; and certain joint venture interests.

Prior to April 30, 2013, we owned 66 percent of Canadian Fertilizers Limited (CFL), a joint venture nitrogen manufacturing facility in Alberta, Canada. On April 30, 2013, CF Industries acquired all of the outstanding interests in CFL that it did not already own and CFL became our wholly owned subsidiary.

In March 2014, we exited our phosphate mining and manufacturing business, which was located in Florida, through a sale to The Mosaic Company. As a result, we became focused solely on nitrogen manufacturing and distribution.

In July 2015, we acquired the remaining 50% equity interest in CF Fertilisers UK Group Limited (formerly known as GrowHow UK Group Limited) (CF Fertilisers UK) not previously owned by us, and CF Fertilisers UK became wholly owned by us. This transaction added CF Fertilisers UK's nitrogen manufacturing complexes in Ince, United Kingdom and Billingham, United Kingdom to our consolidated manufacturing capacity. Due to the substantial increase in the cost of natural gas in the United Kingdom, in 2022, we closed the Ince facility, and in 2023, we ceased operations of the ammonia plant at the Billingham facility, which was idled in September 2022. Since that time, we have imported ammonia for upgrade into AN and other nitrogen products at the Billingham facility.

In February 2016, our strategic venture with CHS commenced, at which time CHS made a capital contribution of $2.8 billion to CFN in exchange for membership interests in CFN, which represented approximately 11% of the total membership interests of CFN.

In late 2015 and 2016, we completed capacity expansion projects at our Donaldsonville complex in Louisiana and our Port Neal complex in Iowa. These projects, originally announced in 2012, included the construction of new ammonia, urea, and UAN plants at our Donaldsonville complex and new ammonia and urea plants at our Port Neal complex. These plants increased our overall production capacity by approximately 25%, improved our product mix flexibility at Donaldsonville, and improved our ability to serve upper-Midwest urea customers from our Port Neal location. The total capital cost of the capacity expansion projects was $5.2 billion.

Prior to April 2, 2018, Terra Nitrogen, Limited Partnership, which owns and operates our Verdigris nitrogen manufacturing facility in Oklahoma, was a subsidiary of TNCLP. On April 2, 2018, Terra Nitrogen GP Inc., the sole general partner of TNCLP and an indirect wholly owned subsidiary of CF Holdings, completed its purchase of all of the publicly traded common units of TNCLP (the Purchase). Upon completion of the Purchase, CF Holdings owned, through its subsidiaries, 100 percent of the general and limited partnership interests of TNCLP.

On December 1, 2023, we acquired an ammonia production facility located in Waggaman, Louisiana from Dyno Nobel Louisiana Ammonia, LLC (DNLA), a U.S. subsidiary of Australia-based Incitec Pivot Limited (IPL), pursuant to an asset purchase agreement with DNLA and IPL. The facility has a nameplate capacity of 880,000 tons of ammonia annually. In connection with the closing of the acquisition on December 1, 2023, we entered into a long-term ammonia offtake agreement providing for us to supply up to 200,000 tons of ammonia per year to IPL's Dyno Nobel, Inc. subsidiary. Under the terms of the asset purchase agreement, $425 million of the $1.675 billion purchase price, subject to adjustment, was allocated by the parties to the ammonia offtake agreement. We funded the balance of the purchase price with $1.223 billion of cash on hand.

Product Tons and Nutrient Tons

Unless otherwise stated, we measure our production and sales volume in this Annual Report on Form 10-K in product tons, which represents the weight of the product measured in short tons (one short ton is equal to 2,000 pounds). References to UAN product tons assume a 32% nitrogen content basis for production volume.

We also provide certain supplementary volume information measured in nutrient tons. Nutrient tons represent the weight of the product's nitrogen content, which varies by product. Ammonia represents 82% nitrogen content, granular urea represents 46% nitrogen content, UAN represents between 28% and 32% nitrogen content and AN represents between 29% and 35% nitrogen content.

Reportable Segments

Our reportable segments consist of the following segments: Ammonia, Granular Urea, UAN, AN and Other. These segments are differentiated by products. We use gross margin to evaluate segment performance and allocate resources. Total other operating costs and expenses (consisting primarily of selling, general and administrative expenses and other operating—net) and non-operating expenses (consisting primarily of interest and income taxes), are centrally managed and are not included in the measurement of segment profitability reviewed by management. See Note 22—Segment Disclosures for additional information.

Our Products

Our primary nitrogen products are ammonia, granular urea, UAN and AN. Our historical sales of nitrogen products by segment are shown in the following table. Net sales do not reflect amounts used internally, such as ammonia, in the manufacture of other products.

	2023		2022		2021	
	Sales Volume (tons)	Net Sales	Sales Volume (tons)	Net Sales	Sales Volume (tons)	Net Sales
	(tons in thousands; dollars in millions)					
Ammonia	3,546	$ 1,679	3,300	$ 3,090	3,589	$ 1,787
Granular Urea	4,570	1,823	4,572	2,892	4,290	1,880
UAN	7,237	2,068	6,788	3,572	6,584	1,788
AN	1,571	497	1,594	845	1,720	510
Other[1]	2,206	564	2,077	787	2,318	573
Total	19,130	$ 6,631	18,331	$ 11,186	18,501	$ 6,538

(1) Other segment products primarily include DEF, urea liquor, nitric acid and aqua ammonia.

Gross margin was $2.55 billion, $5.86 billion and $2.39 billion for the years ended December 31, 2023, 2022 and 2021, respectively.

We own and operate eight nitrogen manufacturing facilities in North America, including six nitrogen manufacturing facilities in the United States, and two in Canada. As of December 31, 2023, the combined production capacity of these eight facilities represented approximately 40%, 42%, 44% and 19% of North American ammonia, granular urea, UAN and AN production capacity, respectively. Each of our nitrogen manufacturing facilities in North America has on-site storage to provide flexibility to manage the flow of outbound shipments without impacting production. Our United Kingdom nitrogen manufacturing facility produces AN and serves primarily the agricultural and industrial markets in the United Kingdom.

The following table shows the production capacities as of December 31, 2023 at each of our nitrogen manufacturing facilities:

	Average Annual Capacity[1]					
	Gross Ammonia[2]	Net Ammonia[2]	UAN[3]	Urea[4]	AN[5]	Other[6]
	(tons in thousands)					
Donaldsonville (Louisiana)[7][8]	4,335	1,390	3,255	2,635	—	445
Medicine Hat (Alberta)	1,230	770	—	810	—	—
Port Neal (Iowa)	1,230	65	800	1,350	—	290
Verdigris (Oklahoma)[8]	1,210	430	1,955	—	—	—
Waggaman (Louisiana)	880	880	—	—	—	—
Woodward (Oklahoma)	480	130	810	—	—	115
Yazoo City (Mississippi)[8][9]	570	—	160	—	1,035	125
Courtright (Ontario)[8][10]	500	265	345	—	—	400
Billingham (U.K.)[8]	—	—	—	—	595	410
	10,435	3,930	7,325	4,795	1,630	1,785
Unconsolidated Affiliate						
PLNL (Trinidad)[11]	360	360	—	—	—	—
Total	10,795	4,290	7,325	4,795	1,630	1,785

(1) Average annual capacity includes allowance for normal outages and planned maintenance shutdowns.
(2) Gross ammonia capacity includes ammonia used to produce upgraded products. Net ammonia capacity is gross ammonia capacity less ammonia used to produce upgraded products based on the product mix shown in the table.
(3) Measured in tons of UAN containing 32% nitrogen by weight.
(4) Reflects granular urea capacity from the Donaldsonville, Medicine Hat, and Port Neal facilities. Urea liquor and DEF production capacities are included in Other.
(5) AN includes prilled products (Amtrate and industrial-grade AN, or IGAN) and AN solution produced for sale.
(6) Includes product tons of: urea liquor and DEF from the Donaldsonville, Port Neal, Woodward, Yazoo City, and Courtright facilities; nitric acid from the Billingham facility. Production of DEF can be increased by reducing urea and/or UAN production.
(7) The Donaldsonville facility capacities present an estimated production mix. This facility is capable of producing between 2.4 million and 3.3 million tons of granular urea and between 1.2 million and 4.3 million tons of UAN annually. The facility is also capable of producing up to 1.2 million product tons of 32.5% DEF.
(8) Reduction of UAN or AN production at the Yazoo City, Courtright, Verdigris, Donaldsonville and Billingham facilities can allow more merchant nitric acid to be made available for sale.
(9) The Yazoo City facility's production capacity depends on product mix. With the facility maximizing the production of AN products, 160,000 tons of UAN can be produced. UAN production can be increased to 450,000 tons by reducing the production of AN to 900,000 tons.
(10) Production of urea liquor and DEF at the Courtright facility can be increased by reducing UAN production.
(11) Represents our 50% interest in the capacity of PLNL.

The following table summarizes our production volume for the last three years:

	December 31,		
	2023	2022	2021
	(tons in thousands)		
Ammonia[1]	9,496	9,807	9,349
Granular urea	4,544	4,561	4,123
UAN (32%)	6,852	6,706	6,763
AN	1,520	1,517	1,646

(1) Gross ammonia production, including amounts subsequently upgraded on-site into granular urea, UAN or AN.

Nitrogen Manufacturing Facilities

Donaldsonville, Louisiana

The Donaldsonville facility is the world's largest and most flexible nitrogen complex. It has six ammonia plants, five urea plants, four nitric acid plants, three UAN plants, and one DEF plant. The complex, which is located on the Mississippi River, includes deep-water docking facilities, access to an ammonia pipeline, and truck and railroad loading capabilities. The complex has on-site storage for 139,000 tons of ammonia, 202,000 tons of UAN (measured on a 32% nitrogen content basis) and 130,000 tons of granular urea.

Medicine Hat, Alberta, Canada

The Medicine Hat facility, located in southeast Alberta, is the largest nitrogen complex in Canada. It has two ammonia plants and one urea plant. The complex has on-site storage for 55,000 tons of ammonia and 55,000 tons of granular urea.

Sergeant Bluff, Iowa (the Port Neal facility)

The Port Neal facility is located approximately 12 miles south of Sioux City, Iowa, on the Missouri River in Sergeant Bluff, Iowa. The facility consists of two ammonia plants, three urea plants, two nitric acid plants and one UAN plant. The location has on-site storage for 82,000 tons of ammonia, 130,000 tons of granular urea, and 100,000 tons of 32% UAN.

Claremore, Oklahoma (the Verdigris facility)

The Verdigris facility is located northeast of Tulsa, Oklahoma, near the Verdigris River, in Claremore, Oklahoma. It is the second largest UAN production facility in North America. The facility consists of two ammonia plants, two nitric acid plants, two UAN plants and a port terminal. We lease the port terminal from the Tulsa-Rogers County Port Authority. The complex has on-site storage for 57,000 tons of ammonia and 102,000 tons of 32% UAN.

Waggaman, Louisiana

The Waggaman facility is located near New Orleans, Louisiana on the Mississippi River approximately 60 miles southeast of the Donaldsonville facility. The facility consists of one ammonia plant, has access to an ammonia pipeline, and has on-site storage for approximately 39,000 tons of ammonia.

Woodward, Oklahoma

The Woodward facility is located in rural northwest Oklahoma and consists of one ammonia plant, two nitric acid plants, two urea plants and two UAN plants. The facility has on-site storage for 34,000 tons of ammonia and 84,000 tons of 32% UAN.

Yazoo City, Mississippi

The Yazoo City facility is located in central Mississippi and includes one ammonia plant, four nitric acid plants, one AN plant, two urea plants, one UAN plant and a dinitrogen tetroxide production and storage facility. The site has on-site storage for 48,000 tons of ammonia, 47,000 tons of 32% UAN and 10,000 tons of AN and related products.

Courtright, Ontario, Canada

The Courtright facility is located south of Sarnia, Ontario near the St. Clair River. The facility consists of an ammonia plant, a UAN plant, a nitric acid plant and a urea plant. The location has on-site storage for 61,000 tons of ammonia and 16,000 tons of 32% UAN.

Billingham, United Kingdom

The Billingham facility, located in the Teesside chemical area in northeastern England, is geographically split among three primary locations: the main site, which contains three nitric acid plants; the Portrack site, approximately two miles away, which contains an AN fertilizer plant; and the North Tees site, approximately seven miles away, which contains an ammonia storage area. These locations collectively have on-site storage for 37,000 tons of ammonia and 138,000 tons of AN. In addition, we used to operate an ammonia plant at the main site of the Billingham facility. However, in September 2022, we idled ammonia production at the facility. Since that time, we have imported ammonia for upgrade at the facility into AN and other nitrogen products. In the third quarter of 2023, we approved our plan to permanently close the ammonia plant at the Billingham facility.

Point Lisas, Trinidad

The Point Lisas Nitrogen facility in Trinidad is owned jointly through a 50/50 venture with Koch Fertilizer LLC. This facility has the capacity to produce 720,000 tons of ammonia annually from natural gas supplied under a contract with The National Gas Company of Trinidad and Tobago Limited (NGC).

Nitrogen Product Raw Materials

Natural gas is the principal raw material and primary fuel source used in the ammonia production process at our nitrogen manufacturing facilities. In 2023, natural gas accounted for approximately 40% of our total production costs for nitrogen products. Our nitrogen manufacturing facilities have access to abundant, competitively-priced natural gas through a reliable network of pipelines that are connected to major natural gas trading hubs. Our facilities utilize the following natural gas hubs: Henry Hub, SONAT and TETCO ELA in Louisiana; ONEOK in Oklahoma; AECO in Alberta; Ventura in Iowa; Demarcation in Kansas; Welcome in Minnesota; Dawn and Parkway in Ontario; and the National Balancing Point (NBP) in the United Kingdom.

In 2023, our nitrogen manufacturing facilities consumed, in the aggregate, approximately 340 million MMBtus of natural gas. We employ a combination of daily spot and term purchases from a variety of quality suppliers to maintain a reliable, competitively-priced supply of natural gas. We also use certain financial instruments to hedge natural gas prices. See Note 17—Derivative Financial Instruments for additional information about our natural gas hedging activities.

Nitrogen Product Distribution

The safe, efficient and economical distribution of nitrogen products is critical for successful operations. Our nitrogen production facilities have access to multiple transportation modes by which we ship products to terminals, warehouses and customers. Each of our production facilities has a unique distribution pattern based on its production capacity and location.

Our North American nitrogen production facilities can ship products via truck and rail to customers and to our storage facilities in the U.S. and Canada, with access to our leased railcar fleet of approximately 4,900 tank and hopper cars, as well as railcars provided by rail carriers. Our United Kingdom nitrogen production facility mainly ships products via truck.

The North American waterway system is also used extensively to ship products from our Donaldsonville, Verdigris and Yazoo City facilities. To ship ammonia and UAN, we employ a fleet of up to eleven tow boats and thirty-six river barges, which are primarily leased. We also utilize contract marine services to move granular urea and AN. We can also export nitrogen products via seagoing vessels from our Donaldsonville and Billingham manufacturing facilities.

The Donaldsonville and Waggaman facilities are connected to the 2,000-mile long Nustar pipeline through which we have the ability to transport ammonia to ten terminals and shipping points in the Midwestern U.S. corn belt.

Storage Facilities and Other Properties

As of December 31, 2023, we owned or leased space at 48 in-market storage terminals and warehouses located in a 22-state region of the United States, Canada and the United Kingdom. Including storage at our production facilities, we have an aggregate storage capacity for approximately 3.0 million tons of product. Our storage capabilities are summarized in the following table:

	Ammonia		Granular Urea		UAN[1]		AN	
	Number of Facilities	Capacity (000 Tons)	Number of Facilities	Capacity (000 Tons)	Number of Facilities	Capacity (000 Tons)	Number of Facilities	Capacity (000 Tons)
Plants	9	552	3	315	6	551	2	148
Terminal and Warehouse Locations								
Owned[2]	22	766	—	—	9	244	—	—
Leased[3]	5	69	3	35	19	279	—	—
Total In-Market	27	835	3	35	28	523	—	—
Total Storage Capacity		1,387		350		1,074		148

(1) Capacity is expressed as the equivalent volume of UAN measured on a 32% nitrogen content basis.
(2) The owned facilities that store UAN also can store ammonia.
(3) Our lease agreements are typically for periods of one to five years and commonly contain provisions for automatic renewal that can extend the lease term unless cancelled by either party.

Customers

The principal customers for our nitrogen products are cooperatives, independent fertilizer distributors, traders, wholesalers and industrial users. Sales are generated by our internal marketing and sales force. CHS was our largest customer in 2023 and accounted for approximately 13% of our consolidated net sales. We have a strategic venture with CHS under which CHS has a minority equity interest in CFN. See Note 19—Noncontrolling Interest for additional information on our strategic venture with CHS.

Competition

Our markets are global and intensely competitive, based primarily on delivered price and, to a lesser extent, on reliability, customer service and product quality. During the peak demand periods, product availability and delivery time also play a role in the buying decisions of customers.

Our primary competitors with North American operations include Nutrien Ltd., Koch Fertilizer LLC, N-7 LLC (a joint venture between OCI N.V. and Dakota Gasification Company) and Yara International. There is also significant competition from products sourced from other regions of the world, including some with lower natural gas or other feedstock costs, which may include the benefit of government subsidies. Because ammonia, urea and UAN are widely-traded fertilizer products and there are limited barriers to entry, we experience competition from foreign-sourced products continuously. Producers of nitrogen-based fertilizers located in the Middle East, Trinidad, Africa and Russia have been major exporters to North America in recent years.

Our primary United Kingdom competition comes from imported products supplied by companies including Yara International, Origin Fertilisers, Ameropa and Thomas Bell & Sons Ltd. Urea and UAN are not produced in the United Kingdom, but along with AN are widely-traded fertilizer products with limited barriers to entry.

Seasonality

The fertilizer business is seasonal. The degree of seasonality of our business can change significantly from year to year due to weather conditions in the agricultural industry and other factors. The strongest demand for our products in North America occurs during the spring planting season, with a second period of strong demand following the fall harvest. In contrast, we and other fertilizer producers generally manufacture and distribute products throughout the year. As a result, we and/or our customers generally build inventories during the low demand periods of the year to ensure timely product availability during the peak sales seasons. Seasonality is greatest for ammonia due to the short application season and the limited ability of our customers and their customers to store significant quantities of this product. The seasonality of fertilizer demand generally results in our sales volumes and net sales being the highest during the spring and our working capital requirements being the highest just prior to the start of the spring planting season. Our quarterly financial results can vary significantly from one year to the next due to weather-related shifts in planting and application schedules and purchasing patterns as well as import timing, import and distribution costs and logistical limitations, such as river conditions.

Environmental, Health and Safety

We are subject to numerous environmental, health and safety laws and regulations in the United States, Canada, the United Kingdom, the European Union (EU) and Trinidad, including laws and regulations relating to the generation and handling of hazardous substances and wastes; the introduction of new chemicals or substances into a market; the cleanup of hazardous substance releases; the discharge of regulated substances to air or water; and the demolition of existing plant sites upon permanent closure. In the United States, these laws include the Clean Air Act, the Clean Water Act, the Resource Conservation and Recovery Act (RCRA), the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA), the Toxic Substances Control Act (TSCA), the Occupational Safety and Health Act (OSHA) and various other federal, state and local statutes. Violations of environmental, health and safety laws can result in substantial penalties, court orders to install pollution-control equipment, civil and criminal sanctions, permit revocations and facility shutdowns. In addition, environmental, health and safety laws and regulations may impose joint and several liability, without regard to fault, for cleanup costs on potentially responsible parties who have released or disposed of hazardous substances into the environment. We may be subject to more stringent enforcement of existing or new environmental, health and safety laws in the future.

Environmental, Health and Safety Expenditures

Our environmental, health and safety capital expenditures in 2023 totaled approximately $36 million. We estimate that we will have approximately $47 million of environmental, health and safety capital expenditures in 2024. In addition, to support safe and reliable operations at our continuous process manufacturing facilities, we conduct scheduled inspections, replacements and overhauls of our plant machinery and equipment, which are referred to as turnarounds. A further description of turnaround activities is included in Note 8—Property, Plant and Equipment—Net in the notes to consolidated financial statements included in Item 8 of this report. Environmental, health and safety laws and regulations are complex, change frequently and have tended to become more stringent over time. We expect that continued government and public emphasis on environmental issues will result in increased future expenditures for environmental controls at our manufacturing and distribution facilities. Such expenditures could have a material adverse effect on our business, financial condition, results of operations and cash flows. Additionally, future environmental, health and safety laws and regulations or reinterpretation of current laws and regulations may require us to make substantial expenditures. Our costs to comply with, or any liabilities under, these laws and regulations could have a material adverse effect on our business, financial condition, results of operations and cash flows.

CERCLA/Remediation Matters

From time to time, we receive notices from governmental agencies or third parties alleging that we are a potentially responsible party at certain cleanup sites under CERCLA or other environmental cleanup laws. In 2011, we received a notice from the Idaho Department of Environmental Quality (IDEQ) that alleged that we were a potentially responsible party for the cleanup of a former phosphate mine site we owned in the late 1950s and early 1960s located in Georgetown Canyon, Idaho. The current owner of the property and a former mining contractor received similar notices for the site. In 2014, we and the current property owner entered into a Consent Order with IDEQ and the U.S. Forest Service to conduct a remedial investigation and feasibility study of the site. The remedial investigation was submitted to the agencies in 2021. The next step will be a risk assessment, followed by a feasibility study. In 2015, we and several other parties received a notice that the U.S. Department of the Interior and other trustees intended to undertake a natural resource damage assessment for 18 former phosphate mines and three former processing facilities in southeast Idaho. The Georgetown Canyon former mine and processing facility was included in the group of former mines and processing facilities identified by the trustees. In June 2021, we received another notice from the U.S. Department of the Interior that the natural resource damage trustees were commencing a ‘subsequent’ phase of the natural resource damage assessment, but no further details were provided with respect to said assessment. Because the former Georgetown Canyon mine site is still in the risk assessment and feasibility study stage, we are not able to estimate at this time our potential liability, if any, with respect to the cleanup of the site or a possible claim for natural resource damages. However, based on the results of the site investigation conducted to date, we do not expect the remedial or financial obligations to which we may be subject involving this or other cleanup sites will have a material adverse effect on our consolidated financial position, results of operations or cash flows.

Regulation of Greenhouse Gases

Our production facilities emit greenhouse gases (GHGs), such as carbon dioxide and nitrous oxide. Natural gas, a fossil fuel, is a primary raw material used in our nitrogen production process. We are subject to GHG regulations in the United Kingdom, Canada and the United States.

Our U.K. manufacturing plant is required to report GHG emissions annually to the United Kingdom Environment Agency pursuant to its site Environmental Permits and Climate Change Agreement, which specifies energy efficiency targets. Failure to meet efficiency targets may require the plant to purchase CO_2 emissions allowances. Our U.K. manufacturing plant is subject to the UK Emissions Trading Scheme (UK ETS), which generally requires us to hold or obtain emission allowances to offset GHG emissions from those aspects of our operations that are subject to regulation under this program.

In Canada, we are required to conduct an annual review of our operations with respect to compliance with Environment Canada's National Pollutant Release Inventory, Ontario's Mandatory Monitoring and Reporting Regulation, and the GHG Reporting Regulation. In addition, our manufacturing plants in Alberta and Ontario are subject to provincial or federal laws that impose a price on excess GHG emissions. Each of these laws establishes carbon dioxide equivalent (CO_2e) emissions standards applicable to our facilities in terms of emissions per unit of production, with the provincial laws and the federal law using different formulas for establishing the intensity-based limits and the reductions in these limits over time. The federal law is the Greenhouse Gas Pollution Pricing Act, which came into effect in 2018 and is intended to function only as a backstop to the provincial programs if such programs do not meet minimum federal criteria. In 2022, the federal government found that both the Alberta and Ontario programs for 2023-2030 met such minimum criteria, and therefore, the provincial laws apply. Effective January 1, 2023, these provincial regulations increased in stringency from 2022 levels, and we expect that the regulations will continue to increase in stringency going forward as Canada continues to work toward its stated goal of net zero GHG emissions by 2050. If a facility's CO_2e emissions exceed the applicable limit, the excess emissions must be offset, either through obtaining qualifying emission credits or by making a payment for each ton of excess emissions. For calendar year 2024, the excess emissions fee under the federal, Alberta and Ontario regulatory programs is CAD $80 per tonne, which fee will increase by CAD $15 per tonne per year, reaching CAD $170 per tonne by 2030.

In the United States, GHG regulation is evolving at state, regional and federal levels, although some of the more significant developments to date, including efforts of the United States Environmental Protection Agency (EPA) to regulate GHG emissions from fossil fuel-fired power plants, do not directly impose obligations on our facilities. The EPA issued a mandatory GHG reporting rule that required all of our U.S. manufacturing facilities, which are considered large emitters of GHGs, to commence monitoring GHG emissions beginning on January 1, 2010 and reporting the previous year's emissions annually starting in 2011. In addition, if we seek to modify or expand any of our major facilities and as a result, are required to obtain a Prevention of Significant Deterioration (PSD) construction permit applicable to such facilities, we could be subject to pollution control requirements applicable to GHGs in addition to requirements applicable to conventional air pollutants. Such requirements may result in increased costs or delays in completing such projects. Other than the states' implementation of this permitting requirement, none of the states where our U.S. production facilities are located – Iowa, Louisiana, Mississippi and Oklahoma – has proposed control regulations limiting GHG emissions.

Increasing concern over the impacts of climate change is driving countries to establish ever more ambitious GHG reduction targets. Approximately 200 countries, including the United States, Canada, the United Kingdom and the members of the EU have joined the Paris Agreement, an international agreement intended to provide a framework pursuant to which the parties to the agreement will attempt to hold the increase in global average temperatures to below 2°C above pre-industrial levels and to pursue efforts to limit the temperature increase to 1.5°C above pre-industrial levels. Each signatory is required to develop its own national plan to attain this objective. In December 2020, the United Kingdom announced a target to reduce GHG emissions 68% from the baseline year of 1990 by 2030. Canada has increased its emissions reduction target under the Paris Agreement to 40-45% below 2005 levels by 2030, up from 30%. In April 2021, the United States increased its goal to reduce GHG emissions to 50-52% below 2005 levels by 2030. Executive orders issued by the Biden administration, including in particular an executive order issued on January 27, 2021 focusing on climate change, evidence the administration's intent to undertake numerous initiatives in an effort to reduce GHG emissions, including promoting renewable energy development, limiting new oil and gas leases on federal lands and, in general, making climate change considerations a critical component of federal policy.

Regulatory Permits and Approvals

We hold numerous environmental and other governmental permits and approvals authorizing operations at each of our facilities. A decision by a government agency to deny or delay issuing a new or renewed regulatory material permit or approval, or to revoke or substantially modify an existing material permit or approval, could have a material adverse effect on our ability to continue operations at the affected facility. Any future expansion of our existing operations is also predicated upon securing the necessary environmental or other permits or approvals. More stringent environmental standards may impact our ability to obtain such permits.

Human Capital Resources

Our long-term success depends on our people. We are dedicated to creating a workplace where employees are proud to work and grow and where everyone feels empowered to do their best work. We do this by investing in extensive recruitment, training and professional development opportunities for our employees and fostering diversity and inclusion in our culture.

Employee Population. We employed approximately 2,700 employees at December 31, 2023, of which 78% were located in the United States, 15% in Canada, and 7% in the United Kingdom. As of December 31, 2023, 11% of our employees have worked for the Company more than 20 years, 20% of our employees have worked for the Company between 11 and 20 years, 27% of our employees have worked for the Company between 6 and 10 years, and 42% of our employees have worked at the Company for less than 6 years. Full-time employees represented nearly 100% of our workforce as of December 31, 2023 and approximately 5% were covered by collective bargaining agreements. We supplement our workforce with contractors with specialized skill sets during periods of peak activity, such as during turnarounds and maintenance events.

Culture, Inclusion and Diversity. Our core values and their underlying principles reflect our commitment to a diverse and inclusive culture, treating one another with respect. Across the Company, all employees complete training to learn to recognize and address the effects of unconscious bias by challenging assumptions; encouraging diversity of experience, opinion, and expression; and supporting a workplace culture that actively strives to be more inclusive. As of December 31, 2023, approximately 16% of our global workforce was female and 17% of the Company's employees in frontline managerial roles were female. Underrepresented groups account for approximately 18% of the Company's U.S. workforce and 17% of our U.S. employees in managerial roles. In order to continue to improve the inclusiveness and diversity of our company and culture, our comprehensive Environmental, Social and Governance (ESG) goals announced in 2020 include goals to increase the representation of females and persons of color in senior leadership roles and to implement a program designed to increase the hiring and promotion of minority and female candidates. As of December 31, 2023, we had exceeded our representation goal with approximately 37% of senior leadership roles held by females and persons of color.

Workforce Health and Safety. Operating in a safe and responsible manner is a core value and an integral part of what sets the Company apart. We believe that focusing on leading indicators — such as the behavioral safety practices we have incorporated into our annual incentive plan — to drive and measure activities that prevent safety incidents, results in our industry-leading safety record. As of December 31, 2023, our employee 12-month rolling average recordable incident rate (RIR) was 0.36 incidents per 200,000 work hours, and during the year ended December 31, 2023, our total recordable injury count was ten. For the year ended December 31, 2023, our days away, restricted or transferred (DART) incident rate was 0.11 injuries per 200,000 work hours, and our lost time incident rate was 0.04 injuries per 200,000 work hours.

Talent Development. A core aspect of our culture is our commitment to identifying and developing talent to help employees accelerate growth and achieve their career goals. We invest in extensive assessment, training and professional development opportunities for our employees through a robust set of formal and informal programs, including targeted job movements, key experiences, and training, with an emphasis on creating a culture of inclusion. Leadership is the quality that drives our values and sets us apart. To help foster leadership, we have developed a set of leadership competencies that provide a common language for how to demonstrate leadership at every level of the organization and have embedded them into all talent management processes, including selection, performance management and succession planning. We view training and development programs as being a key part of talent management, allowing us to grow a stronger company today and in the future.

ITEM 1A. RISK FACTORS.

In addition to the other information contained in this Annual Report on Form 10-K, you should carefully consider the factors discussed below before deciding to invest in any of our securities. These risks and uncertainties, individually or in combination, could materially and adversely affect our business, financial condition, results of operations and cash flows. References to tons refer to short tons and references to tonnes refer to metric tons.

Market Risks

Our business is cyclical, resulting in periods of industry oversupply during which our business, financial condition, results of operations and cash flows tend to be negatively affected.

Historically, selling prices for our products, which are generally global commodities, have fluctuated in response to periodic changes in supply and demand conditions. Demand for nitrogen is affected by planted acreage, crop selection and fertilizer application rates, driven by population growth, gross domestic product growth, changes in dietary habits and non-food use of crops, such as production of ethanol and other biofuels among other things. Demand also includes industrial uses of nitrogen, for example chemical manufacturing and emissions reductants such as diesel exhaust fluid (DEF). Supply is affected primarily by available production capacity and operating rates, raw material costs and availability, energy prices, government policies and global trade.

Periods of strong demand, high capacity utilization and increasing operating margins tend to stimulate global investment in production capacity. In the past, nitrogen manufacturers, including CF Holdings, have built new production facilities or expanded capacity of existing production assets, or announced plans to do so. The construction of new nitrogen manufacturing capacity in the industry, plus improvements to increase output from the existing production assets, increase nitrogen supply availability and affect the balance of supply and demand and nitrogen selling prices. In certain years, global nitrogen capacity has increased faster than global nitrogen demand, creating a surplus of global nitrogen capacity, which has led to lower nitrogen selling prices. For example, in the two-year period ended December 31, 2017, additional production capacity came online and, at the same time, the average selling price for our products declined 34%, from $314 per ton in 2015 to $207 per ton in 2017.

Additional nitrogen production capacity is expected to come online over the next 12 months outside of North America. In addition, plans for building new facilities for green and low-carbon ammonia have been announced by other companies and CF Holdings, such as our proposed plans for an export-oriented greenfield low-carbon ammonia production facility in the southeastern United States. We cannot predict the impact of this additional capacity on nitrogen selling prices. Also, global or local economic, political and financial conditions or changes in such conditions, or other factors, may cause acceleration of announced and/or ongoing projects. Similarly, lower energy prices can spur increases in production in high cost regions, which would result in increased supply and pressure on selling prices. Additionally, if imports increase into an oversupplied region, lower prices in that region could result.

During periods of industry oversupply, our financial condition, results of operations and cash flows tend to be affected negatively as the price at which we sell our products typically declines, resulting in possible reduced profit margins, write-downs in the value of our inventory and temporary or permanent curtailments of production. From time to time, we have experienced periods of industry oversupply, which impacted our financial performance, credit ratings and the trading price for our common stock. Due to the cyclical nature of our industry, we cannot predict the timing or duration of such periods of industry oversupply or the degree to which oversupply conditions would impact our business, financial condition, results of operations and cash flows.

Nitrogen products are global commodities, and we face intense global competition from other producers.

We are subject to intense price competition from our competitors. The nitrogen products that we produce are global commodities, with little or no product differentiation, and customers make their purchasing decisions principally on the basis of delivered price and, to a lesser extent, customer service and product quality. As a consequence, conditions in the international market for nitrogen products significantly influence our operating results.

We compete with many producers, including state-owned and government-subsidized entities. Some of our competitors have greater total resources and are less dependent on earnings from fertilizer sales, which make them less vulnerable to fertilizer industry downturns and better positioned to pursue new expansion and development opportunities. Furthermore, certain governments, in some cases as owners of some of our competitors, may be willing to accept lower prices and profitability on their products or subsidize production inputs or consumption in order to support domestic employment or other political or social goals. Our competitive position could suffer as a result of these factors, including if we are not able to expand our own resources to a similar extent, either through investments in new or existing operations or through acquisitions or joint ventures.

China, the world's largest producer and consumer of nitrogen fertilizers, currently has surplus capacity and many high-cost plants. As a result, the domestic nitrogen industry in China is operating at less than full capacity. In addition, the Chinese government is currently limiting exports through a variety of measures. A number of factors could encourage China to increase product capacity utilization, including changes in Chinese government policy, devaluation of the Chinese renminbi, the relaxation of Chinese environmental standards or decreases in Chinese producers' underlying costs such as the price of Chinese coal. Any resulting increase in export volume could adversely affect the balance between global supply and demand and put downward pressure on global fertilizer prices, which could materially adversely affect our business, financial condition, results of operations and cash flows.

From time to time, certain of our competitors with significant nitrogen fertilizer export capacity have benefited from non-market pricing of natural gas, which has resulted in significant volumes of exports to the United States. For example, the 2016 revocations of U.S. antidumping measures on solid urea and fertilizer grade ammonium nitrate from Russia allowed for increased imports from that country into the United States in recent years. In addition, in recent years, high volumes of urea ammonium nitrate solution (UAN) imports from Russia and The Republic of Trinidad and Tobago (Trinidad) have negatively affected U.S. producers' UAN profitability.

We also face competition from other fertilizer producers in the Middle East, Europe, Latin America and Africa. These producers, depending on market conditions, fluctuating input prices, geographic location and freight economics, may take actions at times with respect to price or selling volumes that adversely affect our business, financial condition, results of operations and cash flows. Some of these producers also benefit from non-market or government-set rates for natural gas pricing. Government policies in these regions may also stimulate future ammonia or hydrogen investments. Recently, many proposed green and low-carbon ammonia projects have been announced or considered, and future hydrogen, energy, or environmental/carbon policies may support development of additional nitrogen production in locations outside North America, including Europe, Australia, India, and the Middle East.

In addition, the international market for nitrogen products is influenced by such factors as currency exchange rates, including the relative value of the U.S. dollar and its impact on the cost of importing nitrogen products into the United States, foreign agricultural policies, the existence of, or changes in, import or foreign currency exchange barriers in certain foreign markets and the laws and policies of the markets in which we operate, including the imposition of new duties, tariffs or quotas, that affect foreign trade and investment. For example, the imposition of duties, tariffs or quotas in a region can directly impact product pricing in that region, which can lead to changes in global trade flows and impact the global supply and demand balance and pricing. Market participants customarily move product between regions of the world, or adjust trade flows, in response to these factors. North America, where we manufacture and sell most of our products, is one of the largest and most accessible nitrogen trading regions in the world. As a result, other manufacturers, traders and other market participants can move nitrogen products to North America when there is uncertainty associated with the supply and demand balance in other regions or when duties, tariffs or quotas impact prices or trade flows in other regions. Thus, duties, tariffs and quotas can lead to uncertainty in the global marketplace and impact the supply and demand balance in many regions, which could adversely affect our business, financial condition, results of operations and cash flows. In October 2019, the European Commission (the Commission) imposed definitive anti-dumping duties on imports to the European Union (EU) of UAN manufactured in Russia, Trinidad and the United States. For imports of UAN manufactured in the United States, the fixed duty rate is €29.48 per tonne (or €26.74 per ton). The duties are expected to remain in place for an initial five-year period and to expire in October 2024 unless there is a request submitted by July 10, 2024 to renew and continue the measures, which request would be investigated by the Commission. How long and at what level these duties will remain in effect and their long-term impact on the international market for nitrogen products are uncertain.

A decline in agricultural production, limitations on the use of our products for agricultural purposes or developments in crop technology could materially adversely affect the demand for our products.

Conditions in the United States, Europe, India, Brazil, China and other countries and regions of global significance in agricultural production significantly impact our operating results. Agricultural planted areas and production can be affected by a number of factors, including weather patterns and field conditions, current and projected grain inventories and prices, crop disease and/or livestock disease, demand for agricultural products and governmental policies regarding production of or trade in agricultural products. These factors are outside of our control.

Governmental policies, including farm and biofuel subsidies, commodity support programs and tariffs, environmental and greenhouse gas policies, as well as the prices of fertilizer products, may also directly or indirectly influence the number of acres planted, the mix of crops planted and the use of fertilizers for particular agricultural applications. Ethanol production in the United States contributes significantly to corn demand, representing approximately 40% of total U.S. corn demand, due in part to federal legislation mandating use of renewable fuels. The resulting increase in ethanol production has led to an increase in the

amount of corn grown in the United States and to increased fertilizer usage on both corn and other crops that have also benefited from improved farm economics. While the current Renewable Fuel Standard encourages continued high levels of corn-based ethanol production, various interested parties have called to eliminate or reduce the renewable fuel mandate, or to eliminate or reduce corn-based ethanol as part of the renewable fuel mandate. Other factors that drive the ethanol market include the prices of ethanol, gasoline and corn. Lower gasoline prices and fewer aggregate miles, driven by increased automobile fuel efficiency, the continued expansion of electric vehicle use or the impact of decreased travel, such as the decreased travel experienced during the coronavirus disease 2019 (COVID-19) pandemic, may put pressure on ethanol prices that could result in reduced profitability and lower production for the ethanol industry. This could have an adverse effect on corn-based ethanol production, planted corn acreage and fertilizer demand. Additionally, government incentives and other policies and recent increased investment in renewable biodiesel and associated soybean crush capacity may drive higher soybean oil prices, resulting in more planted acres allocated to soybeans and other oil crops and displacing some acreage traditionally planted to more nitrogen intensive crops such as grains and cotton.

Developments in crop technology, such as nitrogen fixation, the conversion of atmospheric nitrogen into compounds that plants can assimilate, or nitrogen-efficient varieties, or developments in alternatives to traditional animal feed or alternative proteins, could also reduce the use of nitrogen fertilizers and adversely affect the demand for our products. Widespread adoption of emerging application technologies or alternative farming techniques could disrupt traditional application practices, affecting the volume or types of fertilizer products used and timing of applications. In addition, from time to time various foreign governments and U.S. state legislatures have considered limitations on the use and application of nitrogen fertilizers due to concerns about the negative impact that the application of these products can have on the environment. For example, the United Kingdom has worked with industry to develop an assurance scheme to limit the use of unprotected or uninhibited urea products between January and March of every year, beginning in 2024. While CF Fertilisers UK Limited does not sell solid urea fertilizer in the United Kingdom, limitations on fertilizer use have been and may be considered by other jurisdictions, such as the EU, which has announced its Farm to Fork Strategy and Biodiversity Strategy. In addition, Canada has announced a target of reducing emissions from fertilizers by 30% below 2020 levels by 2030, and is supporting implementation through improved nitrogen management and optimizing fertilizer use. These or other more stringent limitations on greenhouse gas emissions applicable to farmers, the end-users of our nitrogen fertilizers, could reduce the demand for our fertilizer products to the extent their use of our products increases farm-level emissions. Any reduction in the demand for our nitrogen fertilizer products, including as a result of technological developments and/or limitations on the use and application of nitrogen fertilizers, could have a material adverse effect on our business, financial condition, results of operations and cash flows.

Our business is dependent on natural gas, the prices of which are subject to volatility.

Nitrogen from the atmosphere and hydrogen from natural gas, coal and other carbon energy feedstocks, or from the electrolysis of water, are the fundamental building blocks of nitrogen products. Energy feedstock costs comprise a significant portion of the total production cost of nitrogen products and, relative to the industry's marginal producers that set the global price of nitrogen, generally determine profitability for nitrogen producers. Our manufacturing processes utilize natural gas as the principal raw material used in our production of nitrogen products. We use natural gas both as a chemical feedstock and as a fuel to produce ammonia, granular urea, UAN, AN and other nitrogen products.

Most of our nitrogen manufacturing facilities are located in the United States and Canada. As a result, North American natural gas comprises a significant portion of the total production cost of our products. The price of natural gas in North America has historically been volatile. The price has declined on average due in part to the development of significant natural gas reserves, including shale gas, and the rapid improvement in shale gas extraction techniques, such as hydraulic fracturing and horizontal drilling. However, future production of natural gas from shale formations could be reduced by regulatory changes that restrict drilling or hydraulic fracturing or increase its cost or by reduction in oil exploration and development prompted by lower oil prices resulting in production of less associated gas. Changes in the supply of and demand for natural gas can lead to extended periods of higher natural gas prices.

In recent years, the cost of North American natural gas for the production of nitrogen fertilizers has been significantly lower than the cost of natural gas in other parts of the world where the industry's marginal nitrogen producers are located. Any increases in the volume of liquefied natural gas (LNG) exported from the United States to other regions, or increases in natural gas development outside the United States, particularly in regions where nitrogen products are produced, could increase our natural gas costs and/or lower natural gas costs for our competitors. In recent years, LNG export capabilities of the United States have expanded and LNG exports from the United States have increased, resulting in the United States being a leading exporter of LNG globally in 2023, and such expanded capabilities and increases in exports are expected to continue. If natural gas prices outside of North America were to decrease or North American natural gas prices were to increase, our favorable energy cost differentials relative to the industry's marginal nitrogen producers could significantly erode, which could have a material adverse effect on our business, financial condition, results of operations and cash flows.

During 2023, the daily closing price at the Henry Hub, the most heavily-traded natural gas pricing point in North America, reached a low of $1.72 per MMBtu on three consecutive days in June 2023 and a high of $3.81 per MMBtu on January 5, 2023. During the three-year period ended December 31, 2023, the daily closing price at the Henry Hub reached a low of $1.72 per MMBtu on three consecutive days in June 2023 and a high of $23.61 per MMBtu on February 18, 2021.

Certain of our operating facilities are located near natural gas hubs that have experienced increased natural gas development and have favorable basis differences as compared to other North American hubs. Favorable basis differences in certain regions may dissipate over time due to increases in natural gas pipeline or storage capacity in those regions. Additionally, basis differentials may become materially unfavorable due to a lack of inbound gas pipeline or storage capacity in other regions during periods of unusually high demand. Increased demand for natural gas, particularly in the Gulf Coast Region, due to increased industrial demand and increased natural gas exports, could result in increased natural gas prices. If reduced production, increased demand or changes in basis were to occur, or if other developments adversely impact the supply and demand balance for natural gas in North America or elsewhere, natural gas prices could rise, which could have a material adverse effect on our business, financial condition, results of operations and cash flows.

Adverse weather conditions may decrease demand for our fertilizer products, increase the cost of natural gas or materially disrupt our operations. Adverse weather conditions could become more frequent and/or more severe as a result of climate change.

Weather conditions that delay or disrupt field work during the planting, growing, harvesting or application periods may cause agricultural customers to use different forms of nitrogen fertilizer, which may adversely affect demand for the forms that we sell or may impede farmers from applying our fertilizers until the following application period, resulting in lower seasonal demand for our products.

Adverse weather conditions during or following harvest may delay or eliminate opportunities to apply fertilizer in the fall. Weather can also have an adverse effect on crop yields, which could lower the income of growers and impair their ability to purchase fertilizer from our customers. Adverse weather conditions could also impact transportation of fertilizer, which could disrupt our ability to deliver our products to customers on a timely basis. Our quarterly financial results can vary significantly from one year to the next due to weather-related shifts in fertilizer applications, planting schedules and purchasing patterns. Over the longer-term, changes in weather patterns may shift the periods of demand for products and even the regions to which our products are distributed, which could require us to evolve our distribution system.

In addition, we use the North American waterway system extensively to ship products from some of our manufacturing facilities to our distribution facilities and our customers. We also export nitrogen fertilizer products via seagoing vessels from deep-water docking facilities at certain of our manufacturing sites on the U.S. river system through the U.S. Gulf of Mexico. Therefore, persistent significant changes in river or ocean water levels (either up or down, such as a result of flooding, drought or climate change, for example), may require changes to our operating and distribution activities and/or significant capital improvements to our facilities. For example, recent low water levels on the U.S. river system and in the Panama Canal have delayed shipping in these locations, resulting in an increase in shipping costs.

Weather conditions or, in certain cases, weather forecasts, also can disrupt our operations and can affect the price of natural gas, the principal raw material used to make our nitrogen products. Colder and/or longer than normal winters and warmer than normal summers increase the demand for natural gas for residential and industrial use and for power generation, which can increase the cost and/or decrease the availability of natural gas. In addition, adverse weather events, such as storms, hurricanes, tornadoes, or floods, not only can cause loss of power or other impacts to our facilities or damage to or delays in logistics capabilities disrupting our operations, but also can impact the supply of natural gas and utilities and cause prices to rise.

All of the adverse weather conditions described above, including those impacting our customers and our operations, such as the physical risk from storms, hurricanes, tornadoes, or floods could become more frequent and/or more severe as a result of climate change. Our Donaldsonville and Waggaman complexes are located in an area of the United States that experiences a relatively high level of hurricane or high wind activity and several of our complexes are located in areas that experience extreme weather events. In the last several years, there has been an increase in the frequency and severity of adverse weather conditions, including in the geographic areas where we have operations. Any significant adverse weather event or combination of adverse weather events could decrease demand for our fertilizer products, increase the cost of natural gas or materially disrupt our operations—any of which could have a material adverse impact on our business, financial condition, results of operations and cash flows.

Our operating results fluctuate due to seasonality. Our inability to predict future seasonal fertilizer demand accurately could result in our having excess inventory, potentially at costs in excess of market value.

The fertilizer business is seasonal. The degree of seasonality of our business can change significantly from year to year due to conditions in the agricultural industry and other factors. The strongest demand for our products in North America occurs during the spring planting season, with a second period of strong demand following the fall harvest. In contrast, we and other fertilizer producers generally manufacture and distribute products throughout the year. As a result, we and/or our customers generally build inventories during the low demand periods of the year to ensure timely product availability during the peak demand periods. Seasonality is greatest for ammonia due to the short application seasons and the limited ability of our customers and their customers to store significant quantities of this product. The seasonality of fertilizer demand generally results in our sales volumes and net sales being the highest during the spring and our working capital requirements to build inventory being the highest just prior to the start of the spring planting season.

If seasonal demand is less than we expect, we may be left with excess inventory that will have to be stored (in which case our results of operations would be negatively affected by any related increased storage costs) or liquidated (in which case the selling price could be below our production, procurement and storage costs). The risks associated with excess inventory and product shortages are exacerbated by the volatility of nitrogen fertilizer prices, the constraints of our storage capacity, and the relatively brief periods during which farmers can apply nitrogen fertilizers. If prices for our products rapidly decrease, we may be subject to inventory write-downs, adversely affecting our operating results.

A change in the volume of products that our customers purchase on a forward basis, or the percentage of our sales volume that is sold to our customers on a forward basis, could increase our exposure to fluctuations in our profit margins and working capital and materially adversely affect our business, financial condition, results of operations and cash flows.

We offer our customers the opportunity to purchase products from us on a forward basis at prices and delivery dates we propose. Under our forward sales programs, customers generally make an initial cash down payment at the time of order and pay the remaining portion of the contract sales value in advance of the shipment date. Forward sales improve our liquidity by reducing our working capital needs due to the cash payments received from customers in advance of shipment of the product and allow us to improve our production scheduling and planning and the utilization of our manufacturing and distribution assets.

Any cash payments received in advance from customers in connection with forward sales are reflected on our consolidated balance sheets as a current liability until the related orders are shipped, which can take up to several months.

We believe the ability to purchase products on a forward basis is most appealing to our customers during periods of generally increasing prices for nitrogen fertilizers. Our customers may be less willing or even unwilling to purchase products on a forward basis during periods of generally decreasing or stable prices or during periods of relatively high fertilizer prices due to the expectation of lower prices in the future. In addition, our customers may be unwilling to purchase products on a forward basis due to their limited capital resources. Fixing the selling prices of our products, often months in advance of their ultimate delivery to customers, typically causes our reported selling prices and margins to differ from spot market prices and margins available at the time of shipment. In periods of rising fertilizer prices, selling our nitrogen fertilizers on a forward basis may result in lower profit margins than if we had not sold fertilizer on a forward basis.

Operational Risks

Our operations are dependent upon raw materials and utilities provided by third parties, and any delay or interruption in the delivery of raw materials or utilities may adversely affect our business.

We use raw materials, primarily natural gas, and utilities, such as electricity, in the manufacture of our nitrogen products. We purchase raw materials and utilities from third party suppliers. Our natural gas is transported by pipeline to our facilities by third party transportation providers or through the use of facilities owned by third parties. Delays or interruptions in the delivery of raw materials and utilities may be caused by, among other things, extreme weather or natural disasters, unscheduled downtime, labor difficulties or shortages, insolvency of our suppliers or their inability to meet existing contractual arrangements, deliberate sabotage and terrorist incidents, unplanned maintenance or mechanical failures. In addition, the transport of natural gas by pipeline is subject to additional risks, including delays or interruptions caused by capacity constraints, leaks or ruptures. Any delay or interruption in the delivery of raw materials or utilities, even for a limited period, could have a material adverse effect on our business, financial condition, results of operations and cash flows.

Our transportation and distribution activities rely on third party providers and are subject to environmental, safety and regulatory oversight. This exposes us to risks and uncertainties beyond our control that may adversely affect our operations and exposes us to additional liability.

We rely on natural gas pipelines to transport raw materials to our manufacturing facilities. In addition, we rely on railroad, barge, truck, vessel and pipeline companies to coordinate and deliver finished products to our distribution system and to ship finished products to our customers. We also lease rail cars in order to ship finished products. These transportation operations, equipment and services are subject to various hazards and other sources of disruption, including adverse operating conditions on the inland waterway system or on the seas with respect to oceangoing vessels, extreme weather conditions, system failures, unscheduled downtime, labor difficulties or shortages, shutdowns, delays, accidents such as spills and derailments, vessel groundings and other accidents and operating hazards. Additionally, due to the aging infrastructure of certain rail lines, bridges, roadways, pipelines, river locks, and equipment that our third party service providers utilize, we may experience delays in both the receipt of raw materials or the shipment of finished product while repairs, maintenance or replacement activities are conducted. Also, certain third party service providers, such as railroads, have from time to time experienced service delays or shutdowns due to capacity constraints in their systems, operational and maintenance difficulties, blockades, organized labor strikes, weather or safety-related embargoes and delays, and other events, which could impact the shipping of our products and cause disruption in our operations and supply chain.

These transportation operations, equipment and services are also subject to environmental, safety, and regulatory oversight. Due to concerns related to accidents, discharges or other releases of hazardous substances, terrorism or the potential use of fertilizers as explosives, governmental entities could implement new or more stringent regulatory requirements affecting the transportation of raw materials or finished products.

If shipping of our products is delayed or we are unable to obtain raw materials as a result of these transportation companies' failure to operate properly, or if new and more stringent regulatory requirements were implemented affecting transportation operations or equipment, or if there were significant increases in the cost of these services or equipment, our revenues and cost of operations could be adversely affected. In addition, increases in our transportation costs, or changes in such costs relative to transportation costs incurred by our competitors, could have a material adverse effect on our business, financial condition, results of operations and cash flows.

In the United States and Canada, the railroad industry continues various efforts to limit the railroads' potential liability stemming from the transportation of Toxic Inhalation Hazard materials, such as the anhydrous ammonia we transport to and from our manufacturing and distribution facilities. For example, various railroads shift liability to shippers by contract, purport to shift liability to shippers by tariff, or otherwise seek to require shippers to indemnify and defend the railroads from and against liabilities (including in negligence, strict liability, or statutory liability) that may arise from certain acts or omissions of the railroads, third parties that may have insufficient resources, or the Company or from unknown causes or acts of god. These initiatives could materially and adversely affect our operating expenses and potentially our ability to transport anhydrous ammonia and increase our liability for releases of our anhydrous ammonia while in the care, custody and control of the railroads, third parties or us, for which our insurance may be insufficient or unavailable. New or more stringent regulatory requirements also could be implemented affecting the equipment used to ship our raw materials or finished products. Restrictions on service, increases in transportation costs, or changes in such costs relative to transportation costs incurred by our competitors, and any railroad industry initiatives that may impact our ability to transport our products, could have a material adverse effect on our business, financial condition, results of operations and cash flows.

We are reliant on a limited number of key facilities.

Our nitrogen manufacturing facilities are located at nine separate nitrogen complexes, the largest of which is the Donaldsonville complex, which represented approximately 40% of our ammonia production capacity as of December 31, 2023. The suspension of operations at any of these complexes could adversely affect our ability to produce our products and fulfill our commitments, and could have a material adverse effect on our business, financial condition, results of operations and cash flows. Operational disruptions could occur for many reasons, including natural disasters, weather, unplanned maintenance and other manufacturing problems, disease, strikes or other labor unrest or transportation interruptions. For example, our Donaldsonville and Waggaman complexes are located in an area of the United States that experiences extreme weather events, including a relatively high level of hurricane or high wind activity, and several of our other complexes are also located in areas that experience extreme weather events. Extreme weather events, including temperature extremes, depending on their severity and location, have the potential not only to damage our facilities and disrupt our operations, but also to affect adversely the shipping and distribution of our products. Moreover, our facilities may be subject to failure of equipment that may be difficult to replace or have long delivery lead times, due in part to a limited number of suppliers, and could result in operational disruptions.

We are subject to risks relating to our information technology systems, and any technology disruption or cybersecurity incident could negatively affect our operations.

We rely on internal and third-party information technology and computer control systems in many aspects of our business, including internal and external communications, the management of our accounting, financial and supply chain functions and plant operations. If we do not allocate and effectively manage the resources necessary to build, implement and sustain the proper technology infrastructure, we could be subject to transaction errors, inaccurate financial reporting, processing inefficiencies, the loss of customers, business disruptions, or the loss of or damage to our confidential business information due to a security breach. In addition, our information technology systems may be damaged, disrupted or shut down due to attacks by computer hackers, computer viruses, employee error or malfeasance, power outages, hardware failures, telecommunication or utility failures, catastrophes or other unforeseen events, and in any such circumstances our system redundancy and other disaster recovery planning may be ineffective or inadequate. Security breaches of our systems (or the systems of our customers, suppliers or other business partners) could result in the misappropriation, destruction or unauthorized disclosure of confidential information or personal data belonging to us or to our employees, business partners, customers or suppliers, and may subject us to legal liability.

As with most large systems, our information technology systems (and those of our suppliers) have in the past been, and in the future likely will be, subject to computer viruses, malicious codes, unauthorized access and other cyberattacks, and we expect the sophistication and frequency of such attacks to continue to increase. To date, we are not aware of any significant impact on our operations or financial results from such attempts; however, unauthorized access or other types of cyberattacks could disrupt our business operations, result in the loss of assets, and have a material adverse effect on our business, financial condition, or results of operations. Any of the attacks, breaches or other disruptions or damage described above could: interrupt our operations at one or more sites; delay production and shipments; result in the theft of our and our customers' intellectual property and trade secrets; damage customer and business partner relationships and our reputation; result in legal claims and proceedings, liability and penalties under privacy or other laws, or increased costs for security and remediation; or raise concerns regarding our accounting for transactions. Each of these consequences could have an adverse effect on our business, reputation and our financial statements, some of which could be material.

Our business involves the use, storage, and transmission of information about our employees, customers, and suppliers. The protection of such information, as well as our proprietary information, is critical to us. The regulatory environment surrounding information security and privacy is increasingly demanding, with frequent imposition of new requirements and changes to existing requirements. Breaches of our security measures or the accidental loss, inadvertent disclosure, or unapproved dissemination of proprietary information or sensitive or confidential data about us or our employees, customers or suppliers, including the potential loss or disclosure of such information or data as a result of fraud, trickery, or other forms of deception, could expose us or our employees, customers, suppliers or other individuals or entities affected to a risk of loss or misuse of this information, which could ultimately result in litigation and potential legal and financial liability. These events could also damage our reputation or otherwise harm our business.

Acts of terrorism and regulations to combat terrorism could negatively affect our business.

Like other companies with major industrial facilities, we may be targets of terrorist activities. Many of our plants and facilities store significant quantities of ammonia and other materials that can be dangerous if mishandled. Any damage to infrastructure facilities, such as electric generation, transmission and distribution facilities, or injury to employees, who could be direct targets or indirect casualties of an act of terrorism, may affect our operations. Any disruption of our ability to produce or distribute our products could result in a significant decrease in revenues and significant additional costs to replace, repair or insure our assets, which could have a material adverse effect on our business, financial condition, results of operations and cash flows.

Due to concerns related to terrorism or the potential use of certain nitrogen products as explosives, we are subject to various security laws and regulations. In the United States, these security laws include the Maritime Transportation Security Act of 2002 and the Chemical Facility Anti-Terrorism Standards (although this legislation is currently expired as Congress works to reauthorize it). In addition, President Obama issued in 2013 Executive Order 13650 Improving Chemical Facility Safety and Security to improve chemical facility safety in coordination with owners and operators. Governmental entities could implement new or impose more stringent regulations affecting the security of our plants, terminals and warehouses or the transportation and use of fertilizers and other nitrogen products. These regulations could result in higher operating costs or limitations on the sale of our products and could result in significant unanticipated costs, lower revenues and reduced profit margins. We manufacture and sell certain nitrogen products that can be used as explosives. It is possible that governmental entities in the United States or elsewhere could impose additional limitations on the use, sale or distribution of nitrogen products, thereby limiting our ability to manufacture or sell those products, or that illicit use of our products could result in liability for us.

We are subject to risks associated with international operations.

Our international business operations are subject to numerous risks and uncertainties, including difficulties and costs associated with complying with a wide variety of complex laws, treaties and regulations; unexpected changes in regulatory environments; currency fluctuations; tax rates that may exceed those in the United States; earnings that may be subject to withholding requirements; and the imposition of tariffs, exchange controls or other restrictions.

Changes in governmental trade policies can lead to the imposition of new taxes, levies, duties, tariffs or quotas affecting agricultural commodities, fertilizer or industrial products. These can alter or impact costs, trade flows, demand for our products, access to raw materials and other supplies, and regional supply and demand balances for our products.

Our principal reporting currency is the U.S. dollar and our business operations and investments outside the United States increase our risk related to fluctuations in foreign currency exchange rates. The main currencies to which we are exposed, besides the U.S. dollar, are the Canadian dollar, the British pound and the euro. These exposures may change over time as business practices evolve and economic conditions change. We may selectively reduce some foreign currency exchange rate risk by, among other things, requiring contracted purchases of our products to be settled in, or indexed to, the U.S. dollar or a currency freely convertible into U.S. dollars, or hedging through foreign currency derivatives. These efforts, however, may not be effective and could have a material adverse effect on our business, financial condition, results of operations and cash flows.

We are subject to anti-corruption laws and regulations and economic sanctions programs in various jurisdictions, including the U.S. Foreign Corrupt Practices Act of 1977, the United Kingdom Bribery Act 2010, the Canadian Corruption of Foreign Public Officials Act; economic sanctions programs administered by the United Nations (UN), the EU and the Office of Foreign Assets Control of the U.S. Department of the Treasury; and regulations under the Comprehensive Iran Sanctions, Accountability, and Divestment Act of 2010. As a result of doing business internationally, we are exposed to a risk of violating anti-corruption laws and sanctions regulations applicable in those countries where we, our partners or our agents operate. Violations of anti-corruption and sanctions laws and regulations are punishable by civil penalties, including fines, denial of export privileges, injunctions, asset seizures, debarment from government contracts (and termination of existing contracts) and revocations or restrictions of licenses, as well as criminal fines and imprisonment. The violation of applicable laws by our employees, consultants, agents or partners could subject us to penalties and could have a material adverse effect on our business, financial condition, results of operations and cash flows.

We are subject to antitrust and competition laws in various countries throughout the world. We cannot predict how these laws or their interpretation, administration and enforcement will change over time. Changes in antitrust laws globally, or in their interpretation, administration or enforcement, may limit our existing or future operations and growth.

Financial Risks

Our operations and the production and handling of our products involve significant risks and hazards. We are not fully insured against all potential hazards and risks incident to our business. Therefore, our insurance coverage may not adequately cover our losses.

Our operations are subject to hazards inherent in the manufacture, transportation, storage and distribution of chemical products, including ammonia, which is highly toxic and can be corrosive, and ammonium nitrate, which is explosive. These hazards include: explosions; fires; extreme weather and natural disasters; train derailments, collisions, vessel groundings and other transportation and maritime incidents; leaks and ruptures involving storage tanks, pipelines and rail cars; spills, discharges and releases of toxic or hazardous substances or gases; deliberate sabotage and terrorist incidents; mechanical failures; unscheduled plant downtime; labor difficulties and other risks. Some of these hazards can cause bodily injury and loss of life, severe damage to or destruction of property and equipment and environmental damage, and may result in suspension of operations for an extended period of time and/or the imposition of civil or criminal penalties and liabilities.

Our exposure to these risks and hazards is exemplified by a fire and explosion that occurred at a fertilizer storage and distribution facility in West, Texas, in 2013. The fire and explosion resulted in 15 fatalities and claims of injuries to approximately 200 people, and damaged or destroyed a number of homes and buildings around the facility. We did not own or operate the facility or directly sell our products to the facility, but products that we manufactured and sold to others were delivered to the facility and may have been stored at the facility at the time of the incident. We were named as defendants along with other companies in lawsuits, in which the claims against us have since been resolved, alleging various theories of negligence, strict liability, and breach of warranty in connection with the incident.

We maintain property, business interruption, casualty and liability insurance policies, but we are not fully insured against all potential hazards and risks incident to our business. If we were to incur significant liability for which we were not fully insured, it could have a material adverse effect on our business, financial condition, results of operations and cash flows. We are

subject to various self-insured retentions, deductibles and limits under these insurance policies. The policies also contain exclusions and conditions that could have a material adverse impact on our ability to receive indemnification thereunder. Our policies are generally renewed annually. As a result of market conditions, our premiums, self-insured retentions and deductibles for certain insurance policies can increase substantially and, in some instances, certain insurance may become unavailable or available only for reduced amounts of coverage. In addition, significantly increased costs could lead us to decide to reduce, or possibly eliminate, coverage. There can be no assurance that we will be able to buy and maintain insurance with adequate limits and reasonable pricing terms and conditions.

Our indebtedness could adversely affect our cash flow, prevent us from fulfilling our obligations and impair our ability to pursue or achieve other business objectives.

As of December 31, 2023, we had approximately $3.0 billion of total funded indebtedness, consisting primarily of unsecured senior notes with varying maturity dates between 2026 and 2044, or approximately 26% of our total capitalization (total debt plus total equity), and an additional $750 million of unsecured senior borrowing availability (reflecting no outstanding borrowings and no outstanding letters of credit) for general corporate purposes under our revolving credit agreement (the Revolving Credit Agreement). Our debt service obligations will have an impact on our earnings and cash flow for so long as the indebtedness is outstanding.

Our indebtedness could, as a result of our debt service obligations or through the operation of the financial and other restrictive covenants to which we are subject under the agreements and instruments governing that indebtedness and otherwise, have important consequences. For example, it could:

- make it more difficult for us to pay or refinance our debts as they become due during adverse economic and industry conditions because any related decrease in revenues could cause us not to have sufficient cash flows from operations to make our scheduled debt payments;
- cause us to be less able to take advantage of significant business opportunities, such as acquisition opportunities, and to react to changes in market or industry conditions;
- cause us to use a portion of our cash flow from operations for debt service, reducing the availability of cash to fund working capital and capital expenditures, and other business activities;
- cause us to be more vulnerable to general adverse economic and industry conditions;
- expose us to the risk of increased interest rates because certain of our borrowings, including borrowings under the Revolving Credit Agreement, could be at variable rates of interest;
- make us more leveraged than some of our competitors, which could place us at a competitive disadvantage;
- restrict our ability to pay dividends on our common stock or utilize excess cash to repurchase shares of our common stock;
- limit our ability to borrow additional amounts to fund working capital, capital expenditures and other general corporate purposes; and
- result in our credit ratings being downgraded, which could increase the cost of further borrowings.

We expect to consider options to refinance our outstanding indebtedness from time to time. Our ability to obtain any financing, whether through the issuance of new debt securities or otherwise, and the terms of any such financing are dependent on, among other things, our financial condition, financial market conditions within our industry and generally, credit ratings and numerous other factors, including factors beyond our control. Consequently, in the event that we need to access the credit markets, including to refinance our debt, there can be no assurance that we will be able to obtain financing on acceptable terms or within an acceptable timeframe, if at all. An inability to obtain financing with acceptable terms when needed could have a material adverse effect on our business, financial condition, results of operations and cash flows.

The terms of our existing indebtedness allow us to incur significant additional debt. If we incur additional indebtedness, the risks that we face as a result of our leverage could intensify. If our financial condition or operating results deteriorate, our relations with our creditors, including the holders of our outstanding debt securities, the lenders under the Revolving Credit Agreement and our suppliers, may be materially and adversely affected.

A failure to satisfy the financial maintenance covenant under the Revolving Credit Agreement or a breach of the covenants under any of the agreements governing our indebtedness could limit the borrowing availability under the Revolving Credit Agreement or result in an event of default under such agreements.

Our ability to comply with the covenants in the agreements and instruments governing our indebtedness, including the consolidated net leverage ratio maintenance covenant contained in the Revolving Credit Agreement, will depend upon our future performance and various other factors, such as market prices for our nitrogen products, natural gas prices and other business, competitive and regulatory factors, many of which are beyond our control. We may not be able to maintain compliance with all of these covenants. In that event, we may not be able to access the borrowing availability under the

Revolving Credit Agreement and we would need to seek an amendment to our debt agreements or would need to refinance our indebtedness. There can be no assurance that we can obtain future amendments or waivers of our debt agreements and instruments, or refinance our debt, and, even if we were able to do so, such relief might only last for a limited period, potentially necessitating additional amendments, waivers or refinancings. Any noncompliance by us with the covenants under our debt agreements and instruments could result in an event of default under those debt agreements and instruments. An event of default under an agreement or instrument governing any of our indebtedness may allow our creditors to accelerate the related debt and may result in the acceleration of any other debt to which a cross-acceleration or cross-default provision applies. If our lenders or holders of our debt securities accelerate the repayment of borrowings, we may be forced to liquidate certain assets to repay all or part of our indebtedness, which could materially and adversely impair our business operations. An event of default under the Revolving Credit Agreement would permit the lenders thereunder to terminate all commitments to extend further credit under the Revolving Credit Agreement. In the event our creditors accelerate the repayment of our indebtedness, we cannot assure that we would have sufficient assets to make such repayment.

Potential future downgrades of our credit ratings could adversely affect our access to capital, cause vendors to change their credit terms for doing business with us, and could otherwise have a material adverse effect on us.

As of February 12, 2024, our corporate credit rating by S&P Global Ratings is BBB with a stable outlook; our corporate credit rating by Moody's Investor Services, Inc. is Baa3 with a positive outlook; and our corporate credit rating with Fitch Ratings, Inc. is BBB with a stable outlook. These ratings and our current credit condition affect, among other things, our ability to access new capital, especially debt, as well as the payment terms that vendors are willing to provide us. Negative changes in these ratings may result in more stringent covenants and higher interest rates under the terms of any new debt, and could cause vendors to shorten our payment terms, require us to pay in advance for materials or services, or provide letters of credit, security, or other credit enhancements in order to do business with us.

Tax matters, including changes in tax laws or rates, adverse determinations by taxing authorities and imposition of new taxes could adversely affect our results of operations and financial condition.

We are subject to taxes in the United States, where most of our operations are located, and in several foreign jurisdictions where our subsidiaries are organized or conduct business. Tax laws or rates in the various jurisdictions in which we operate may be subject to significant change. Our future effective tax rate could also be affected by changes in our mix of earnings from jurisdictions with differing statutory tax rates and tax systems, changes in valuation of deferred tax assets and liabilities or changes in tax laws or their interpretation.

We are also subject to regular reviews, examinations and audits by the Internal Revenue Service (IRS) and other taxing authorities in jurisdictions where we conduct business. Although we believe our tax estimates are reasonable, if a taxing authority disagrees with the positions we have taken, we could face additional tax liabilities, including interest and penalties. There can be no assurance that payment of such additional amounts upon final adjudication of any disputes will not have a material impact on our financial condition, results of operations and cash flows.

We have used the cash we generate outside the United States primarily to fund development of our business in non-U.S. jurisdictions. If the funds generated by our U.S. business are not sufficient to meet our need for cash in the United States or if cash generated outside the United States exceeds the needs of such non-U.S. operations, we may repatriate a portion of our future international earnings to the United States. Under the tax laws of the foreign countries in which we operate, those international earnings could be subject to withholding taxes when repatriated; therefore, the repatriation of those earnings could result in an increase in our worldwide effective tax rate and an increase in our use of cash to pay these taxes.

We also need to comply with other new, evolving or revised tax laws and regulations. The enactment of, or increases in, carbon taxes, tariffs or value added taxes, or other changes in the application of existing taxes, in markets in which we are currently active, or may be active in the future, or on specific products that we sell or with which our products compete, could have an adverse effect on our financial condition and results of operations.

Our business is subject to risks involving derivatives and the risk that our hedging activities might not prevent losses.

From time to time, we utilize natural gas derivatives to hedge our financial exposure to the price volatility of natural gas, the principal raw material we use in the production of nitrogen-based products. We may use natural gas futures, swaps and option contracts traded in over-the-counter markets or on exchanges. In addition, from time to time, we use fixed-price, physical purchase and sales contracts to hedge our exposure to natural gas price volatility. In order to manage our exposure to changes in foreign currency exchange rates, we may from time to time use foreign currency derivatives (primarily forward exchange contracts).

Our use of derivatives can result in volatility in reported earnings due to the unrealized mark-to-market adjustments that occur from changes in the value of the derivatives that do not qualify for, or to which we do not apply, hedge accounting. To the extent that our derivative positions lose value, we may be required to post collateral with our counterparties, adversely affecting our liquidity.

Hedging arrangements are imperfect and unhedged risks will always exist. In addition, our hedging activities may themselves give rise to various risks that could adversely affect us. For example, we are exposed to counterparty credit risk when our derivatives are in a net asset position. The counterparties to our derivatives are multi-national commercial banks, major financial institutions or large energy companies.

Our liquidity could be negatively impacted by a counterparty default on settlement of one or more of our derivative financial instruments or by the triggering of any cross default provisions or credit support requirements against us. Additionally, the International Swaps and Derivative Association master netting arrangements for most of our derivative instruments contain credit-risk-related contingent features, such as cross default provisions and credit support requirements. In the event of certain defaults or a credit ratings downgrade, our counterparty may request early termination and net settlement of certain derivative trades or may require us to collateralize derivatives in a net liability position.

At other times we may not utilize derivatives or derivative strategies to hedge certain risks or to reduce the financial exposure of price volatility. As a result, we may not prevent certain material adverse impacts that could have been mitigated through the use of derivative strategies.

Environmental and Regulatory Risks

We are subject to numerous environmental, health and safety laws, regulations and permitting requirements, as well as potential environmental liabilities, which may require us to make substantial expenditures or modify business plans.

We are subject to numerous environmental, health and safety laws and regulations in the United States, Canada, the United Kingdom, the EU, Trinidad and other locations, including laws and regulations relating to the generation and handling of hazardous substances and wastes; the introduction of new chemicals or substances into a market; the cleanup of hazardous substance releases; the discharge of regulated substances to land, air or water; and the demolition and cleanup of existing plant sites upon permanent closure. In the United States, these laws include the Clean Air Act, the Clean Water Act, the Resource Conservation and Recovery Act, the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA), the Toxic Substances Control Act and various other federal, state, provincial, local and international laws. In November 2021, the Infrastructure Investment and Jobs Act reinstated and doubled the Superfund tax on chemicals, including ammonia and nitric acid. These taxes were put in place from July 1, 2022 through December 31, 2031 and apply to both the domestic production and import of ammonia and nitric acid products, except to the extent such products are used in fertilizer or animal feed, used as fuel or exported.

As a producer of nitrogen products working with hazardous substances, our business faces risks of spills, discharges or other releases of those substances into the environment. Certain environmental laws, including CERCLA, impose joint and several liability, without regard to fault, for cleanup costs on persons who have disposed of or released hazardous substances into the environment. Given the nature of our business, we have incurred, are incurring currently, and are likely to incur periodically in the future, liabilities under CERCLA and other environmental cleanup laws at our current facilities or facilities previously owned or operated by us or other acquired businesses, adjacent or nearby third-party facilities or offsite disposal locations. The costs associated with future cleanup activities that we may be required to conduct or finance may be material. Additionally, we may become liable to third parties for damages, including personal injury and property damage, resulting from the disposal or release of hazardous substances into the environment.

Violations of environmental, health and safety laws can result in substantial penalties, court orders to install pollution-control equipment, civil and criminal sanctions, permit revocations and facility shutdowns. Environmental, health and safety laws change regularly and have tended to become more stringent over time. As a result, we have not always been and may not always be in compliance with all environmental, health and safety laws and regulations. We may be subject to more stringent enforcement of existing or new environmental, health and safety laws in the future. Additionally, future environmental, health and safety laws and regulations or reinterpretation of or changes to current laws and regulations may require us to make substantial expenditures or modify business plans. Our costs to comply with, or any liabilities under, these laws and regulations could have a material adverse effect on our business, financial condition, results of operations and cash flows.

From time to time, our production, distribution or storage of anhydrous ammonia and other hazardous or regulated substances has resulted in accidental releases that have temporarily disrupted our operations and/or resulted in liability for administrative penalties, cleanup costs, and/or claims for personal injury. To date, our costs to resolve these liabilities have not

been material. However, we could incur significant costs if our liability coverage is not sufficient to pay for all or a large part of any judgments against us, or if our insurance carrier refuses coverage for these losses.

We hold numerous environmental and other governmental permits and approvals authorizing operations at each of our facilities. Expansion or modification of our operations is predicated upon securing necessary environmental or other permits or approvals, and in some cases, the ability of our partners, lessors and other third-party providers, as applicable, to secure such permits or approvals. More stringent environmental, health and safety laws and regulations, a reinterpretation of or changes to current laws and regulations, or community opposition to permits and approvals could make it more difficult to obtain necessary governmental permits (including renewals of our existing permits) or approvals. In addition, a focus on the cumulative impact of industrial operations on minority, lower income, and other historically underrepresented and/or disadvantaged communities could impact decisions relating to the issuance of new or renewal of existing permits to the extent that our operations are located in the vicinity of such communities. A decision by a government agency to deny or delay issuing a new or renewed regulatory permit or approval, or to revoke or substantially modify an existing permit or approval, or a determination that we have violated a law or permit could have a material adverse effect on our ability to continue operations at our facilities and on our business, financial condition, results of operations and cash flows.

Future regulatory or legislative restrictions on greenhouse gas (GHG) emissions in the jurisdictions in which we operate or conduct business could materially adversely affect our business, financial condition, results of operations and cash flows.

Our production facilities emit GHGs, such as carbon dioxide and nitrous oxide, and natural gas, a fossil fuel that releases methane when extracted from the earth, is a primary raw material used in our nitrogen production process. Because conventional ammonia production generates CO_2 as an unavoidable chemical byproduct, ammonia production globally is considered an emissions- and energy-intensive industry. We are subject to GHG regulations in the United Kingdom, Canada and the United States. In the United States, our existing facilities, which are considered large emitters of GHGs, currently are only subject to GHG emissions reporting obligations. New facilities that we build, or existing facilities that we modify in the future, could also be subject to GHG emissions standards included in their air permits.

Our manufacturing plants in the Alberta and Ontario provinces of Canada are subject to regulations that impose a price on excess GHG emissions. These regulations establish carbon dioxide equivalent (CO_2e) emissions standards applicable to our facilities in terms of emissions per unit of production, with each province using different formulas for establishing these intensity limits and changes in these limits over time (and federal law applying if provincial plans are not considered sufficiently stringent). If CO_2e emissions exceed the applicable limits, the excess emissions must be offset, either through obtaining qualifying emission credits or offsets or by making a payment for each ton of excess emissions. Pursuant to Canadian regulations, emissions are subject to an annual increase in price on CO_2e through 2030, and these GHG regulations became more stringent effective January 1, 2023 and became subject to an increasing carbon price of CAD $80/tonne on January 1, 2024.

Increasing concern over the effects of climate change is driving countries to establish ever more ambitious GHG reduction targets. Approximately 200 countries, including the United States, Canada, the United Kingdom and the members of the EU, have joined the Paris Agreement, an international agreement intended to provide a framework pursuant to which the parties to the agreement will attempt to hold the increase in global average temperatures to below 2°C above pre-industrial levels and to pursue efforts to limit the temperature increase to 1.5ºC above pre-industrial levels. The United States, Canada and the United Kingdom have also announced national targets to reduce GHG emissions in each case by 40% or more by 2030 as compared to 2005 levels and have led or joined other initiatives to spur faster reductions related to carbon dioxide, methane and other GHGs.

In addition, the Biden administration has issued several executive orders focused on climate change to promote more active management of these issues across the executive branch, including by the EPA and the Departments of Energy, Agriculture, Interior, Transportation and Treasury, and has issued proposed and final regulations related to methane and other GHG reduction efforts. In late December 2023, the Internal Revenue Service issued proposed guidance on the 45V hydrogen production tax credit created by the 2022 Inflation Reduction Act (IRA). Final implementation of the IRA may impact the market for green and low-carbon hydrogen and associated ammonia products. In addition, pursuant to the IRA, the EPA will begin to assess a methane fee on certain oil and natural gas facilities for methane emissions that exceed a designated threshold. This fee will apply to methane emissions from 2024 onward, and, under the proposed rules issued by the EPA in January 2024, will be assessed beginning in 2025. In May 2023, the EPA proposed new regulations requiring certain types of power plants to change their operations and/or install emissions control equipment to reduce GHG emissions. While these proposed regulations do not apply to us, they could inform future EPA regulations that may apply to us and require us to reduce our GHG emissions.

The EU finalized its overall carbon border adjustment mechanism in May 2023. During the interim phase, covering imports, including nitrogenous fertilizers, entering the EU from the fourth quarter of 2023 through the fourth quarter of 2025,

importers must file quarterly reports on the emissions intensity of covered products. For imports that enter the EU starting in 2026, charges will be required for emissions over certain thresholds, with the EU still to set forth additional details. Other governments are also considering border adjustment mechanisms for carbon intensive products. The imposition of any carbon border adjustment taxes may impact investment and trade flows, which could adversely impact our business.

More stringent GHG regulations, if they are enacted, are likely to have a significant impact on us, because our production facilities emit GHGs such as carbon dioxide and nitrous oxide and because natural gas, a fossil fuel that releases methane when extracted from the earth, is a primary raw material used in our nitrogen production process. Regulation of GHGs may require us to make changes in our operating activities that would increase our operating costs, reduce our efficiency, limit our output, require us to make capital improvements to our facilities, increase our costs for or limit the availability of energy, raw materials or transportation, or otherwise materially adversely affect our business, financial condition, results of operations and cash flows. Changes could also be made to tax or other regulatory policies related to decarbonization, electricity generation or clean energy that could impact our business and investment decisions. In addition, to the extent that GHG restrictions are not imposed in countries where our competitors operate or are less stringent than regulations that may be imposed in the United States, Canada or the United Kingdom, our competitors may have cost or other competitive advantages over us.

Strategic Risks

The market for green and low-carbon (blue) ammonia may be slow to develop, may not develop to the size expected or may not develop at all. Moreover, we may not be successful in the development and implementation of our green and low-carbon ammonia projects in a timely or economic manner, or at all, due to a number of factors, many of which are beyond our control.

The market for green and low-carbon ammonia is developing and evolving, may not develop to the size or at the rate we expect, and is dependent in part on the developing market for green and low-carbon hydrogen, for which ammonia can serve as a transport and storage mechanism. These markets are heavily influenced by demand for clean energy, technology evolution and federal, state and local government laws, regulations and policies concerning carbon emissions, renewable electricity, clean energy, and corporate accountability in the United States and abroad. These factors may also affect the market criteria for green and low-carbon ammonia, including the degree of reduction of direct GHG emissions and the requirements of renewable electricity.

We believe the demand for green and low-carbon ammonia could take several years to materialize and then ten or more years to fully develop and mature, and we cannot be certain that this market or the market for green and low-carbon hydrogen will grow to the size or at the rate we expect or at all. Hydrogen currently accounts for less than 1% of the world's energy needs.

The recognition and acceptance of green and low-carbon ammonia as a transport and storage mechanism for green and low-carbon hydrogen, the use of green and low-carbon ammonia as a fuel in its own right, the use of green and low-carbon ammonia as a fertilizer, and the development and growth of end market demand and applications for green and low-carbon hydrogen and green and low-carbon ammonia are uncertain and dependent on a number of factors outside of our control. These factors include, among others, the extent to which and rate at which cost competitive global renewable energy capacity increases, the pricing of traditional and alternative sources of energy, the realization of technological improvements required to increase the efficiency and lower the costs of production of green and low-carbon ammonia, the regulatory environment, the rate and extent of infrastructure investment and development which may be affected by the relevant parties' ability to obtain permits for these investments, the availability of tax benefits and other incentives, the implementation of policy in foreign jurisdictions providing economic support for or otherwise mandating decarbonization and our ability to provide green and low-carbon ammonia offerings cost-effectively. In addition, further development of alternative decarbonization technologies may result in viable alternatives to the use of low-carbon ammonia for many potential decarbonization applications, resulting in lower than expected market demand growth relative to our current expectations. If a sustainable market for green or low-carbon ammonia or hydrogen fails to develop, develops more slowly than we anticipate, or develops in a way that is not viable to serve with our assets and capabilities, we may decide not to implement, or may not be successful in implementing, one or more elements of our multi-year strategic plan.

Our clean energy strategy also depends on the realization of certain technical improvements required to increase the efficiency and lower the costs of production of green and low-carbon ammonia. Over time, as we seek to convert additional existing facilities to green and low-carbon production and further expand our green and low-carbon ammonia production capacity, we may face operational difficulties and execution risks related to the design, development and construction. If our assumptions about the engineering and project execution requirements necessary to successfully build or convert the facility capacity that we are contemplating and to scale up to larger production quantities prove to be incorrect, we may be unable to produce substantial quantities of green or low-carbon ammonia, and the cost to construct such green and low-carbon ammonia

facilities, or the production costs associated with the operation of such facilities, may be higher than we project. The production of low-carbon ammonia depends to a large extent upon the ability of third parties to develop class VI carbon sequestration wells and carbon dioxide transportation pipelines, which currently do not exist at large scale and are subject to a permitting process and operational risks, which may result in delays, impact viability in some or all situations, or create long-term liabilities.

Recently, many proposed green and low-carbon ammonia projects have been announced or considered, and future hydrogen, energy, or environmental/carbon policies may support development of additional nitrogen production in locations outside North America, including Europe, Australia, and the Middle East. In the event that the growth in supply of green and low-carbon ammonia and green and low-carbon hydrogen exceeds the growth in demand for those products, the resulting unfavorable supply and demand balance could lead to lower selling prices than we expect for many of our products, which could negatively affect our business, financial condition, results of operations and cash flows.

We may not be successful in the expansion of our business.

We routinely consider possible expansions of our business, both within the United States and elsewhere. Major investments in our business, including acquisitions, partnerships, joint ventures, business combination transactions or other major investments, such as our green and low-carbon ammonia projects, require significant managerial resources, the diversion of which from our other activities or opportunities may negatively affect the existing operations of our business. We may be unable to identify or successfully compete for certain acquisition targets, which may hinder or prevent us from acquiring a target or completing other transactions. The risks of any expansion of our business through investments, acquisitions, partnerships, joint ventures or business combination transactions may increase due to the significant capital and other resources that we may have to commit to any such expansion, which may not be recoverable if the expansion initiative to which they were devoted is ultimately not implemented. In addition, these efforts may require capital resources that could otherwise be used for the improvement and expansion of our existing business. As a result of these and other factors, including general economic risk, we may not be able to realize our projected returns or other expected benefits from acquisitions, partnerships, joint ventures, business combination transactions or other major investments. Among the risks associated with the pursuit and consummation of acquisitions, partnerships, joint ventures or other major investments or business combinations are those involving:

- difficulties in integrating the parties' operations, systems, technologies, products, cultures, and personnel;
- incurrence of significant transaction-related expenses;
- potential integration or restructuring costs;
- potential impairment charges related to the goodwill, intangible assets or other assets to which any such transaction relates, in the event that the economic benefits of such transaction prove to be less than anticipated;
- other unanticipated costs associated with such transactions;
- our ability to achieve operating and financial efficiencies, synergies and cost savings;
- our ability to obtain the desired financial or strategic benefits from any such transaction;
- the parties' ability to retain key business relationships, including relationships with employees, customers, partners and suppliers;
- potential loss of key personnel;
- entry into markets or involvement with products with which we have limited current or prior experience or in which competitors may have stronger positions;
- assumption of contingent liabilities, including litigation;
- exposure to unanticipated liabilities, including litigation;
- differences in the parties' internal control environments, which may require significant time and resources to resolve in conformity with applicable legal and accounting standards;
- increased scope, geographic diversity and complexity of our operations;
- the tax effects of any such transaction; and
- the potential for costly and time-consuming litigation, including stockholder lawsuits.

Moreover, legal proceedings or other risks from acquisitions and other business combinations may arise years after a transaction has been completed and may involve matters unrelated to the business acquired. For example, in 2022, we were named along with other parties in certain product liability actions relating to a product containing the herbicide paraquat, which was allegedly sold, manufactured, distributed and/or marketed by Terra Industries Inc. (Terra) before it exited such lines of business, which exit occurred more than ten years before CF Holdings acquired Terra in April 2010.

In addition, most major capital projects are dependent on the availability and performance of engineering firms, construction firms, equipment and material suppliers, transportation providers and other vendors necessary to design and implement those projects on a timely basis and on acceptable terms. Major investments such as capital improvements at our

facilities are subject to a number of risks, any of which could prevent us from completing capital projects in a timely or economic manner or at all, including, without limitation, cost overruns, non-performance of third parties, the inability to obtain necessary permits or other permitting matters, adverse weather, defects in materials and workmanship, labor and raw material shortages, transportation constraints, changes to international trade-related policy, engineering and construction change orders, errors in design, construction or start-up, and other unforeseen difficulties.

International acquisitions, partnerships, joint ventures, investments or business combinations and other international expansions of our business involve additional risks and uncertainties, including, but not limited to:

- the impact of particular economic, tax, currency, political, legal and regulatory risks associated with specific countries;
- challenges caused by distance and by language and cultural differences;
- difficulties and costs of complying with a wide variety of complex laws, treaties and regulations;
- unexpected changes in regulatory environments;
- political and economic instability, including the possibility for civil unrest;
- nationalization of properties by foreign governments;
- tax rates that may exceed those in the United States, and earnings that may be subject to withholding requirements;
- the imposition of tariffs, exchange controls or other restrictions; and
- the impact of currency exchange rate fluctuations.

If we finance acquisitions, partnerships, joint ventures, business combination transactions or other major investments by issuing equity or convertible or other debt securities or loans, our existing stockholders may be diluted or we could face constraints under the terms of, and as a result of the repayment and debt-service obligations under, the additional indebtedness. A business combination transaction between us and another company could result in our stockholders receiving cash or shares of another entity on terms that such stockholders may not consider desirable. Moreover, the regulatory approvals associated with a business combination may result in divestitures or other changes to our business, the effects of which are difficult to predict.

FORWARD LOOKING STATEMENTS

From time to time, in this Annual Report on Form 10-K as well as in other written reports and oral statements, we make forward-looking statements that are not statements of historical fact and may involve a number of risks and uncertainties. These statements relate to analyses and other information that are based on forecasts of future results and estimates of amounts not yet determinable. These statements may also relate to our prospects, future developments and business strategies. We have used the words "anticipate," "believe," "could," "estimate," "expect," "intend," "may," "plan," "predict," "project," "will" or "would" and similar terms and phrases, including references to assumptions, to identify forward-looking statements in this document. These forward-looking statements are made based on currently available competitive, financial and economic data, our current expectations, estimates, forecasts and projections about the industries and markets in which we operate and management's beliefs and assumptions concerning future events affecting us. These statements are not guarantees of future performance and are subject to risks, uncertainties and factors relating to our operations and business environment, all of which are difficult to predict and many of which are beyond our control. Therefore, our actual results may differ materially from what is expressed in or implied by any forward-looking statements. We want to caution you not to place undue reliance on any forward-looking statements. We do not undertake any responsibility to release publicly any revisions to these forward-looking statements to take into account events or circumstances that occur after the date of this document. Additionally, we do not undertake any responsibility to provide updates regarding the occurrence of any unanticipated events which may cause actual results to differ from those expressed or implied by the forward-looking statements contained in this document.

Important factors that could cause actual results to differ materially from our expectations are disclosed under "Risk Factors" and elsewhere in this Annual Report on Form 10-K. Such factors include, among others:

- the cyclical nature of our business and the impact of global supply and demand on our selling prices;
- the global commodity nature of our nitrogen products, the conditions in the international market for nitrogen products, and the intense global competition from other producers;
- conditions in the United States, Europe and other agricultural areas, including the influence of governmental policies and technological developments on the demand for our fertilizer products;
- the volatility of natural gas prices in North America and the United Kingdom;
- weather conditions and the impact of adverse weather events;
- the seasonality of the fertilizer business;
- the impact of changing market conditions on our forward sales programs;
- difficulties in securing the supply and delivery of raw materials or utilities, increases in their costs or delays or interruptions in their delivery;
- reliance on third party providers of transportation services and equipment;
- our reliance on a limited number of key facilities;
- risks associated with cybersecurity;
- acts of terrorism and regulations to combat terrorism;
- risks associated with international operations;
- the significant risks and hazards involved in producing and handling our products against which we may not be fully insured;
- our ability to manage our indebtedness and any additional indebtedness that may be incurred;
- our ability to maintain compliance with covenants under our revolving credit agreement and the agreements governing our indebtedness;
- downgrades of our credit ratings;
- risks associated with changes in tax laws and disagreements with taxing authorities;
- risks involving derivatives and the effectiveness of our risk management and hedging activities;
- potential liabilities and expenditures related to environmental, health and safety laws and regulations and permitting requirements;
- regulatory restrictions and requirements related to greenhouse gas emissions;
- the development and growth of the market for green and low-carbon (blue) ammonia and the risks and uncertainties relating to the development and implementation of our green and low-carbon ammonia projects; and
- risks associated with expansions of our business, including unanticipated adverse consequences and the significant resources that could be required.

ITEM 1B. UNRESOLVED STAFF COMMENTS.

None.

ITEM 1C. CYBERSECURITY.

Cybersecurity risk management, including our processes for assessing, identifying and managing material risks from cybersecurity threats, is an integral part of our overall enterprise risk management (ERM) program. The ERM program includes an annual assessment process designed to identify risks, including those from cybersecurity threats, that could affect us and the achievement of our objectives; to understand, assess, and prioritize those risks; and to facilitate the implementation of risk management strategies and processes across the company that are responsive to the company's risk profile, business strategies, and specific material risk exposures. The ERM program seeks to integrate consideration of risk and risk management into business decision-making throughout the company, including through the implementation of policies and procedures intended to ensure that necessary information with respect to material risks, including material risks from cybersecurity threats, is transmitted to senior executives and, as appropriate, to the Board of Directors (Board) or relevant committees. The Board regularly reviews and discusses with the key members of management responsible for management of risk the guidelines and policies governing the ERM process, the key risks identified in the ERM process, the likelihood of occurrence and the potential impact assigned to those risks by management, and the risk mitigation strategies in each instance.

The Audit Committee of the Board provides oversight in connection with management's cybersecurity efforts. The Audit Committee receives periodic reports summarizing threat detection and mitigation plans, audits of internal controls, training and certification and other cyber priorities and initiatives, as well as timely updates from senior leaders on material incidents relating to cybersecurity. The Audit Committee also receives regular reports on the efficacy of our cybersecurity risks and related policies and procedures from our chief information officer and other members of senior management who are tasked with monitoring cybersecurity risks. Our chief information officer oversees a dedicated team of certified cybersecurity professionals, with an average of over 12 years of relevant experience.

Our cybersecurity strategy prioritizes protection, detection, analysis, and response to known, anticipated or unexpected cyber threats, effective management of cyber risks and resilience against cyber incidents. We maintain a formal cybersecurity program structured around the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF), a voluntary framework created by industry and the U.S. government to promote the protection of our infrastructure from cybersecurity risks. We contract with an external auditing firm to assess our cybersecurity controls relative to industry peers using the NIST CSF, which has five functions: identify, protect, detect, respond and recover. We consistently evaluate the threat landscape, adopting a multifaceted approach to cybersecurity risks through a zero-trust strategy focusing on prevention, detection, and mitigation, which includes the following programs and practices:

- Our cybersecurity team conducts an annual review of cybersecurity risks at the ERM level, integrating significant cybersecurity risks into our overall ERM program. We remain committed to increasing investments in cybersecurity, which includes providing additional training for end-users, adopting a zero-trust methodology, identifying and safeguarding critical assets, and reinforcing monitoring and alerting capabilities. Our proactive approach involves regular testing of defenses through simulations and penetration tests, both technically and through a comprehensive review of operational policies and procedures. At the managerial level, our cybersecurity team consistently monitors alerts and holds regular meetings to discuss threat levels, trends, and remediation strategies. Additionally, we conduct periodic external penetration tests and maturity testing to assess the effectiveness of our security controls, including processes, procedures, and our readiness to face the evolving threat landscape.
- We consider and assess the cybersecurity risks associated with the utilization of third-party service providers under our third-party risk management program. Pursuant to the program, we evaluate security and data privacy controls prior to sharing or authorizing the hosting of sensitive data in computing environments managed by third parties. In addition, our standard terms and conditions with third-party service providers feature contractual provisions mandating specific security protections.
- Our cybersecurity incident response plan is designed to detect and address potential threats that may impact the confidentiality, integrity, and availability of our technology systems. The response plan includes coordinated processes for handling security and data privacy incidents, encompassing communication and effective response.
- Our global business continuity program includes information technology disaster recovery, supporting resilience in both our business and information technology.

To date, we have not identified any risks from cybersecurity threats, including as a result of any previous cybersecurity incidents, that have materially affected or that we believe are reasonably likely to materially affect our business strategy, results

of operations, or financial condition. We cannot, however, eliminate all risks from cybersecurity threats or provide assurances that we have not experienced an undetected cybersecurity incident. For more information about these risks, see the disclosure in Item 1A. Risk Factors under "Operational Risks—We are subject to risks relating to our information technology systems, and any technology disruption or cybersecurity incident could negatively affect our operations."

ITEM 2. PROPERTIES.

Information regarding our facilities and properties is included in Item 1. Business—Nitrogen Manufacturing Facilities and Item 1. Business—Storage Facilities and Other Properties.

ITEM 3. LEGAL PROCEEDINGS.

For information on pending proceedings relating to environmental remediation matters, see Item 1. Business—Environmental, Health and Safety—CERCLA/Remediation Matters.

ITEM 4. MINE SAFETY DISCLOSURES.

Not applicable.

PART II

ITEM 5. MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES.

Our common stock is traded on the New York Stock Exchange under the symbol "CF." As of February 12, 2024, there were 680 stockholders of record.

The following table sets forth share repurchases, on a trade date basis, for each of the three months of the quarter ended December 31, 2023:

	Issuer Purchases of Equity Securities			
Period	Total number of shares (or units) purchased	Average price paid per share (or unit)[1]	Total number of shares (or units) purchased as part of publicly announced plans or programs[2]	Maximum number (or approximate dollar value) of shares (or units) that may yet be purchased under the plans or programs (in thousands)[2]
October 1, 2023 - October 31, 2023	1,298 [3]	$ 80.27	—	$ 2,800,052
November 1, 2023 - November 30, 2023	2,870,066 [4]	78.40	2,870,023	2,575,052
December 1, 2023 - December 31, 2023	—	—	—	2,575,052
Total	2,871,364	78.40	2,870,023	

(1) Average price paid per share of CF Industries Holdings, Inc. (CF Holdings) common stock repurchased under the 2022 Share Repurchase Program, as defined below, is the execution price, excluding commissions paid to brokers and excise taxes.

(2) On November 2, 2022, we announced that our Board of Directors authorized the repurchase of up to $3 billion of CF Holdings common stock, which is effective through December 31, 2025 (the 2022 Share Repurchase Program). This share repurchase program is discussed in Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations—Liquidity and Capital Resources—Share Repurchase Programs and in Note 20—Stockholders' Equity, in the notes to consolidated financial statements included in Item 8. Financial Statements and Supplementary Data.

(3) Represents shares withheld to pay employee tax obligations upon the lapse of restrictions on restricted stock units.

(4) Includes 43 shares withheld to pay employee tax obligations upon the lapse of restrictions on restricted stock units.

ITEM 6. [RESERVED]

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS.

You should read the following discussion and analysis in conjunction with the consolidated financial statements and related notes included in Item 8. Financial Statements and Supplementary Data. All references to "CF Holdings," "we," "us," "our" and "the Company" refer to CF Industries Holdings, Inc. and its subsidiaries, except where the context makes clear that the reference is to CF Industries Holdings, Inc. only and not its subsidiaries. All references to "CF Industries" refer to CF Industries, Inc., a 100% owned subsidiary of CF Industries Holdings, Inc. References to tons refer to short tons and references to tonnes refer to metric tons. Notes referenced in this discussion and analysis refer to the notes to consolidated financial statements that are found in Item 8. Financial Statements and Supplementary Data—Notes to Consolidated Financial Statements. For a discussion and analysis of the year ended December 31, 2022 compared to the year ended December 31, 2021, see Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations in our 2022 Annual Report on Form 10-K filed with the Securities and Exchange Commission (SEC) on February 23, 2023. The following is an outline of the discussion and analysis included herein:

- *Overview of CF Holdings*
- *Market Conditions*
- *Financial Executive Summary*
- *Acquisition of Waggaman Ammonia Production Facility*
- *Items Affecting Comparability of Results*
- *Consolidated Results of Operations*
- *Operating Results by Business Segment*
- *Liquidity and Capital Resources*
- *Critical Accounting Estimates*
- *Recent Accounting Pronouncements*

Overview of CF Holdings

Our Company

Our mission is to provide clean energy to feed and fuel the world sustainably. With our employees focused on safe and reliable operations, environmental stewardship, and disciplined capital and corporate management, we are on a path to decarbonize our ammonia production network – the world's largest – to enable green and low-carbon hydrogen and nitrogen products for energy, fertilizer, emissions abatement, and other industrial activities. Our nitrogen manufacturing complexes in the United States, Canada and the United Kingdom, an extensive storage, transportation and distribution network in North America, and logistics capabilities enabling a global reach underpin our strategy to leverage our unique capabilities to accelerate the world's transition to clean energy. Our principal customers are cooperatives, independent fertilizer distributors, traders, wholesalers and industrial users. Our core product is anhydrous ammonia (ammonia), which contains 82% nitrogen and 18% hydrogen. Our nitrogen products that are upgraded from ammonia are granular urea, urea ammonium nitrate solution (UAN) and ammonium nitrate (AN). Our other nitrogen products include diesel exhaust fluid (DEF), urea liquor, nitric acid and aqua ammonia, which are sold primarily to our industrial customers.

Our principal assets as of December 31, 2023 include:

- six U.S. nitrogen manufacturing facilities, located in Donaldsonville, Louisiana (the largest nitrogen complex in the world); Sergeant Bluff, Iowa (our Port Neal complex); Yazoo City, Mississippi; Claremore, Oklahoma (our Verdigris complex); Woodward, Oklahoma; and Waggaman, Louisiana. The Waggaman facility is wholly owned by us, and the other five U.S. nitrogen manufacturing facilities are wholly owned directly or indirectly by CF Industries Nitrogen, LLC (CFN), of which we own approximately 89% and CHS Inc. (CHS) owns the remainder (see Note 19—Noncontrolling Interest for additional information on our strategic venture with CHS);
- two Canadian nitrogen manufacturing facilities, located in Medicine Hat, Alberta (the largest nitrogen complex in Canada) and Courtright, Ontario;
- a United Kingdom nitrogen manufacturing facility located in Billingham;
- an extensive system of terminals and associated transportation equipment located primarily in the Midwestern United States; and

- a 50% interest in Point Lisas Nitrogen Limited (PLNL), an ammonia production joint venture located in the Republic of Trinidad and Tobago (Trinidad) that we account for under the equity method.

Our Strategy

Our strategy is to leverage our unique capabilities to accelerate the world's transition to clean energy. We believe this strategy builds upon the Company's leadership in ammonia production to capture emerging opportunities available to ammonia produced with a lower carbon intensity than that of ammonia produced through traditional processes. These opportunities include traditional applications in agriculture to help reduce the carbon footprint of food production and the life cycle carbon intensity of ethanol production, enabling its use for sustainable aviation fuel, among other purposes. They also include new applications, such as power generation and marine shipping, that would use the hydrogen component of the ammonia molecule for clean energy given that ammonia does not contain or emit carbon when combusted.

We execute our strategy across four dimensions: decarbonizing our existing network to accelerate the availability of low-carbon ammonia; evaluating new low-carbon ammonia capacity growth; forging partnerships to accelerate our timeline and bridge gaps in areas where we do not have expertise; and collaborating to build understanding of ammonia's clean energy capability, safety track record and regulatory environment.

Decarbonization projects in our existing network include our green ammonia project at our Donaldsonville, Louisiana complex. Green ammonia refers to ammonia produced with hydrogen sourced through an electrolysis process that produces no carbon emissions. In April 2021, we signed an engineering and procurement contract with thyssenkrupp to supply a 20 MW alkaline water electrolysis plant to produce green hydrogen at our Donaldsonville complex. We will integrate the green hydrogen generated by the electrolysis plant into existing ammonia synthesis loops to enable the production of approximately 20,000 tons per year of green ammonia. The green hydrogen production facility is mechanically complete, and commissioning activities began in early 2024. We believe that the Donaldsonville green ammonia project will be the largest of its kind in North America at the time of its startup.

Decarbonization projects in our existing network also include the production of low-carbon ammonia. Low-carbon ammonia is ammonia produced by conventional processes but with approximately 60-98% of the process and flue gas CO_2 generated by ammonia production removed through carbon capture and sequestration (CCS). We are executing a project also at our Donaldsonville complex that will enable us to produce a significant volume of low-carbon ammonia. At an estimated cost of $200 million, we are constructing a CO_2 dehydration and compression facility to enable CCS at the facility. Engineering activities for the construction of the dehydration and compression unit continue to advance, all major equipment for the facility has been procured, fabrication of the CO_2 compressors is proceeding, and construction activities at the Donaldsonville site are underway. Once the dehydration and compression unit is in service and sequestration is initiated, we expect that the Donaldsonville complex will have the capacity to dehydrate and compress up to 2 million tons per year of process CO_2, thereby converting a portion of our existing ammonia production to low-carbon ammonia. In October 2022, we announced that we had entered into a definitive CO_2 offtake agreement with ExxonMobil to transport and permanently sequester the CO_2 from Donaldsonville. Start-up for the project is planned for 2025. Under current regulations, the project would be expected to qualify for tax credits under Section 45Q of the Internal Revenue Code, which provides a credit per tonne of CO_2 sequestered.

Alongside these projects, we are also evaluating the construction of greenfield low-carbon ammonia capacity in Louisiana. In the fourth quarter of 2023, we and Mitsui & Co., Ltd. (Mitsui) completed a front-end engineering and design (FEED) study on a greenfield steam methane reforming (SMR) ammonia facility with CCS technologies. The FEED study estimates the cost of a project with these attributes to be in the range of $3 billion, with approximately $2.5 billion allocated to the ammonia facility and CCS technologies and approximately $500 million allocated to scalable common infrastructure for the site, such as ammonia storage and vessel loading docks.

We and Mitsui are progressing two additional FEED studies focused on technologies with the potential to further reduce the carbon intensity of the proposed low-carbon ammonia facility, including a FEED study evaluating autothermal reforming (ATR) ammonia production technology and a FEED study assessing the cost and viability of adding flue gas capture to an SMR ammonia facility. We expect to complete both FEED studies in the second half of 2024.

We and Mitsui are targeting the second half of 2024 for the final investment decision on the proposed greenfield low-carbon ammonia facility. Should the companies agree to move forward, the ammonia facility would be constructed at our new Blue Point complex. We own the land for the complex, which is located on the west bank of the Mississippi river in Ascension Parish, Louisiana. Construction and commissioning of a new world-scale ammonia plant typically takes approximately four years from the time construction begins.

In addition to ongoing discussions with existing customers who have interest in forthcoming availability of low-carbon ammonia for traditional applications, we are engaged in advanced discussions regarding the supply of low-carbon ammonia for

new applications. In the first quarter of 2023, we signed a memorandum of understanding (MOU) with JERA Co., Inc. (JERA), Japan's largest energy generator, regarding the long-term supply of up to 500,000 tonnes per year of clean ammonia beginning in 2027. The execution of the MOU was the result of a supplier comparison and evaluation process for the procurement of clean ammonia that JERA initiated in February 2022 for the world's first commercial scale ammonia co-firing operations that JERA is developing. The MOU establishes a framework for JERA and us to assess how we would best supply JERA with clean ammonia, which will be required to be produced with at least 60% lower carbon emissions than conventionally produced ammonia, under a long-term offtake agreement. We and JERA are evaluating a range of potential supply options, including JERA making an equity investment with us to develop a clean ammonia facility in Louisiana and a supplementary long-term offtake agreement.

We are also evaluating and in various stages of developing discussions and agreements with other companies for clean ammonia long-term offtake opportunities related to new applications of ammonia.

Industry Factors

We operate in a highly competitive, global industry. Our operating results are influenced by a broad range of factors, including those outlined below.

Global Supply and Demand Factors

Our products are globally traded commodities and are subject to price competition. The customers for our products make their purchasing decisions principally on the basis of delivered price and, to a lesser extent, on reliability, customer service and product quality. The selling prices of our products fluctuate in response to global market conditions, changes in supply and demand and cost factors.

Historically, global fertilizer demand has been driven primarily by population growth, gross domestic product growth, changes in dietary habits, planted acreage, and application rates, among other things. We expect these key variables to continue to have major impacts on long-term fertilizer demand for the foreseeable future. Short-term fertilizer demand growth may depend on global economic conditions, farm sector income, weather patterns, the level of global grain stocks relative to consumption, fertilizer application rates, and governmental regulations, including fertilizer subsidies or requirements mandating increased use of bio-fuels or industrial nitrogen products, such as DEF. Geopolitical factors such as temporary disruptions in fertilizer trade related to government intervention, shipping delays and/or cost increases resulting from regional conflicts or changes in the buying/selling patterns of key exporting/consuming countries, including China, India, Russia and Brazil, among others, often play a major role in shaping near-term market fundamentals. The economics of nitrogen-based fertilizer manufacturing play a key role in decisions to increase or reduce production capacity. Supply of fertilizers is generally driven by available capacity and operating rates, raw material costs and availability, government policies and global trade. Raw materials are dependent on energy sources such as natural gas or coal; therefore, supply costs are affected by the supply of and demand for those commodities.

Global Trade in Fertilizer

Profitability of our products within a particular geographic region is determined not only by the relationship between global supply and demand, but also by the supply/demand balance within that region. Regional supply and demand can be influenced significantly by factors affecting trade within regions. Some of these factors include the relative cost to produce and deliver product, relative currency values, the availability of credit, agricultural supply and demand, industrial product demand and policies such as emissions abatement, government support for manufacturers or purchasers and governmental nitrogen product trade policies, including the imposition of duties, tariffs or quotas, that affect foreign trade or investment. Government energy or carbon policies may also affect regional nitrogen supply and demand. The development of additional natural gas reserves in North America has decreased natural gas costs in North America relative to the rest of the world, making North American nitrogen fertilizer producers more competitive. Changes in currency values may also alter our cost competitiveness relative to producers in other regions of the world.

The North American nitrogen fertilizer market for certain nitrogen products is dependent on imports to balance supply and demand, and imports traditionally account for a significant portion of nitrogen fertilizer products consumed in North America. Producers of nitrogen-based fertilizers located in the Middle East, Trinidad, Africa and Russia have been major exporters to North America in recent years.

Farmers' Economics

The demand for fertilizer is affected by the aggregate crop planting decisions and fertilizer application rate decisions of individual farmers. Individual farmers make planting decisions based largely on prospective profitability of a harvest, while the specific varieties and amounts of fertilizer they apply depend on factors like their current liquidity, soil conditions, weather patterns, crop and fertilizer prices, fertilizer products used and timing of applications, expected yields and the types of crops planted.

Market Conditions

Geopolitical Environment

Changes in the geopolitical environment can have significant effects on our financial results. Russia's invasion of Ukraine in February 2022, and the resulting war between Russia and Ukraine, disrupted global markets for certain commodities, including natural gas, nitrogen fertilizers and certain commodity grains and oilseeds, leading to production curtailments, export reductions and logistical complications involving these commodities. Additionally, energy, financial and transportation sanctions were announced by U.S., Canadian, European and other governments against Russia in response to the war. During 2022 and 2023, market participants continued to adjust trade flows and manufacturers have continued to adjust production levels in response to changing conditions resulting from these factors. As of the date of filing of this report, nitrogen fertilizers have largely been explicitly exempted from sanctions against Russia by the United States and certain other governments, and as a result, there has been an increase in Russian fertilizer exports into the United States and other parts of the world.

As further described below, natural gas is the principal raw material used to produce our nitrogen products. Natural gas is a globally traded commodity that experiences price fluctuations based on supply and demand balances and has been impacted by geopolitical events, including the war between Russia and Ukraine. European energy markets, which have historically sourced a substantial portion of their natural gas from Russia, were disrupted by Russia's invasion of Ukraine and the subsequent reduction of Russian natural gas supply to Europe during 2022. This led to further increases in natural gas prices and natural gas price volatility, which in turn have led to disruptions in manufacturing and distribution activities at other nitrogen manufacturers and suppliers in our industry, resulting in changes in nitrogen product trade flows and reductions in global fertilizer supply. In September 2022, in response to the high prices for natural gas in the United Kingdom, we idled ammonia production at our Billingham complex and since that time have been importing ammonia from one of our other manufacturing facilities or from purchases in the open market for upgrade into AN and other nitrogen products at our Billingham complex. In the third quarter of 2023, we approved our plan to permanently close the ammonia plant at our Billingham complex.

The geopolitical developments that began in 2022 relating to the war in Ukraine also led to some supply chain disruptions for Russian and other producers of fertilizer, contributing to reduced global nitrogen fertilizer supply. In particular, disruptions in nitrogen exports from Russia, a leading exporter of nitrogen fertilizer products globally, and nitrogen production in Europe, as a result of higher and more volatile natural gas prices as Russian-sourced natural gas supply declined, contributed to reduced nitrogen supply globally. In addition, Russia and Ukraine have been large exporters of commodity grains such as wheat, corn and soybeans. The direct and indirect impacts of the war in Ukraine, and the related uncertainty, resulted in reduced commodity grain supply from Russia and Ukraine, causing increased prices for grains globally. The increase in commodity grain prices in turn supported strong demand for nitrogen fertilizer in 2022.

These geopolitical developments further contributed to an already tight global supply and demand balance for nitrogen fertilizers, causing changes in global trade flows as both manufacturers and customers reacted to the changing market dynamics. As a result, global nitrogen fertilizer prices remained high and experienced significant volatility in 2022.

As Russian-sourced natural gas supply declined due to geopolitical factors, European purchasers of natural gas increased imports of liquefied natural gas (LNG) to build storage levels leading up to the winter 2022/2023 peak demand season. As storage levels of natural gas increased due to the increased LNG imports, in conjunction with a warmer than normal winter in Europe, prices for natural gas fell during the first half of 2023 and then stabilized during the third quarter of 2023. As natural gas prices decreased, certain nitrogen producers restarted previously idled production, leading to an increase in global nitrogen production operating rates. The increased global nitrogen product supply availability resulting from the increase in operating rates, in addition to new global production coming on line and an increase in imports of Russian UAN into the United States, resulted in an increase in supply and lower average selling prices in 2023 compared to average selling prices realized in 2022. Russian exports of ammonia, which had been curtailed due to reduced pipeline flow, may also increase in 2024 as Russian producers develop alternative avenues to export product.

We expect that these geopolitical events, and any further government-imposed sanctions or other government actions affecting food or energy security, will continue to have an impact on the supply and demand balance of nitrogen fertilizer products globally and selling prices for our nitrogen fertilizer products.

In addition, beginning in the fourth quarter of 2023 and continuing into 2024, there have been escalating conflicts in the Middle East. This has led to shipping disruptions as shippers avoid conflicts in the Red Sea, which increases shipping times and costs as both producers and customers adjust shipping routes and trade flows. These regional conflicts could lead to more widespread geopolitical disruption.

Nitrogen Selling Prices and Sales Volume

Our nitrogen products are globally traded commodities with selling prices that fluctuate in response to global market conditions, changes in supply and demand, and other cost factors including domestic and local conditions. Intense global competition—reflected in import volumes and prices—strongly influences delivered prices for nitrogen fertilizers. In general, the prevailing global prices for nitrogen products must be at a level to incent the high cost marginal producer to produce product at a breakeven or above price, or else they would cease production and leave a portion of global demand unsatisfied.

Average selling prices for all of our major products were lower in 2023 than in 2022, as lower global energy costs reduced the global market clearing price required to meet global demand. In addition, the higher average selling prices in 2022 were driven in part by the geopolitical environment, as described above. The average selling price for our products was $347 per ton in 2023 compared to $610 per ton in 2022. The 43% decrease in the average selling prices for our products in 2023 compared to 2022 resulted in a year-over-year decrease in net sales of approximately $4.98 billion.

Our total sales volume was 4% higher in 2023 than in 2022, due primarily to higher sales volume in our UAN, Ammonia and Other segments. We shipped 19.1 million tons of product in 2023 compared to 18.3 million tons in 2022. The increase in total sales volume resulted in an increase in net sales of approximately $426 million. The impact of the Waggaman acquisition, which is included in our operating results since the acquisition on December 1, 2023, increased our sales volume by 57,000 tons and our net sales by $28 million.

Sales volume for our products in 2023, 2022 and 2021 is shown in the table below.

	2023		2022		2021	
	Sales Volume (tons)	Net Sales	Sales Volume (tons)	Net Sales	Sales Volume (tons)	Net Sales
	(tons in thousands; dollars in millions)					
Ammonia	3,546	$ 1,679	3,300	$ 3,090	3,589	$ 1,787
Granular Urea	4,570	1,823	4,572	2,892	4,290	1,880
UAN	7,237	2,068	6,788	3,572	6,584	1,788
AN	1,571	497	1,594	845	1,720	510
Other	2,206	564	2,077	787	2,318	573
Total	19,130	$ 6,631	18,331	$ 11,186	18,501	$ 6,538

Natural Gas

Natural gas is the principal raw material used to produce our nitrogen products. Natural gas is both a chemical feedstock and a fuel used to produce nitrogen products. Natural gas is a significant cost component of our manufactured nitrogen products, representing approximately 40% and 50%, respectively, of our production costs in 2023 and 2022.

The following table presents the average daily market price of natural gas at the Henry Hub, the most heavily-traded natural gas pricing point in North America:

	Year ended December 31,														
	2023		2022		2021		2023 v. 2022				2022 v. 2021				
Average daily market price of natural gas Henry Hub (Louisiana)	$	2.53	$	6.38	$	3.82	$	(3.85)	(60)	%	$	2.56	67	%	

Most of our nitrogen manufacturing facilities are located in the United States and Canada. As a result, the price of natural gas in North America directly impacts a substantial portion of our operating expenses. Throughout 2023, there were several factors that led to lower average North American natural gas prices compared to 2022. Warmer-than-normal temperatures in the

first quarter of 2023 drove lower heating demand for natural gas, while North American supply remained strong, as there were few weather-related production disruptions. In addition, although the higher cost for natural gas outside of North America incentivized liquefaction facilities in the United States to export domestic natural gas during 2023, the outage at the Freeport LNG facility in the first quarter of 2023 as well as planned maintenance in 2023 at several other LNG facilities limited total gas exports, resulting in greater domestic supply availability. In the fourth quarter of 2023, a warmer-than-average start to the winter season led to lower-than-expected heating demand, while North American natural gas production during the fourth quarter of 2023 was at historical highs. As a result, injection rates into natural gas storage in the United States during the fourth quarter of 2023 exceeded historical averages, leading to end-of-year storage levels that were 19% higher than one year earlier and 13% higher than the five-year average.

Our Billingham U.K. nitrogen manufacturing facility, prior to its ceasing ammonia production beginning in September 2022, was subject to fluctuations associated with the price of natural gas in Europe. Russia's invasion of Ukraine in February 2022 disrupted European energy markets and threatened security of supply, driving natural gas prices in Europe upward to unprecedented levels. The average daily market price of natural gas at the National Balancing Point (NBP), the major trading point for natural gas in the United Kingdom, was $24.56 for 2022. In September 2022, as a result of extremely high and volatile natural gas prices and the lack of a corresponding increase in global nitrogen product market prices, we idled ammonia production at our Billingham complex. Since that time, we have imported ammonia for upgrade into AN and other nitrogen products at our Billingham complex. In the third quarter of 2023, we approved our plan to permanently close the ammonia plant at our Billingham complex. As a result, we continue to import ammonia for upgrade.

The total cost of natural gas used for production at our manufacturing facilities, which includes the impact of realized natural gas derivatives, decreased 49% to $3.67 per MMBtu in 2023 from $7.18 per MMBtu in 2022. The decrease in natural gas costs in 2023 as compared to 2022 resulted in an increase in gross margin of approximately $1.16 billion.

Financial Executive Summary

We reported net earnings attributable to common stockholders of $1.53 billion in 2023 compared to $3.35 billion in 2022, a decrease in net earnings of 54%, or $1.82 billion. The decrease in net earnings reflects a decrease in gross margin partially offset by the following: a lower income tax provision, lower net interest expense, a decrease in net earnings attributable to the noncontrolling interest, and impairment charges related to our U.K. operations in 2022 that did not recur in 2023.

Gross margin decreased by $3.32 billion to $2.55 billion for the year ended December 31, 2023 compared to the year ended December 31, 2022. The decrease in gross margin was due primarily to a 43% decrease in average selling prices to $347 per ton in 2023 from $610 per ton in 2022, which decreased gross margin by $4.98 billion. The impact of lower selling prices was partially offset by lower natural gas costs, which increased gross margin by $1.16 billion.

Lower gross margin was partially offset by the following: a decrease in the income tax provision of $748 million due primarily to lower pre-tax earnings in 2023, a $287 million decrease in net interest expense, and a decrease of $278 million in net earnings attributable to the noncontrolling interest. In addition, pre-tax impairment and restructuring charges of $258 million related to our U.K. operations were recognized in 2022 compared to $10 million in 2023.

Diluted net earnings per share attributable to common stockholders decreased $8.51 per share, to $7.87 per share in 2023 compared to $16.38 per share in 2022 due primarily to lower net earnings, partially offset by lower weighted-average common shares outstanding as a result of shares repurchased under our share repurchase programs.

Acquisition of Waggaman Ammonia Production Facility

On December 1, 2023, we acquired an ammonia production facility located in Waggaman, Louisiana from Dyno Nobel Louisiana Ammonia, LLC (DNLA), a U.S. subsidiary of Australia-based Incitec Pivot Limited (IPL), pursuant to an asset purchase agreement with DNLA and IPL. The facility has a nameplate capacity of 880,000 tons of ammonia annually.

In connection with the acquisition, we entered into a long-term ammonia offtake agreement providing for us to supply up to 200,000 tons of ammonia per year to IPL's Dyno Nobel, Inc. subsidiary. Under the terms of the asset purchase agreement, $425 million of the purchase price of $1.675 billion, subject to adjustment, was allocated by the parties to the ammonia offtake agreement. We funded the balance of the purchase price with $1.223 billion of cash on hand.

The consideration transferred reflects an estimated net working capital adjustment and other adjustments to the purchase price, which is subject to further adjustment pursuant to the terms of the asset purchase agreement. We expect any further purchase price adjustments required under the asset purchase agreement will be completed in 2024.

The financial results of the Waggaman facility are included in our consolidated statement of operations and in our

Ammonia segment from the acquisition date of December 1, 2023. For the year ended December 31, 2023, the amount of net sales and net earnings of the Waggaman facility since the acquisition date was $28 million and $7 million, respectively.

In connection with recording the acquisition, we recognized, among other items, goodwill, intangible assets and a supply contract liability. See Note 6—Acquisition of Waggaman Ammonia Production Facility for additional information.

Items Affecting Comparability of Results

During the years ended December 31, 2023 and 2022, we reported net earnings attributable to common stockholders of $1.53 billion and $3.35 billion, respectively. In addition to the impact of market conditions and the acquisition of the Waggaman ammonia production facility discussed above, certain items affected the comparability of our financial results during the years ended December 31, 2023 and 2022. The following table and related discussion outline these items and their impact on the comparability of our financial results for these periods. The descriptions of items below that refer to amounts in the table refer to the pre-tax amounts unless otherwise noted.

	2023		2022	
	Pre-Tax	After-Tax[1]	Pre-Tax	After-Tax[1]
	(in millions)			
Unrealized net mark-to-market (gain) loss on natural gas derivatives[2]	$ (39)	$ (30)	$ 41	$ 31
Loss on foreign currency transactions, including intercompany loans[3]	—	—	28	21
U.K. operations:				
U.K. long-lived and intangible asset impairment	—	—	239	180
U.K. operations restructuring	10	8	19	14
Acquisition and integration costs	39	29	—	—
Unrealized gain on embedded derivative liability[3]	—	—	(14)	(11)
Impairment of equity method investment in PLNL[4]	43	32	—	—
Pension settlement loss and curtailment gains—net[5]	—	—	17	13
Canada Revenue Agency Competent Authority Matter and Transfer pricing positions:				
Interest expense	—	—	170	168
Interest income	—	—	(29)	(22)
Income tax provision[6]	—	—	—	65
Loss on debt extinguishment	—	—	8	6

(1) The tax impact is calculated utilizing a marginal effective rate of 23.5% in both 2023 and 2022, except for acquisition and integration costs related to the Waggaman acquisition, which reflects a 26.2% marginal effective rate, and U.K. long-lived and intangible asset impairment, which reflects the amount of income tax benefit recognized. An income tax benefit for the U.K. goodwill impairment was not recorded as it is nondeductible for income tax purposes.

(2) Included in cost of sales in our consolidated statements of operations.

(3) Included in other operating—net in our consolidated statements of operations.

(4) Included in equity in (loss) earnings of operating affiliate in our consolidated statements of operations.

(5) Included in other non-operating—net in our consolidated statement of operations.

(6) The after-tax income tax provision amount of $65 million reflects an income tax provision of $70 million, consisting of the $78 million income tax provision referenced below under "Canada Revenue Agency Competent Authority Matter" and the $8 million of income tax benefit referenced below under "Transfer pricing positions," net of $5 million of income tax provision that is reflected in the after-tax interest expense and interest income amounts shown in this table.

Unrealized net mark-to-market (gain) loss on natural gas derivatives

Natural gas is the largest and most volatile single component of the manufacturing cost for our nitrogen-based products. At certain times, we have managed the risk of changes in natural gas prices through the use of derivative financial instruments. The derivatives that we use for this purpose are primarily natural gas fixed price swaps, basis swaps and options. We use natural gas derivatives as an economic hedge of natural gas price risk, but without the application of hedge accounting. This can result in volatility in reported earnings due to the unrealized mark-to-market adjustments that occur from changes in the value of the derivatives, which are reflected in cost of sales in our consolidated statements of operations. In 2023 and 2022, we recognized an unrealized net mark-to-market (gain) loss on natural gas derivatives of $(39) million and $41 million, respectively.

Loss on foreign currency transactions, including intercompany loans

In 2022, we recognized a loss on foreign currency transactions of $28 million, which consists of foreign currency exchange rate impacts on foreign currency denominated transactions, including the impact of changes in foreign currency exchange rates on intercompany loans that were not permanently invested.

U.K. operations

In 2023, we recognized total charges of $10 million, consisting primarily of the recognition of an asset retirement obligation and post-employment benefits related to contractual and statutory obligations due to employees as a result of our approved plan to permanently close the ammonia plant at our Billingham complex.

In 2022, we recognized total charges of $258 million, consisting primarily of asset impairment charges related to property, plant and equipment at our Billingham and Ince facilities and definite-lived intangible assets. See Note 7—United Kingdom Operations Restructuring and Impairment Charges for further information.

Acquisition and integration costs

On December 1, 2023, we acquired an ammonia production facility located in Waggaman, Louisiana, as described above under "Acquisition of Waggaman Ammonia Production Facility." In 2023, we incurred $39 million of acquisition and integration costs related to the Waggaman acquisition.

Unrealized gain on embedded derivative liability

Under the terms of our strategic venture with CHS, if our credit rating as determined by two of three specified credit rating agencies is below certain levels, we are required to make a non-refundable yearly payment of $5 million to CHS until the earlier of the date that our credit rating is upgraded to above such levels by two of the three specified credit rating agencies or February 1, 2026. This obligation is recorded at fair value and has been recognized on our consolidated balance sheets as an embedded derivative. Beginning in 2016, our credit ratings were below such levels and, as a result, under the terms of the strategic venture, we made an annual payment of $5 million to CHS in the fourth quarter of each year from 2016 through 2021. Our credit rating was upgraded above certain levels in July 2022 by one of the specified credit rating agencies and in October 2022 by another one of the specified credit rating agencies. As a result of these upgrades, in the fourth quarter of 2022, there was a reduction in the fair value of the embedded derivative liability, and we recognized an unrealized gain of $14 million.

Impairment of equity method investment in PLNL

PLNL, our joint venture in Trinidad, operates an ammonia plant that relies on natural gas supplied, under a gas sales contract (the NGC Contract), by The National Gas Company of Trinidad and Tobago Limited (NGC). The NGC Contract had an expiration date of September 2023. In the third quarter of 2023, PLNL entered into a new gas sales contract with NGC (the New NGC Contract), which is effective October 2023 through December 2025.

In the third quarter of 2023 and due to the terms of the New NGC Contract, we assessed our investment in PLNL for impairment and determined that the carrying value of our equity method investment in PLNL exceeded its fair value. As a result, we recorded an impairment of our equity method investment in PLNL of $43 million, which is reflected in equity in (loss) earnings of operating affiliate in our consolidated statement of operations for the year ended December 31, 2023.

Pension settlement loss and curtailment gains—net

On July 15, 2022, we entered into an agreement with an insurance company to purchase a non-participating group annuity contract and transfer approximately $375 million of our primary U.S. defined benefit pension plan's projected benefit obligation. The transaction closed on July 22, 2022 and was funded with plan assets. Under the transaction, the insurance company assumed responsibility for pension benefits and annuity administration for approximately 4,000 retirees or their beneficiaries. As a result of this transaction, in the third quarter of 2022, we remeasured the plan's projected benefit obligation and plan assets, and we recognized a non-cash pre-tax pension settlement loss of $24 million, reflecting the unamortized net unrecognized postretirement benefit costs related to the settled obligations, with a corresponding offset to accumulated other comprehensive loss. In the fourth quarter of 2022, the final settlement of the non-participating group annuity contract resulted in a refund of $4 million to us, which decreased the non-cash pre-tax pension settlement loss recognized by $3 million to $21 million. The settlement loss is reflected in other non-operating—net in our consolidated statement of operations for the year ended December 31, 2022.

In October 2022, we remeasured certain of our U.S. and Canadian defined benefit pension plans due to plan amendments resulting from a revision to our North American retirement plan strategy. As a result of these plan amendments, we recognized $4 million of curtailment gains, which is reflected in other non-operating—net in our consolidated statement of operations for the year ended December 31, 2022. See Note 13—Pension and Other Postretirement Benefits for further information.

Canada Revenue Agency Competent Authority Matter

In 2016, the Canada Revenue Agency (CRA) and Alberta Tax and Revenue Administration (Alberta TRA) issued Notices of Reassessment for tax years 2006 through 2009 to one of our Canadian subsidiaries asserting a disallowance of certain patronage deductions. We filed Notices of Objection with respect to the Notices of Reassessment with the CRA and Alberta TRA and posted letters of credit in lieu of paying the additional tax liability assessed. The letters of credit served as security until the matter was resolved. In 2018, the matter, including the related transfer pricing topic regarding the allocation of profits between Canada and the United States, was accepted for consideration under the bilateral settlement provisions of the U.S.-Canada tax treaty (the Treaty) by the United States and Canadian competent authorities, and included tax years 2006 through 2011. In the second quarter of 2021, the Company submitted the transfer pricing aspect of the matter into the arbitration process under the terms of the Treaty.

In February 2022, we were informed that a decision was reached by the arbitration panel for tax years 2006 through 2011. In March 2022, we received further details of the results of the arbitration proceedings and the settlement provisions between the United States and Canadian competent authorities, and we accepted the decision of the arbitration panel. Under the terms of the arbitration decision, additional income for tax years 2006 through 2011 was subject to tax in Canada, resulting in our having additional Canadian tax liability for those tax years. See Note 12—Income Taxes for additional information.

In 2022, as a result of the impact of these events on our Canadian and U.S. federal and state income taxes, we recognized an income tax provision of $78 million, reflecting the net impact of $129 million of accrued income taxes payable to Canada for tax years 2006 through 2011, partially offset by net income tax receivables of approximately $51 million in the United States, and we accrued net interest of $102 million, primarily reflecting the estimated interest payable to Canada.

Transfer pricing positions

As a result of the outcome of the arbitration decision discussed above, we also evaluated our transfer pricing positions between Canada and the United States for open years 2012 and after. Based on this evaluation, we recorded the following in 2022:

- liabilities for unrecognized tax benefits of $159 million, with a corresponding income tax provision, and accrued interest of $59 million related to the liabilities for unrecognized tax benefits, and
- noncurrent income tax receivables of $188 million, with a corresponding income tax benefit, and accrued interest income of $20 million related to the noncurrent income tax receivables.

In 2022, the impact of these evaluations of transfer pricing positions on our consolidated statement of operations, including $21 million of net deferred income tax provision for other transfer pricing tax effects, was $8 million of income tax benefit and $39 million of net interest expense before tax ($44 million after tax).

Loss on debt extinguishment

On April 21, 2022, we redeemed in full all of the $500 million outstanding principal amount of the 3.450% senior notes due June 2023 (the 2023 Notes) in accordance with the optional redemption provisions in the indenture governing the 2023 Notes. The total aggregate redemption price paid in connection with the April 2022 redemption of the 2023 Notes was $513 million, including accrued interest. As a result, we recognized a loss on debt extinguishment of $8 million, consisting primarily of the premium paid on the redemption of the $500 million principal amount of the 2023 Notes prior to their scheduled maturity.

Consolidated Results of Operations

The following table presents our consolidated results of operations for the years ended December 31, 2023, 2022 and 2021:

	Year ended December 31,						
	2023	2022	2021	2023 v. 2022		2022 v. 2021	
	(in millions, except as noted)						
Net sales	$ 6,631	$ 11,186	$ 6,538	$ (4,555)	(41)%	$ 4,648	71 %
Cost of sales (COS)	4,086	5,325	4,151	(1,239)	(23)%	1,174	28 %
Gross margin	2,545	5,861	2,387	(3,316)	(57)%	3,474	146 %
Gross margin percentage	38.4 %	52.4 %	36.5 %	(14.0)%		15.9 %	
Selling, general and administrative expenses	289	290	223	(1)	— %	67	30 %
U.K. goodwill impairment	—	—	285	—	— %	(285)	(100)%
U.K. long-lived and intangible asset impairment	—	239	236	(239)	(100)%	3	1 %
U.K. operations restructuring	10	19	—	(9)	(47)%	19	N/M
Acquisition and integration costs	39	—	—	39	N/M	—	— %
Other operating—net	(31)	10	(39)	(41)	N/M	49	N/M
Total other operating costs and expenses	307	558	705	(251)	(45)%	(147)	(21)%
Equity in (loss) earnings of operating affiliate	(8)	94	47	(102)	N/M	47	100 %
Operating earnings	2,230	5,397	1,729	(3,167)	(59)%	3,668	212 %
Interest expense	150	344	184	(194)	(56)%	160	87 %
Interest income	(158)	(65)	(1)	(93)	(143)%	(64)	N/M
Loss on debt extinguishment	—	8	19	(8)	(100)%	(11)	(58)%
Other non-operating—net	(10)	15	(16)	(25)	N/M	31	N/M
Earnings before income taxes	2,248	5,095	1,543	(2,847)	(56)%	3,552	230 %
Income tax provision	410	1,158	283	(748)	(65)%	875	309 %
Net earnings	1,838	3,937	1,260	(2,099)	(53)%	2,677	212 %
Less: Net earnings attributable to noncontrolling interest	313	591	343	(278)	(47)%	248	72 %
Net earnings attributable to common stockholders	$ 1,525	$ 3,346	$ 917	$ (1,821)	(54)%	$ 2,429	265 %
Diluted net earnings per share attributable to common stockholders	$ 7.87	$ 16.38	$ 4.24	$ (8.51)	(52)%	$ 12.14	286 %
Diluted weighted-average common shares outstanding	193.8	204.2	216.2	(10.4)	(5)%	(12.0)	(6)%
Dividends declared per common share	$ 1.60	$ 1.50	$ 1.20	$ 0.10	7 %	$ 0.30	25 %

N/M—Not Meaningful

The following table presents certain supplemental data for the years ended December 31, 2023, 2022 and 2021:

	Year ended December 31,						
	2023	2022	2021	2023 v. 2022		2022 v. 2021	
Natural gas supplemental data (per MMBtu)							
Natural gas costs in COS[1]	$ 3.26	$ 7.16	$ 4.21	$ (3.90)	(54)%	$ 2.95	70 %
Realized derivatives loss in COS[2]	0.41	0.02	—	0.39	N/M	0.02	N/M
Cost of natural gas used for production in COS	$ 3.67	$ 7.18	$ 4.21	$ (3.51)	(49)%	$ 2.97	71 %
Average daily market price of natural gas Henry Hub (Louisiana)	$ 2.53	$ 6.38	$ 3.82	$ (3.85)	(60)%	$ 2.56	67 %
Unrealized net mark-to-market (gain) loss on natural gas derivatives	$ (39)	$ 41	$ 25	$ (80)	N/M	$ 16	64 %
Depreciation and amortization	$ 869	$ 850	$ 888	$ 19	2 %	$ (38)	(4)%
Capital expenditures	$ 499	$ 453	$ 514	$ 46	10 %	$ (61)	(12)%
Sales volume by product tons (000s)	19,130	18,331	18,501	799	4 %	(170)	(1)%
Production volume by product tons (000s):							
Ammonia[3]	9,496	9,807	9,349	(311)	(3)%	458	5 %
Granular urea	4,544	4,561	4,123	(17)	— %	438	11 %
UAN (32%)	6,852	6,706	6,763	146	2 %	(57)	(1)%
AN	1,520	1,517	1,646	3	— %	(129)	(8)%

N/M—Not Meaningful

(1) Includes the cost of natural gas and related transportation that is included in cost of sales during the period under the first-in, first-out inventory cost method.

(2) Includes realized gains and losses on natural gas derivatives settled during the period. Excludes unrealized mark-to-market gains and losses on natural gas derivatives.

(3) Gross ammonia production, including amounts subsequently upgraded on-site into granular urea, UAN, or AN.

The following is a discussion and analysis of our consolidated results of operations for the year ended December 31, 2023, compared to the year ended December 31, 2022. For a discussion and analysis of our consolidated results of operations for the year ended December 31, 2022 compared to the year ended December 31, 2021, see Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations in our 2022 Annual Report on Form 10-K filed with the SEC on February 23, 2023.

Net Sales

Our net sales are derived primarily from the sale of nitrogen products and are determined by the quantities of nitrogen products we sell and the selling prices we realize. The volumes, mix and selling prices we realize are determined to a great extent by a combination of global and regional supply and demand factors. Net sales also include shipping and handling costs that are billed to our customers. Sales incentives are reported as a reduction in net sales.

Our net sales decreased $4.56 billion, or 41%, to $6.63 billion in 2023 compared to $11.19 billion in 2022. The decrease in our net sales reflects (i) the impact of the Waggaman acquisition, which is included in our operating results since the acquisition on December 1, 2023, and increased our net sales by $28 million, and (ii) a $4.58 billion, or 41%, decline in our net sales excluding the impact of the Waggaman acquisition, due to a 43% decrease in average selling prices, partially offset by a 4% increase in sales volume.

Our average selling price was $347 per ton in 2023 compared to $610 per ton in 2022, a decrease of 43%, due to lower average selling prices across all of our segments, as lower global energy costs reduced the global market clearing price required to meet global demand. The impact of lower average selling prices was a decrease in net sales of approximately $4.98 billion for 2023 compared to 2022.

Our sales volume of 19.1 million product tons in 2023 was 4% higher compared to 18.3 million product tons in 2022, due primarily to higher sales volume in our UAN, Ammonia and Other segments. The impact of the Waggaman acquisition increased our ammonia sales volume by 57,000 tons.

Gross ammonia production for 2023 decreased to approximately 9.5 million tons compared to 9.8 million tons in 2022, due primarily to our Billingham ammonia plant, which was idled in late 2022 and permanently closed in 2023. We expect gross ammonia production for 2024 will be approximately 10 million tons. The expected increase in gross ammonia production for 2024 compared to 2023 reflects the inclusion of the Waggaman facility, which we acquired on December 1, 2023.

Cost of Sales

Our cost of sales includes manufacturing costs, purchased product costs, distribution costs and storage costs. Manufacturing costs, the most significant element of cost of sales, consist primarily of raw materials, realized and unrealized gains and losses on natural gas derivatives, maintenance, direct labor, depreciation and other plant overhead expenses. Purchased product costs primarily include the cost to purchase nitrogen fertilizers to augment or replace production at our facilities. Distribution costs consist of the cost of freight required to transport finished products from our plants to our distribution facilities, which are recognized in cost of sales when the product is sold to our customers. Storage costs consist of costs incurred prior to final shipment to customers.

Our cost of sales decreased $1.24 billion, or 23%, to $4.09 billion in 2023 as compared to $5.33 billion in 2022. The decrease in our cost of sales reflects (i) the impact of the Waggaman acquisition, which increased our cost of sales by $14 million, (ii) lower costs for natural gas including the impact of realized derivatives, excluding the impact of the Waggaman acquisition, which decreased cost of sales by $1.16 billion, and (iii) lower costs for ammonia purchased from PLNL, our joint venture in Trinidad. Lower natural gas costs include the benefit of ceasing ammonia production in the United Kingdom and the corresponding reduction in natural gas purchases. For our U.K. operations, the decrease in our cost of sales from lower natural gas costs was partially offset by the cost to import ammonia for upgrading into AN and other nitrogen products.

Cost of sales also includes the impact of a $39 million unrealized net mark-to-market gain on natural gas derivatives in 2023 compared to a $41 million loss in 2022.

Cost of sales averaged $214 per ton in 2023, a 26% decrease from $290 per ton in 2022. Our cost of natural gas, including the impact of realized derivatives, decreased 49% to $3.67 per MMBtu in 2023 from $7.18 per MMBtu in 2022.

Selling, General and Administrative Expenses

Our selling, general and administrative expenses consist primarily of corporate office expenses such as salaries and other payroll-related costs for our executive, administrative, legal, financial, IT, and sales functions, as well as certain taxes and insurance and other professional service fees, including those for corporate initiatives, and amortization of definite-lived intangible assets.

Selling, general and administrative expenses were $289 million in 2023 as compared to $290 million in 2022. The decrease was due primarily to charitable contributions in 2022 for the initial funding of the CF Industries Foundation formed in December 2022 that did not recur in 2023, partially offset by higher costs related to certain corporate initiatives, including higher amortization due to our recently implemented enterprise resource planning (ERP) system.

U.K. Operations

In 2023, related to our U.K. operations, we recognized total charges of $10 million, consisting primarily of the recognition of an asset retirement obligation and post-employment benefits related to contractual and statutory obligations due to employees as a result of our approved plan to permanently close the ammonia plant at our Billingham complex.

In 2022, related to our U.K. operations, we recognized total charges of $258 million, consisting of $239 million of asset impairment charges primarily related to property, plant and equipment at our Billingham and Ince facilities and definite-lived intangible assets and $19 million of restructuring charges primarily related to post-employment benefits related to contractual and statutory obligations and one-time termination benefits. See Note 7—United Kingdom Operations Restructuring and Impairment Charges for further information.

Acquisition and Integration Costs

On December 1, 2023, we acquired an ammonia production facility located in Waggaman, Louisiana, as described above under “Acquisition of Waggaman Ammonia Production Facility.” In 2023, we incurred $39 million of acquisition and integration costs related to the Waggaman acquisition.

Other Operating—Net

Other operating—net includes administrative costs that do not relate directly to our central operations. Costs included in "other operating costs" can include foreign currency transaction gains and losses, unrealized gains and losses on foreign currency derivatives, litigation expenses, gains and losses on the disposal of fixed assets and FEED study costs related to our clean energy initiatives.

Other operating—net was $31 million of income in 2023 compared to $10 million of expense in 2022. The income in 2023 consists primarily of gains on sales of emission credits, partially offset by FEED study costs for our clean energy initiatives. See "Our Strategy," above, for additional information related to our clean energy initiatives.

The expense in 2022 primarily includes a loss on foreign currency transactions of $28 million, which consists of foreign currency exchange rate impacts on foreign currency denominated transactions, including the impact of changes in foreign currency exchange rates on intercompany loans that were not permanently invested. The loss on foreign currency transactions in 2022 was partially offset by an unrealized gain of $14 million related to an embedded derivative liability. See "Items Affecting Comparability of Results—Unrealized gain on embedded derivative liability," above, for further information.

Equity in (Loss) Earnings of Operating Affiliate

Equity in (loss) earnings of operating affiliate consists of our 50% ownership interest in PLNL. We include our share of the net earnings from our equity method investment in PLNL as an element of earnings from operations because this investment provides additional production and is integrated with our other supply chain and sales activities.

Equity in (loss) earnings of operating affiliate was an $8 million loss in 2023 compared to $94 million of earnings in 2022. The $8 million loss in 2023 includes an impairment of our equity method investment in PLNL of $43 million recognized in the third quarter of 2023. See "Items Affecting Comparability of Results—Impairment of equity method investment in PLNL," above, for further discussion. Lower equity in earnings of operating affiliate also reflects a decrease in the operating results of PLNL as a result of lower ammonia selling prices, partially offset by lower natural gas costs.

Interest Expense

Interest expense includes interest on our long-term debt, amortization of the related fees required to execute financing agreements, annual fees pursuant to our revolving credit agreement and interest on tax liabilities. Capitalized interest relating to the construction of major capital projects reduces interest expense as the interest is capitalized and amortized over the estimated useful lives of the related assets.

Interest expense was $150 million in 2023 compared to $344 million in 2022. The decrease of $194 million was due primarily to $184 million of tax-related interest expense recorded in 2022, including $170 million of interest expense related to income tax matters described under "Items Affecting Comparability of Results—Canada Revenue Agency Competent Authority Matter" and "Items Affecting Comparability of Results—Transfer pricing positions," above. The decrease in interest expense was also due in part to lower interest expense on borrowings due to the redemption in full of all of the $500 million principal amount of the 2023 Notes in April 2022 prior to their scheduled maturity.

Interest Income

Interest income includes amounts earned on our cash, cash equivalents, and investments and any interest earned related to income tax refunds.

Interest income was $158 million in 2023 compared to $65 million in 2022. The increase of $93 million reflects a $102 million increase in interest income on short-term investments in 2023 due primarily to higher interest rates. Partially offsetting this effect of higher interest rates was interest income recorded in 2022 that did not recur in 2023, which was related to income tax matters described under "Items Affecting Comparability of Results—Canada Revenue Agency Competent Authority Matter" and "Items Affecting Comparability of Results—Transfer pricing positions," above.

Loss on Debt Extinguishment

Loss on debt extinguishment of $8 million in 2022 is described under "Items Affecting Comparability of Results—Loss on debt extinguishment," above.

Other Non-Operating—Net

Other non-operating—net was $10 million of income in 2023 compared to $15 million of expense in 2022. The variance was due primarily to the impact of defined benefit pension plan transactions in 2022, including a pension settlement loss of $21 million and curtailment gains of $4 million, which are described under "Items Affecting Comparability of Results—Pension settlement loss and curtailment gains—net," above.

Income Tax Provision

Our income tax provision for 2023 was $410 million on pre-tax income of $2.25 billion, or an effective tax rate of 18.3%, compared to an income tax provision of $1.16 billion on pre-tax income of $5.10 billion, or an effective tax rate of 22.7%, in 2022.

Our effective tax rate is impacted by earnings attributable to the noncontrolling interest in CFN, as our consolidated income tax provision does not include a tax provision on the earnings attributable to the noncontrolling interest. Our effective tax rate for 2023 of 18.3%, which is based on pre-tax income of $2.25 billion, would be 2.9 percentage points higher, or 21.2%, if based on pre-tax income exclusive of the earnings attributable to the noncontrolling interest of $313 million. Our effective tax rate for 2022 of 22.7%, which is based on pre-tax income of $5.10 billion, would be 3.0 percentage points higher, or 25.7%, if based on pre-tax income exclusive of the earnings attributable to the noncontrolling interest of $591 million.

For 2022, our income tax provision includes $22 million of income tax expense to record a valuation allowance in the United Kingdom, $23 million of income tax benefit for the excess tax benefit related to certain share-based compensation activity and $78 million of income tax provision related to the Canada Revenue Agency Competent Authority Matter, which is described above under "Items Affecting Comparability of Results."

Both 2023 and 2022 were impacted by additional discrete tax items. See Note 12—Income Taxes for additional information.

Net Earnings Attributable to Noncontrolling Interest

Net earnings attributable to noncontrolling interest includes the net earnings attributable to the approximately 11% CHS minority equity interest in CFN, a subsidiary of CF Holdings.

Net earnings attributable to noncontrolling interest decreased $278 million, or 47%, to $313 million in 2023 compared to $591 million in 2022 due to lower earnings of CFN driven by lower average selling prices as described above under "Net Sales."

Diluted Net Earnings Per Share Attributable to Common Stockholders

Net earnings per share attributable to common stockholders decreased 52% to $7.87 per diluted share in 2023 from $16.38 per diluted share in 2022. This decrease was due primarily to a decrease in gross margin, driven by lower average selling prices, partially offset by lower costs for natural gas. Lower gross margin was partially offset by a decrease in the income tax provision resulting from lower profitability, a decrease in interest expense due primarily to interest expense on tax liabilities in 2022 related to Canadian-U.S. transfer pricing matters that did not recur in 2023, a decrease in net earnings attributable to the noncontrolling interest of $278 million, and pre-tax impairment and restructuring charges related to our U.K. operations of $258 million in 2022 compared to $10 million in 2023. Additionally, diluted weighted-average common shares outstanding declined 5% from 204.2 million shares for 2022 to 193.8 million shares for 2023, due primarily to repurchases of common shares under our share repurchase programs.

Operating Results by Business Segment

Our reportable segment structure reflects how our chief operating decision maker, as defined in accounting principles generally accepted in the United States (U.S. GAAP), assesses the performance of our reportable segments and makes decisions about resource allocation. These segments are differentiated by products. Our management uses gross margin to evaluate segment performance and allocate resources. Total other operating costs and expenses (consisting primarily of selling, general and administrative expenses and other operating—net) and non-operating expenses (consisting primarily of interest and income taxes), are centrally managed and are not included in the measurement of segment profitability reviewed by management. The following table presents summary operating results by business segment:

	Ammonia[1]	Granular Urea[2]	UAN[2]	AN[2]	Other[2]	Consolidated
			(in millions)			
Year ended December 31, 2023						
Net sales	$ 1,679	$ 1,823	$ 2,068	$ 497	$ 564	$ 6,631
Cost of sales	1,138	1,010	1,251	359	328	4,086
Gross margin	$ 541	$ 813	$ 817	$ 138	$ 236	$ 2,545
Gross margin percentage	32.2 %	44.6 %	39.5 %	27.8 %	41.8 %	38.4 %
Year ended December 31, 2022						
Net sales	$ 3,090	$ 2,892	$ 3,572	$ 845	$ 787	$ 11,186
Cost of sales	1,491	1,328	1,489	597	420	5,325
Gross margin	$ 1,599	$ 1,564	$ 2,083	$ 248	$ 367	$ 5,861
Gross margin percentage	51.7 %	54.1 %	58.3 %	29.3 %	46.6 %	52.4 %
Year ended December 31, 2021						
Net sales	$ 1,787	$ 1,880	$ 1,788	$ 510	$ 573	$ 6,538
Cost of sales	1,162	992	1,119	475	403	4,151
Gross margin	$ 625	$ 888	$ 669	$ 35	$ 170	$ 2,387
Gross margin percentage	35.0 %	47.2 %	37.4 %	6.9 %	29.7 %	36.5 %

(1) The financial results of the Waggaman facility are included in the Ammonia segment from the Waggaman acquisition date of December 1, 2023. See "Acquisition of Waggaman Ammonia Production Facility" for additional information.

(2) The cost of the products that are upgraded into other products is transferred at cost into the upgraded product results.

The following is a discussion and analysis of our operating results by business segment for the year ended December 31, 2023 compared to the year ended December 31, 2022. For a discussion and analysis of our operating results by business segment for the year ended December 31, 2022 compared to the year ended December 31, 2021, see Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations in our 2022 Annual Report on Form 10-K filed with the SEC on February 23, 2023.

Ammonia Segment

Our Ammonia segment produces anhydrous ammonia (ammonia), which is the base product that we manufacture, containing 82% nitrogen and 18% hydrogen. The results of our Ammonia segment consist of sales of ammonia to external customers for its nitrogen content as a fertilizer, in emissions control and in other industrial applications. In addition, we upgrade ammonia into other nitrogen products such as granular urea, UAN and AN.

The following table presents summary operating data for our Ammonia segment:

	Year ended December 31,						
	2023[1]	2022	2021	2023 v. 2022		2022 v. 2021	
	(in millions, except as noted)						
Net sales	$ 1,679	$ 3,090	$ 1,787	$ (1,411)	(46)%	$ 1,303	73 %
Cost of sales	1,138	1,491	1,162	(353)	(24)%	329	28 %
Gross margin	$ 541	$ 1,599	$ 625	$ (1,058)	(66)%	$ 974	156 %
Gross margin percentage	32.2 %	51.7 %	35.0 %	(19.5)%		16.7 %	
Sales volume by product tons (000s)	3,546	3,300	3,589	246	7 %	(289)	(8)%
Sales volume by nutrient tons (000s)[2]	2,908	2,707	2,944	201	7 %	(237)	(8)%
Average selling price per product ton	$ 473	$ 936	$ 498	$ (463)	(49)%	$ 438	88 %
Average selling price per nutrient ton[2]	$ 577	$ 1,141	$ 607	$ (564)	(49)%	$ 534	88 %
Gross margin per product ton	$ 153	$ 485	$ 174	$ (332)	(68)%	$ 311	179 %
Gross margin per nutrient ton[2]	$ 186	$ 591	$ 212	$ (405)	(69)%	$ 379	179 %
Depreciation and amortization	$ 171	$ 166	$ 209	$ 5	3 %	$ (43)	(21)%
Unrealized net mark-to-market (gain) loss on natural gas derivatives	$ (11)	$ 13	$ 7	$ (24)	N/M	$ 6	86 %

N/M—Not Meaningful

(1) The financial results of the Waggaman facility are included in the Ammonia segment from the Waggaman acquisition date of December 1, 2023. See "Acquisition of Waggaman Ammonia Production Facility" for additional information.

(2) Ammonia represents 82% nitrogen content. Nutrient tons represent the tons of nitrogen within the product tons.

Year Ended December 31, 2023 Compared to Year Ended December 31, 2022

Net Sales. Net sales in our Ammonia segment decreased by $1.41 billion, or 46%, to $1.68 billion in 2023 from $3.09 billion in 2022. The decrease in our net sales reflects (i) the impact of the Waggaman acquisition completed on December 1, 2023, which increased net sales by $28 million, and (ii) a $1.44 billion, or 47%, decline in our net sales excluding the impact of the Waggaman acquisition, due to a 49% decrease in average selling prices, partially offset by a 6% increase in sales volume. Average selling prices decreased to $473 per ton in 2023 compared to $936 per ton in 2022 as lower global energy costs reduced the global market clearing price required to meet global demand. Sales volume was higher in 2023 due primarily to higher supply availability resulting from higher beginning inventory.

Cost of Sales. Cost of sales in our Ammonia segment averaged $320 per ton in 2023, a 29% decrease from $451 per ton in 2022. The decrease was due primarily to lower realized natural gas costs, including the impact of realized derivatives, and a lower cost per ton for ammonia purchased from PLNL, our joint venture in Trinidad. The impact of the Waggaman acquisition on our cost of sales per ton in our Ammonia segment was not material.

Gross Margin. Gross margin in our Ammonia segment decreased by $1.06 billion to $541 million in 2023 from $1.60 billion in 2022, and our gross margin percentage was 32.2% in 2023 compared to 51.7% in 2022. These results reflect (i) the impact of the Waggaman acquisition completed on December 1, 2023, which increased gross margin by $14 million, and (ii) a decrease in gross margin of $1.07 billion excluding the impact of the Waggaman acquisition, due primarily to a 49% decrease in average selling prices, which decreased gross margin by $1.60 billion. The decrease in average selling prices was partially offset by a decrease in realized natural gas costs, including the impact of realized derivatives, which increased gross margin by $293 million, a net decrease in manufacturing, maintenance and other costs, which increased gross margin by $167 million, and a 6% increase in sales volume, which increased gross margin by $40 million. Gross margin also includes the impact of an $11 million unrealized net mark-to-market gain on natural gas derivatives in 2023 compared to a $13 million loss in 2022.

Granular Urea Segment

Our Granular Urea segment produces granular urea, which contains 46% nitrogen. Produced from ammonia and carbon dioxide, it has the highest nitrogen content of any of our solid nitrogen fertilizers. Granular urea is produced at our Donaldsonville, Port Neal and Medicine Hat complexes.

The following table presents summary operating data for our Granular Urea segment:

	Year ended December 31,						
	2023	2022	2021	2023 v. 2022		2022 v. 2021	
	(in millions, except as noted)						
Net sales	$ 1,823	$ 2,892	$ 1,880	$ (1,069)	(37)%	$ 1,012	54 %
Cost of sales	1,010	1,328	992	(318)	(24)%	336	34 %
Gross margin	$ 813	$ 1,564	$ 888	$ (751)	(48)%	$ 676	76 %
Gross margin percentage	44.6 %	54.1 %	47.2 %	(9.5)%		6.9 %	
Sales volume by product tons (000s)	4,570	4,572	4,290	(2)	— %	282	7 %
Sales volume by nutrient tons (000s)[1]	2,102	2,103	1,973	(1)	— %	130	7 %
Average selling price per product ton	$ 399	$ 633	$ 438	$ (234)	(37)%	$ 195	45 %
Average selling price per nutrient ton[1]	$ 867	$ 1,375	$ 953	$ (508)	(37)%	$ 422	44 %
Gross margin per product ton	$ 178	$ 342	$ 207	$ (164)	(48)%	$ 135	65 %
Gross margin per nutrient ton[1]	$ 387	$ 744	$ 450	$ (357)	(48)%	$ 294	65 %
Depreciation and amortization	$ 285	$ 272	$ 235	$ 13	5 %	$ 37	16 %
Unrealized net mark-to-market (gain) loss on natural gas derivatives	$ (11)	$ 13	$ 6	$ (24)	N/M	$ 7	117 %

N/M—Not Meaningful

(1) Granular urea represents 46% nitrogen content. Nutrient tons represent the tons of nitrogen within the product tons.

Year Ended December 31, 2023 Compared to Year Ended December 31, 2022

Net Sales. Net sales in our Granular Urea segment decreased $1.07 billion, or 37%, to $1.82 billion in 2023 compared to $2.89 billion in 2022 due primarily to a 37% decrease in average selling prices. Average selling prices decreased to $399 per ton in 2023 compared to $633 per ton in 2022 as lower global energy costs reduced the global market clearing price required to meet global demand.

Cost of Sales. Cost of sales in our Granular Urea segment averaged $221 per ton in 2023, a 24% decrease from $291 per ton in 2022, due primarily to lower realized natural gas costs, including the impact of realized derivatives.

Gross Margin. Gross margin in our Granular Urea segment decreased by $751 million to $813 million in 2023 from $1.56 billion in 2022, and our gross margin percentage was 44.6% in 2023 compared to 54.1% in 2022. The decrease in gross margin was driven by a 37% decrease in average selling prices, which decreased gross margin by approximately $1.10 billion, and a net increase in manufacturing, maintenance and other costs, which decreased gross margin by $3 million. These factors that decreased gross margin were partially offset by lower realized natural gas costs, including the impact of realized derivatives, which increased gross margin by $262 million, and the impact of changes in sales volume and mix, which increased gross margin by $68 million. Gross margin also includes the impact of an $11 million unrealized net mark-to-market gain on natural gas derivatives in 2023 compared to a $13 million loss in 2022.

UAN Segment

Our UAN segment produces urea ammonium nitrate solution (UAN). UAN, a liquid fertilizer product with a nitrogen content that typically ranges from 28% to 32%, is produced by combining urea and ammonium nitrate. UAN is produced at our Courtright, Donaldsonville, Port Neal, Verdigris, Woodward, and Yazoo City complexes.

The following table presents summary operating data for our UAN segment:

	Year ended December 31,						
	2023	2022	2021	2023 v. 2022		2022 v. 2021	
	(in millions, except as noted)						
Net sales	$ 2,068	$ 3,572	$ 1,788	$ (1,504)	(42)%	$ 1,784	100 %
Cost of sales	1,251	1,489	1,119	(238)	(16)%	370	33 %
Gross margin	$ 817	$ 2,083	$ 669	$ (1,266)	(61)%	$ 1,414	211 %
Gross margin percentage	39.5 %	58.3 %	37.4 %	(18.8)%		20.9 %	
Sales volume by product tons (000s)	7,237	6,788	6,584	449	7 %	204	3 %
Sales volume by nutrient tons (000s)[1]	2,283	2,148	2,075	135	6 %	73	4 %
Average selling price per product ton	$ 286	$ 526	$ 272	$ (240)	(46)%	$ 254	93 %
Average selling price per nutrient ton[1]	$ 906	$ 1,663	$ 862	$ (757)	(46)%	$ 801	93 %
Gross margin per product ton	$ 113	$ 307	$ 102	$ (194)	(63)%	$ 205	201 %
Gross margin per nutrient ton[1]	$ 358	$ 970	$ 322	$ (612)	(63)%	$ 648	201 %
Depreciation and amortization	$ 288	$ 269	$ 259	$ 19	7 %	$ 10	4 %
Unrealized net mark-to-market (gain) loss on natural gas derivatives	$ (11)	$ 14	$ 5	$ (25)	N/M	$ 9	180 %

N/M—Not Meaningful

(1) UAN represents between 28% and 32% of nitrogen content, depending on the concentration specified by the customer. Nutrient tons represent the tons of nitrogen within the product tons.

Year Ended December 31, 2023 Compared to Year Ended December 31, 2022

Net Sales. Net sales in our UAN segment decreased $1.50 billion, or 42%, to $2.07 billion in 2023 compared to $3.57 billion in 2022 due primarily to a 46% decrease in average selling prices, partially offset by a 7% increase in sales volume. Average selling prices decreased to $286 per ton in 2023 compared to $526 per ton in 2022 due primarily to lower global energy costs reducing the global market clearing price required to meet global demand and an increase in imports of Russian UAN into the United States. The increase in sales volume was due primarily to strong customer participation in our post-application season sales program as well as higher supply availability resulting from higher production.

Cost of Sales. Cost of sales in our UAN segment averaged $173 per ton in 2023, a 21% decrease from $219 per ton in 2022, due primarily to the impact of lower realized natural gas costs, including the impact of realized derivatives.

Gross Margin. Gross margin in our UAN segment decreased by $1.27 billion to $817 million in 2023 from $2.08 billion in 2022, and our gross margin percentage was 39.5% in 2023 compared to 58.3% in 2022. The decrease in gross margin was due primarily to a 46% decrease in average selling prices, which decreased gross margin by $1.73 billion. This decrease in gross margin was partially offset by lower realized natural gas costs, including the impact of realized derivatives, which increased gross margin by $287 million, a 7% increase in sales volume, which increased gross margin by $146 million and a net decrease in manufacturing, maintenance and other costs, which increased gross margin by $10 million. Gross margin also includes the impact of an $11 million unrealized net mark-to-market gain on natural gas derivatives in 2023 compared to a $14 million loss in 2022.

AN Segment

Our AN segment produces ammonium nitrate (AN). AN, which has a nitrogen content between 29% and 35%, is produced by combining anhydrous ammonia and nitric acid. AN is used as nitrogen fertilizer and is also used by industrial customers for commercial explosives and blasting systems. AN is produced at our Yazoo City and Billingham complexes.

The following table presents summary operating data for our AN segment:

	Year ended December 31,						
	2023	2022	2021	2023 v. 2022		2022 v. 2021	
	(in millions, except as noted)						
Net sales	$ 497	$ 845	$ 510	$ (348)	(41)%	$ 335	66 %
Cost of sales	359	597	475	(238)	(40)%	122	26 %
Gross margin	$ 138	$ 248	$ 35	$ (110)	(44)%	$ 213	N/M
Gross margin percentage	27.8 %	29.3 %	6.9 %	(1.5)%		22.4 %	
Sales volume by product tons (000s)	1,571	1,594	1,720	(23)	(1)%	(126)	(7)%
Sales volume by nutrient tons (000s)[1]	538	545	582	(7)	(1)%	(37)	(6)%
Average selling price per product ton	$ 316	$ 530	$ 297	$ (214)	(40)%	$ 233	78 %
Average selling price per nutrient ton[1]	$ 924	$ 1,550	$ 876	$ (626)	(40)%	$ 674	77 %
Gross margin per product ton	$ 88	$ 156	$ 20	$ (68)	(44)%	$ 136	N/M
Gross margin per nutrient ton[1]	$ 257	$ 455	$ 60	$ (198)	(44)%	$ 395	N/M
Depreciation and amortization	$ 48	$ 61	$ 77	$ (13)	(21)%	$ (16)	(21)%
Unrealized net mark-to-market (gain) loss on natural gas derivatives	$ (2)	$ (2)	$ 4	$ —	— %	$ (6)	N/M

N/M—Not Meaningful

(1) AN represents between 29% and 35% of nitrogen content. Nutrient tons represent the tons of nitrogen within the product tons.

Year Ended December 31, 2023 Compared to Year Ended December 31, 2022

In September 2022, as a result of extremely high and volatile natural gas prices and the lack of a corresponding increase in global nitrogen product market prices, we idled ammonia production at our Billingham complex. Since that time, we have imported ammonia for upgrade at our Billingham complex into AN and other nitrogen products, which are included in our Other segment. In the third quarter of 2023, we approved our plan to permanently close the ammonia plant at our Billingham complex.

Net Sales. Net sales in our AN segment decreased $348 million, or 41%, to $497 million in 2023 from $845 million in 2022 due primarily to a 40% decrease in average selling prices. Average selling prices decreased to $316 per ton in 2023 compared to $530 per ton in 2022 as lower global energy costs reduced the global market clearing price required to meet global demand.

Cost of Sales. Cost of sales in our AN segment averaged $228 per ton in 2023, a 39% decrease from $374 per ton in 2022. The decrease was due primarily to lower production costs for our Billingham complex in 2023, as purchased ammonia was used for upgrading into AN as compared to the higher-cost natural gas used to produce ammonia to upgrade into AN in 2022. In addition, the decrease in cost of sales per ton reflects lower realized natural gas costs in North America in 2023 compared to 2022.

Gross Margin. Gross margin in our AN segment decreased by $110 million to $138 million in 2023 from $248 million in 2022, and our gross margin percentage was 27.8% in 2023 compared to 29.3% in 2022. The decrease in gross margin reflects the following:

- a 40% decrease in average selling prices, which decreased gross margin by $286 million, and a 1% decrease in sales volume, which decreased gross margin by $10 million; and
- a decrease in realized natural gas costs, including the impact of realized derivatives, which increased gross margin by $207 million, due primarily to the closure of the ammonia plant at our Billingham complex, partially offset by a net increase of $21 million in manufacturing, maintenance and other costs, driven by the cost for purchased ammonia for upgrade at our Billingham complex.

Other Segment

Our Other segment primarily includes the following products:

- diesel exhaust fluid (DEF), an aqueous urea solution typically made with 32.5% or 50% high-purity urea and the remainder deionized water;
- urea liquor, a liquid product that we sell in concentrations of 40%, 50% and 70% urea as a chemical intermediate; and
- nitric acid, a nitrogen-based mineral acid that is used in the production of nitrate-based fertilizers, nylon precursors and other specialty chemicals.

The following table presents summary operating data for our Other segment:

	Year ended December 31,						
	2023	2022	2021	2023 v. 2022		2022 v. 2021	
	(in millions, except as noted)						
Net sales	$ 564	$ 787	$ 573	$ (223)	(28)%	$ 214	37 %
Cost of sales	328	420	403	(92)	(22)%	17	4 %
Gross margin	$ 236	$ 367	$ 170	$ (131)	(36)%	$ 197	116 %
Gross margin percentage	41.8 %	46.6 %	29.7 %	(4.8)%		16.9 %	
Sales volume by product tons (000s)	2,206	2,077	2,318	129	6 %	(241)	(10)%
Sales volume by nutrient tons (000s)[1]	434	408	458	26	6 %	(50)	(11)%
Average selling price per product ton	$ 256	$ 379	$ 247	$ (123)	(32)%	$ 132	53 %
Average selling price per nutrient ton[1]	$ 1,300	$ 1,929	$ 1,251	$ (629)	(33)%	$ 678	54 %
Gross margin per product ton	$ 107	$ 177	$ 73	$ (70)	(40)%	$ 104	142 %
Gross margin per nutrient ton[1]	$ 544	$ 900	$ 371	$ (356)	(40)%	$ 529	143 %
Depreciation and amortization	$ 64	$ 67	$ 87	$ (3)	(4)%	$ (20)	(23)%
Unrealized net mark-to-market (gain) loss on natural gas derivatives	$ (4)	$ 3	$ 3	$ (7)	N/M	$ —	— %

N/M—Not Meaningful

(1) Nutrient tons represent the tons of nitrogen within the product tons.

Year Ended December 31, 2023 Compared to Year Ended December 31, 2022

In June 2022, we approved and announced our proposed plan to restructure our U.K. operations, including the planned permanent closure of our Ince facility. In August 2022, the final restructuring plan was approved, and decommissioning activities were initiated. We produced compound fertilizer products (NPKs), which are solid granular fertilizer products for which the nutrient content is a combination of nitrogen, phosphorus and potassium, only at our Ince facility, and closure of this facility resulted in our discontinuation of the NPK product line, which was included in our Other segment.

In September 2022, as a result of extremely high and volatile natural gas prices in the United Kingdom and the lack of a corresponding increase in global nitrogen product market prices, we idled ammonia production at our Billingham complex. Since that time, we have imported ammonia for upgrade at our Billingham complex into AN and other nitrogen products, including nitric acid and aqua ammonia, which are included in our Other segment. In the third quarter of 2023, we approved our plan to permanently close the ammonia plant at our Billingham complex.

Net Sales. Net sales in our Other segment decreased $223 million, or 28%, to $564 million in 2023 from $787 million in 2022 due to a 32% decrease in average selling prices, partially offset by a 6% increase in sales volume. Average selling prices decreased to $256 per ton in 2023 compared to $379 per ton in 2022 as lower global energy costs reduced the global market clearing price required to meet global demand. The increase in sales volume was due primarily to higher DEF sales volume.

Cost of Sales. Cost of sales in our Other segment averaged $149 per ton in 2023, a 26% decrease from $202 per ton in 2022, due primarily to lower production costs for our Billingham complex in 2023, as purchased ammonia was used for upgrading into other nitrogen products as compared to the higher-cost natural gas used to produce ammonia to upgrade into other nitrogen products in 2022. In addition, the decrease in cost of sales per ton reflects lower realized natural gas costs in North America in 2023 compared to 2022.

Gross Margin. Gross margin in our Other segment decreased by $131 million to $236 million in 2023 from $367 million in 2022, and our gross margin percentage was 41.8% in 2023 compared to 46.6% in 2022. The decrease in gross margin reflects the following:

- a 32% decrease in average selling prices, which decreased gross margin by $263 million, partially offset by a 6% increase in sales volume, which increased gross margin by $29 million;
- a decrease in realized natural gas costs, including the impact of realized derivatives, which increased gross margin by $106 million, partially offset by a net increase of $10 million in manufacturing, maintenance and other costs, driven by the cost for purchased ammonia for upgrade at our Billingham complex; and
- the impact of a $4 million unrealized net mark-to-market gain on natural gas derivatives in 2023 compared to a $3 million loss in 2022.

Liquidity and Capital Resources

Our primary uses of cash are generally for operating costs, working capital, capital expenditures, debt service, investments, taxes, share repurchases, and dividends. Our working capital requirements are affected by several factors, including demand for our products, selling prices, raw material costs, freight costs and seasonal factors inherent in the business. We may also utilize our cash to fund acquisitions. In addition, we may from time to time seek to retire or purchase our outstanding debt through cash purchases, in open market or privately negotiated transactions or otherwise. Such repurchases, if any, will depend on prevailing market conditions, our liquidity requirements, contractual restrictions and other factors. The amounts involved may be material.

Generally, our primary source of cash is cash from operations, which includes cash generated by customer advances. We may also from time to time access the capital markets or engage in borrowings under our revolving credit agreement.

On December 1, 2023, we acquired an ammonia production facility located in Waggaman, Louisiana. See “Acquisition of Waggaman Ammonia Production Facility,” below, for additional information.

As of December 31, 2023, our cash and cash equivalents balance was $2.03 billion, a decrease of $291 million from $2.32 billion at December 31, 2022. At December 31, 2023, we were in compliance with all applicable covenant requirements under our revolving credit agreement and senior notes, and unused borrowing capacity under our revolving credit agreement was $750 million.

On January 31, 2024, our Board of Directors (the Board) declared a quarterly dividend of $0.50 per common share, representing an increase from the quarterly dividend of $0.40 per common share that was declared and paid in the fourth quarter of 2023. The dividend will be paid on February 29, 2024 to stockholders of record as of February 15, 2024.

Cash Equivalents

Cash equivalents include highly liquid investments that are readily convertible to known amounts of cash with original maturities of three months or less. Under our short-term investment policy, we may invest our cash balances, either directly or through mutual funds, in several types of investment-grade securities, including notes and bonds issued by governmental entities or corporations. Securities issued by governmental entities include those issued directly by the U.S. and Canadian federal governments; those issued by state, local or other governmental entities; and those guaranteed by entities affiliated with governmental entities.

Capital Spending

We make capital expenditures to sustain our asset base, increase our capacity or capabilities, improve plant efficiency, comply with various environmental, health and safety requirements, and invest in our clean energy strategy. Capital expenditures totaled $499 million in 2023 compared to $453 million in 2022.

Capital expenditures in 2024 are estimated to be in the range of $550 million. Planned capital expenditures are generally subject to change due to delays in regulatory approvals or permitting, unanticipated increases in cost, changes in scope and completion time, performance of third parties, delays in the receipt of equipment, adverse weather, defects in materials and workmanship, labor or material shortages, transportation constraints, acceleration or delays in the timing of the work and other unforeseen difficulties.

Acquisition of Waggaman Ammonia Production Facility

On December 1, 2023, we acquired an ammonia production facility located in Waggaman, Louisiana from Dyno Nobel Louisiana Ammonia, LLC (DNLA), a U.S. subsidiary of Australia-based Incitec Pivot Limited (IPL), pursuant to an asset purchase agreement with DNLA and IPL. The facility has a nameplate capacity of 880,000 tons of ammonia annually.

In connection with the acquisition, we entered into a long-term ammonia offtake agreement providing for us to supply up to 200,000 tons of ammonia per year to IPL's Dyno Nobel, Inc. subsidiary. Under the terms of the asset purchase agreement, $425 million of the purchase price of $1.675 billion, subject to adjustment, was allocated by the parties to the ammonia offtake agreement. We funded the balance of the purchase price with $1.223 billion of cash on hand. The consideration transferred reflects an estimated net working capital adjustment and other adjustments to the purchase price, which is subject to further adjustment pursuant to the terms of the asset purchase agreement. We expect any further purchase price adjustments required under the asset purchase agreement will be completed in 2024.

The financial results of the Waggaman facility are included in our consolidated statement of operations and in our Ammonia segment from the acquisition date of December 1, 2023. See Note 6—Acquisition of Waggaman Ammonia Production Facility for additional information.

Share Repurchase Programs

On November 3, 2021, the Board authorized the repurchase of up to $1.5 billion of CF Holdings common stock through December 31, 2024 (the 2021 Share Repurchase Program). On November 2, 2022, the Board authorized the repurchase of up to $3 billion of CF Holdings common stock commencing upon completion of the 2021 Share Repurchase Program and effective through December 31, 2025 (the 2022 Share Repurchase Program). Repurchases under our share repurchase programs may be made from time to time in the open market, through privately negotiated transactions, through block transactions, through accelerated share repurchase programs, or otherwise. The manner, timing and amount of repurchases will be determined by our management based on the evaluation of market conditions, stock price, and other factors. Shares repurchased, including those repurchased under share repurchase programs, are retired as approved by the Board.

The following table summarizes the share repurchases under the 2022 Share Repurchase Program and 2021 Share Repurchase Program.

	2022 Share Repurchase Program		2021 Share Repurchase Program	
	Shares	Amounts[1]	Shares	Amounts[1]
	(in millions)			
Shares repurchased in 2022:				
First quarter	—	$ —	1.3	$ 100
Second quarter	—	—	5.3	490
Third quarter	—	—	6.1	532
Fourth quarter	—	—	2.2	223
Total shares repurchased in 2022	—	—	14.9	1,345
Shares repurchased as of December 31, 2022	—	$ —	14.9	$ 1,345
Shares repurchased in 2023:				
First quarter	—	$ —	1.1	$ 75
Second quarter	0.8	50	1.2	80
Third quarter	1.9	150	—	—
Fourth quarter	2.9	225	—	—
Total shares repurchased in 2023	5.6	425	2.3	155
Shares repurchased as of December 31, 2023	5.6	$ 425	17.2	$ 1,500

(1) As defined in the share repurchase programs, amounts reflect the price paid for the shares of common stock repurchased, excluding commissions paid to brokers and excise taxes.

In the first half of 2023, we completed the 2021 Share Repurchase Program with the repurchase of approximately 2.3 million shares for $155 million. In 2023, we repurchased 5.6 million shares under the 2022 Share Repurchase Program for $425 million.

Government Policies

The policies or laws of governments around the world can result in the imposition of taxes, duties, tariffs or other restrictions or regulatory requirements on imports and exports of raw materials, finished goods or services from a particular country or region of the world. The policies and laws of governments can also impact the subsidization of natural gas prices, and subsidies or quotas applied to domestic producers or farmers. Due to the critical role that fertilizers play in food production, the construction and operation of fertilizer plants often are influenced by economic, political and social objectives. Additionally, the import or export of fertilizer can be subject to local taxes imposed by governments which can have the effect of either encouraging or discouraging import and export activity. The impact of changes in governmental policies or laws or the political or social objectives of a country could have a material impact on fertilizer demand and selling prices and therefore could impact our liquidity.

Repatriation of Foreign Earnings and Income Taxes

We have operations in Canada, the United Kingdom and a 50% interest in a joint venture in Trinidad. Historically, the estimated additional U.S. and foreign income taxes due upon repatriation of the earnings of these foreign operations to the United States were recognized in our consolidated financial statements as the earnings were recognized, unless the earnings were considered to be permanently reinvested based upon our then current plans. However, the cash payment of the income tax liabilities associated with repatriation of earnings from foreign operations occurred at the time of the repatriation. As a result, the recognition of income tax expense related to foreign earnings, as applicable, and the payment of taxes resulting from repatriation of those earnings could occur in different periods.

In light of changes made by the Tax Cuts and Jobs Act, commencing with the 2018 tax year, the United States no longer taxes earnings of foreign subsidiaries even when such earnings are earned or repatriated to the United States, unless such earnings are subject to U.S. rules on passive income or certain anti-abuse provisions. Foreign subsidiary earnings may still be subject to withholding taxes when repatriated to the United States.

Cash balances held by our joint venture are maintained at sufficient levels to fund local operations as accumulated earnings are repatriated from the joint venture on a periodic basis.

As of December 31, 2023, approximately $205 million of our consolidated cash and cash equivalents balance of $2.03 billion was held by our Canadian and United Kingdom subsidiaries. As of December 31, 2023, we recorded a deferred tax liability of $9 million on the undistributed earnings of our Canadian subsidiaries for which the Company does not have an indefinite reinvestment assertion. We have not provided for deferred taxes on the remainder of undistributed earnings from our foreign subsidiaries because such earnings would not give rise to additional tax liabilities upon repatriation or are considered to be indefinitely reinvested.

Debt

Revolving Credit Agreement

On October 26, 2023, we entered into a new senior unsecured revolving credit agreement (the New Revolving Credit Agreement), which replaced our prior senior unsecured revolving credit agreement (the Prior Revolving Credit Agreement) that was scheduled to mature on December 5, 2024. The Prior Revolving Credit Agreement provided for a revolving credit facility of up to $750 million and included a letter of credit sub-limit of $125 million.

The New Revolving Credit Agreement provides for a revolving credit facility of up to $750 million with a maturity of October 26, 2028 and includes a letter of credit sub-limit of $125 million. Borrowings under the New Revolving Credit Agreement may be used for working capital, capital expenditures, acquisitions, share repurchases and other general corporate purposes. CF Industries is the lead borrower, and CF Holdings is the sole guarantor, under the New Revolving Credit Agreement.

Borrowings under the New Revolving Credit Agreement can be denominated in U.S. dollars, Canadian dollars, euros and British pounds. Borrowings in U.S. dollars bear interest at a per annum rate equal to, at our option, an applicable adjusted term Secured Overnight Financing Rate or base rate plus, in either case, a specified margin. We are required to pay an undrawn commitment fee on the undrawn portion of the commitments under the New Revolving Credit Agreement and customary letter of credit fees. The specified margin and the amount of the commitment fee depend on CF Holdings' credit rating at the time.

The New Revolving Credit Agreement contains representations and warranties and affirmative and negative covenants, including one financial covenant. The financial covenant requires that the total net leverage ratio (as defined in the New Revolving Credit Agreement) be not greater than 3.75:1.00 (the Maximum Total Net Leverage Ratio) as of the last day of each fiscal quarter, provided that, if any borrower or subsidiary consummates a material acquisition during any fiscal quarter, CF Industries may elect to increase the Maximum Total Net Leverage Ratio to 4.25:1.00 for the period of four consecutive fiscal quarters commencing with such fiscal quarter (and no further such election may be made unless and until the Maximum Total Net Leverage Ratio is less than or equal to 3.75:1.00 as of the end of two consecutive fiscal quarters after the end of such period).

As of December 31, 2023, we were in compliance with all covenants under the New Revolving Credit Agreement.

The New Revolving Credit Agreement contains events of default (with notice requirements and cure periods, as applicable) customary for a financing of this type, including, but not limited to, non-payment of principal, interest or fees; inaccuracy of representations and warranties in any material respect; and failure to comply with specified covenants. Upon the occurrence and during the continuance of an event of default under the New Revolving Credit Agreement and after any applicable cure period, subject to specified exceptions, the administrative agent may, and at the request of the requisite lenders is required to, accelerate the loans under the New Revolving Credit Agreement or terminate the lenders' commitments under the New Revolving Credit Agreement.

As of December 31, 2023, we had unused borrowing capacity under the New Revolving Credit Agreement of $750 million and no outstanding letters of credit under the New Revolving Credit Agreement. As of December 31, 2023 and 2022, and during the years then ended, there were no borrowings outstanding under either the Prior Revolving Credit Agreement or the New Revolving Credit Agreement.

Letters of Credit Under Bilateral Agreement

We are party to a bilateral agreement providing for the issuance of up to $425 million (reflecting an increase of $75 million in December 2023) of letters of credit. As of December 31, 2023, approximately $291 million of letters of credit were outstanding under this agreement.

Senior Notes

Long-term debt presented on our consolidated balance sheets as of December 31, 2023 and 2022 consisted of the following debt securities issued by CF Industries:

		December 31, 2023		December 31, 2022	
	Effective Interest Rate	Principal Outstanding	Carrying Amount[1]	Principal Outstanding	Carrying Amount[1]
		(in millions)			
Public Senior Notes:					
5.150% due March 2034	5.293%	$ 750	$ 741	$ 750	$ 741
4.950% due June 2043	5.040%	750	742	750	742
5.375% due March 2044	5.478%	750	741	750	740
Senior Secured Notes:					
4.500% due December 2026[2]	4.783%	750	744	750	742
Total long-term debt		$ 3,000	$ 2,968	$ 3,000	$ 2,965

(1) Carrying amount is net of unamortized debt discount and deferred debt issuance costs. Total unamortized debt discount was $7 million as of both December 31, 2023 and 2022, and total deferred debt issuance costs were $25 million and $28 million as of December 31, 2023 and 2022, respectively.

(2) Effective August 23, 2021, these notes are no longer secured, in accordance with the terms of the applicable indenture.

Public Senior Notes

On April 21, 2022, we redeemed in full all of the $500 million outstanding principal amount of the 2023 Notes in accordance with the optional redemption provisions in the indenture governing the 2023 Notes. The total aggregate redemption price paid in connection with the April 2022 redemption of the 2023 Notes, which was funded with cash on hand, was $513 million, including accrued interest. As a result, we recognized a loss on debt extinguishment of $8 million, consisting primarily of the premium paid on the redemption of the $500 million principal amount of the 2023 Notes prior to their scheduled maturity.

Under the indentures (including the applicable supplemental indentures) governing our senior notes due 2034, 2043 and 2044 identified in the table above (the Public Senior Notes), each series of Public Senior Notes is guaranteed by CF Holdings. Interest on the Public Senior Notes is payable semiannually, and the Public Senior Notes are redeemable at our option, in whole at any time or in part from time to time, at specified make-whole redemption prices.

The indentures governing the Public Senior Notes contain covenants that limit, among other things, the ability of CF Holdings and its subsidiaries, including CF Industries, to incur liens on certain assets to secure debt, to engage in sale and leaseback transactions, to merge or consolidate with other entities and to sell, lease or transfer all or substantially all of the assets of CF Holdings and its subsidiaries to another entity. Each of the indentures governing the Public Senior Notes provides for customary events of default, which include (subject in certain cases to customary grace and cure periods), among others, nonpayment of principal or interest on the applicable Public Senior Notes; failure to comply with other covenants or agreements under the indenture; certain defaults on other indebtedness; the failure of CF Holdings' guarantee of the applicable Public Senior Notes to be enforceable; and specified events of bankruptcy or insolvency. Under each indenture governing the Public Senior Notes, in the case of an event of default arising from one of the specified events of bankruptcy or insolvency, the applicable Public Senior Notes would become due and payable immediately, and, in the case of any other event of default (other than an event of default related to CF Industries' and CF Holdings' reporting obligations), the trustee or the holders of at least 25% in aggregate principal amount of the applicable Public Senior Notes then outstanding may declare all of such Public Senior Notes to be due and payable immediately.

Under each of the indentures governing the Public Senior Notes, specified changes of control involving CF Holdings or CF Industries, when accompanied by a ratings downgrade, as defined with respect to the applicable series of Public Senior Notes, constitute change of control repurchase events. Upon the occurrence of a change of control repurchase event with respect to a series of Public Senior Notes, unless CF Industries has exercised its option to redeem such Public Senior Notes, CF Industries will be required to offer to repurchase them at a price equal to 101% of the principal amount thereof, plus accrued and unpaid interest, if any, to, but not including, the date of repurchase.

Senior Secured Notes

Under the terms of the indenture governing the 4.500% senior secured notes due 2026 (the 2026 Notes), the 2026 Notes are guaranteed by CF Holdings. Interest on the 2026 Notes is payable semiannually, and the 2026 Notes are redeemable at our option, in whole at any time or in part from time to time, at specified make-whole redemption prices.

Under the indenture governing the 2026 Notes, specified changes of control involving CF Holdings or CF Industries, when accompanied by a ratings downgrade, as defined with respect to the 2026 Notes, constitute change of control repurchase events. Upon the occurrence of a change of control repurchase event with respect to the 2026 Notes, unless CF Industries has exercised its option to redeem such notes, CF Industries will be required to offer to repurchase them at a price equal to 101% of the principal amount thereof, plus accrued and unpaid interest, if any, to, but not including, the date of repurchase.

The indenture governing the 2026 Notes contains covenants that limit, among other things, the ability of CF Holdings and its subsidiaries, including CF Industries, to incur liens on certain assets to secure debt, to engage in sale and leaseback transactions, to merge or consolidate with other entities and to sell, lease or transfer all or substantially all of the assets of CF Holdings and its subsidiaries to another entity. The indenture governing the 2026 Notes provides for customary events of default, which include (subject in certain cases to customary grace and cure periods), among others, nonpayment of principal or interest of the 2026 Notes; failure to comply with other covenants or agreements under the indenture; certain defaults on other indebtedness; the failure of CF Holdings' guarantee of the 2026 Notes to be enforceable; and specified events of bankruptcy or insolvency. Under the indenture governing the 2026 Notes, in the case of an event of default arising from one of the specified events of bankruptcy or insolvency, the 2026 Notes would become due and payable immediately, and, in the case of any other event of default (other than an event of default related to CF Industries' and CF Holdings' reporting obligations), the trustee or the holders of at least 25% in aggregate principal amount of the 2026 Notes then outstanding may declare all of such notes to be due and payable immediately.

Forward Sales and Customer Advances

We offer our customers the opportunity to purchase products from us on a forward basis at prices and on delivery dates we propose. Therefore, our reported fertilizer selling prices and margins may differ from market spot prices and margins available at the time of shipment.

Customer advances, which typically represent a portion of the contract's value, are received shortly after the contract is executed, with any remaining unpaid amount generally being collected by the time control transfers to the customer, thereby reducing or eliminating the accounts receivable related to such sales. Any cash payments received in advance from customers in connection with forward sales contracts are reflected on our consolidated balance sheets as a current liability until control transfers and revenue is recognized. As of December 31, 2023 and 2022, we had $130 million and $229 million, respectively, in customer advances on our consolidated balance sheets.

While customer advances are generally a significant source of liquidity, the level of forward sales contracts is affected by many factors including current market conditions, our customers' outlook of future market fundamentals and seasonality. During periods of declining prices, customers tend to delay purchasing fertilizer in anticipation that prices in the future will be lower than the current prices. If the level of sales under our forward sales programs were to decrease in the future, our cash received from customer advances would likely decrease and our accounts receivable balances would likely increase. Additionally, borrowing under the New Revolving Credit Agreement could become necessary. Due to the volatility inherent in our business and changing customer expectations, we cannot estimate the amount of future forward sales activity.

Under our forward sales programs, a customer may delay delivery of an order due to weather conditions or other factors. These delays generally subject the customer to potential charges for storage or may be grounds for termination of the contract by us. Such a delay in scheduled shipment or termination of a forward sales contract due to a customer's inability or unwillingness to perform may negatively impact our reported sales.

Natural Gas

Natural gas is the principal raw material used to produce our nitrogen products. We use natural gas both as a chemical feedstock and as a fuel to produce ammonia, granular urea, UAN, AN and other products. Expenditures on natural gas are a significant portion of our production costs, representing approximately 40% of our total production costs in 2023. As a result of these factors, natural gas prices have a significant impact on our operating expenses and can thus affect our liquidity. Natural gas costs in our cost of sales, including the impact of realized natural gas derivatives, decreased 49% to $3.67 per MMBtu in 2023 from $7.18 per MMBtu in 2022.

We enter into agreements for a portion of our future natural gas supply and related transportation. As of December 31, 2023, our natural gas purchase agreements have terms that range from four months to three years and a total minimum commitment of approximately $736 million, and our natural gas transportation agreements have terms that range from one to ten years and a total minimum commitment of approximately $106 million. Our minimum commitments to purchase and transport natural gas are based on prevailing market-based forward prices excluding reductions for plant maintenance and turnaround activities.

Most of our nitrogen manufacturing facilities are located in the United States and Canada. As a result, the price of natural gas in North America directly impacts a substantial portion of our operating expenses. During 2023, the daily closing price at the Henry Hub, the most heavily-traded natural gas pricing point in North America, reached a low of $1.72 per MMBtu on three consecutive days in June 2023 and a high of $3.81 per MMBtu on January 5, 2023. During the three-year period ended December 31, 2023, the daily closing price at the Henry Hub reached a low of $1.72 per MMBtu on three consecutive days in June 2023 and a high of $23.61 per MMBtu on February 18, 2021.

Derivative Financial Instruments

We use derivative financial instruments to reduce our exposure to changes in prices for natural gas that will be purchased in the future. Natural gas is the largest and most volatile component of our manufacturing cost for nitrogen-based products. From time to time, we may also use derivative financial instruments to reduce our exposure to changes in foreign currency exchange rates. Volatility in reported quarterly earnings can result from the unrealized mark-to-market adjustments in the value of the derivatives. In 2023 and 2022, we recognized an unrealized net mark-to-market (gain) loss on natural gas derivatives of $(39) million and $41 million, respectively, which is reflected in cost of sales in our consolidated statements of operations.

Derivatives expose us to counterparties and the risks associated with their ability to meet the terms of the contracts. For derivatives that are in net asset positions, we are exposed to credit loss from nonperformance by the counterparties. We control our credit risk through the use of multiple counterparties that are multinational commercial banks, other major financial institutions or large energy companies, and the use of International Swaps and Derivatives Association (ISDA) master netting arrangements. The ISDA agreements are master netting arrangements commonly used for over-the-counter derivatives that mitigate exposure to counterparty credit risk, in part, by creating contractual rights of netting and setoff, the specifics of which vary from agreement to agreement.

The ISDA agreements for most of our derivative instruments contain credit-risk-related contingent features, such as cross default provisions. In the event of certain defaults or termination events, our counterparties may request early termination and net settlement of certain derivative trades or may require us to collateralize derivatives in a net liability position. As of December 31, 2023 and 2022, the aggregate fair value of the derivative instruments with credit-risk-related contingent features in net liability positions was $34 million and $73 million, respectively, which also approximates the fair value of the assets that may be needed to settle the obligations if the credit-risk-related contingent features were triggered at the reporting dates.

As of December 31, 2023, our open natural gas derivative contracts consisted of natural gas fixed price swaps, basis swaps and options for 49.0 million MMBtus. As of December 31, 2022, our open natural gas derivative contracts consisted of natural gas fixed price swaps, basis swaps and options for 66.3 million MMBtus. At both December 31, 2023 and 2022, we had no cash collateral on deposit with counterparties for derivative contracts. The credit support documents executed in connection with certain of our ISDA agreements generally provide us and our counterparties the right to set off collateral against amounts owing under the ISDA agreements upon the occurrence of a default or a specified termination event.

Defined Benefit Pension Plans

We contributed $44 million to our pension plans in 2023. In 2024 and 2025, we expect to contribute approximately $27 million and $11 million, respectively, to our U.K. pension plans as currently agreed with the plans' trustees.

Distributions on Noncontrolling Interest in CFN

The CFN Board of Managers approved semi-annual distribution payments for the years ended December 31, 2023, 2022 and 2021, in accordance with CFN's limited liability company agreement, as follows:

Approved and paid	Distribution Period	Distribution Amount (in millions)
First quarter of 2024	Six months ended December 31, 2023	$ 144
Third quarter of 2023	Six months ended June 30, 2023	204
First quarter of 2023	Six months ended December 31, 2022	255
Third quarter of 2022	Six months ended June 30, 2022	372
First quarter of 2022	Six months ended December 31, 2021	247
Third quarter of 2021	Six months ended June 30, 2021	130

Cash Flows

Net cash provided by operating activities in 2023 was $2.76 billion, a decrease of $1.10 billion compared to $3.86 billion in 2022. The decrease in cash flow from operations was due primarily to lower net earnings, partially offset by changes in net working capital. Net earnings in 2023 was $1.84 billion compared to $3.94 billion in 2022. The decrease in net earnings was due primarily to a decrease in gross margin, driven by lower average selling prices, partially offset by lower natural gas costs, a lower income tax provision, lower net interest expense, a decrease in net earnings attributable to noncontrolling interest, and impairment charges related to our U.K. operations in 2022 that did not recur in 2023. During 2023, net changes in working capital increased cash flow from operations by $20 million, while in 2022, net changes in working capital reduced cash flow from operations by $900 million. The difference in the impact of changes in working capital on cash flow from operations in 2023 as compared to 2022 was attributable primarily to favorable changes in 2023 as compared to 2022 in accounts receivable, inventory, customer advances and income tax liabilities.

Net cash used in investing activities was $1.68 billion in 2023 compared to $440 million in 2022, or an increase of $1.24 billion. The increase was due primarily to the consideration paid of $1.223 billion for the Waggaman acquisition. See "Acquisition of Waggaman Ammonia Production Facility," above for additional information. During 2023, capital expenditures totaled $499 million compared to $453 million in 2022.

Net cash used in financing activities was $1.37 billion in 2023 compared to $2.70 billion in 2022. The decrease was due primarily to a decrease in share repurchases and lower distributions to the noncontrolling interest. In 2023, we paid $580 million for share repurchases compared to $1.35 billion for share repurchases in 2022. In 2023, distributions to the noncontrolling interest were $459 million compared to $619 million in 2022. In addition, we paid $507 million in connection with the redemption of the 2023 Notes in 2022.

Critical Accounting Estimates

Our discussion and analysis of our financial condition, results of operations, liquidity and capital resources is based upon our consolidated financial statements, which have been prepared in accordance with U.S. GAAP. U.S. GAAP requires that we select policies and make estimates and judgments that affect the reported amounts of assets, liabilities, revenues, expenses and related disclosure of contingent assets and liabilities. On an ongoing basis, we evaluate our estimates. We base our estimates on historical experience, technological assessment, opinions of appropriate outside experts, and the most recent information available to us. Actual results may differ from these estimates. Changes in estimates that may have a material impact on our results are discussed in the context of the underlying financial statements to which they relate. The following discussion presents information about our most critical accounting estimates.

Acquisition of Waggaman Ammonia Production Facility

On December 1, 2023, we acquired an ammonia production facility located in Waggaman, Louisiana from DNLA, a U.S. Subsidiary of Australia-based IPL, pursuant to an asset purchase agreement with DNLA and IPL. We accounted for the acquisition as a business combination. In connection with the acquisition, we entered into a long-term ammonia offtake agreement providing for us to supply up to 200,000 tons of ammonia per year to IPL's Dyno Nobel, Inc. subsidiary (the Supply Contract). In connection with recording the acquisition, we recognized, among other items, property, plant and equipment, goodwill, intangible assets consisting primarily of customer relationships, and a supply contract liability. See Note 6—Acquisition of Waggaman Ammonia Production Facility for additional information.

The valuation of the assets acquired and liabilities assumed as a result of our acquisition of the Waggaman ammonia production facility on December 1, 2023, requires us to make significant estimates and assumptions.

We estimated the fair value of the customer relationships of $455 million using the multi-period excess earnings method of the income approach, which incorporated the estimated future cash flows associated with the net earnings attributable to the acquired customer relationships. The estimated future cash flows were discounted to their present value using an appropriate risk-adjusted discount rate from the perspective of a market participant. Key assumptions used in estimating future cash flows include forecasted product selling prices, future production and sales volumes, probability of renewal, projected natural gas costs, operating rates, operating expenses, inflation, tax rates, capital spending, and contributory asset charges, among other factors. The discount rate utilized in estimating the fair value of the customer relationships was 9.0% and was derived using a capital asset pricing model and publicly available data for comparable companies to estimate the cost of equity financing. We used Moody's Baa corporate bonds as a benchmark to estimate the cost of debt. Additional assumptions utilized in the valuation of the customer relationships included the duration of the forecasted cash flows and the tax amortization benefit, among other factors.

The terms of the Supply Contract were determined to be unfavorable compared to market as of the acquisition date. Therefore, we estimated the fair value of the Supply Contract liability of $757 million using the with and without method of the income approach. This method included estimating the future cash flows by applying both a market price and the contract price in determining the respective forecasted sales. The difference in the estimated future cash flows using the resulting forecasted sales was then discounted to present value at the date of acquisition using an appropriate risk-adjusted discount rate from the perspective of a market participant. Key assumptions used in estimating future cash flows include forecasted product selling price, specifically the market price in excess of the contract price, future production and sales volumes, probability of renewal, and inflation, among other factors. The discount rate utilized in estimating the fair value of the Supply Contract liability was 6.2% and was derived based on the historical cost of our issued debt and treasury and corporate bond yields as of the acquisition date. Additional assumptions utilized in the valuation of the Supply Contract liability included the duration of the forecasted cash flows, among other factors.

Due to the inherent uncertainties involved in making estimates and assumptions, actual results may differ from those assumed in our forecasts. Of the factors discussed above, the following assumptions are more sensitive than others:

- forecasted product selling prices included in the estimated future cash flows utilized in the valuation of the customer relationships and the Supply Contract liability,

- projected natural gas costs included in the estimated future cash flows utilized in the valuation of the customer relationships, and
- the discount rate utilized in each of these valuations to discount the cash flows in arriving at the fair values.

Assuming that all other assumptions and estimates utilized in our valuation of the customer relationships and Supply Contract liability remain unchanged, a change in each of these inputs would have the following effect on the estimated fair value used to allocate the purchase price in the Waggaman acquisition as of December 1, 2023:

	Increase/(Decrease) in Fair Value			
	Customer Relationships		Supply Contract Liability	
Assumption	(in millions)			
	+$5.00	-$5.00	+$5.00	-$5.00
Average Selling Price per Product Ton	$ 26	$ (25)	$ 16	$ (15)
	+$0.05	-$0.05		
Natural Gas Cost per MMBtu	$ (9)	$ 9	n/a	n/a
	+50 bps	-50 bps	+50 bps	-50 bps
Discount Rate	$ (25)	$ 27	$ (35)	$ 38

n/a—not applicable

Recoverability of Long-Lived Assets, Goodwill and Investment in Unconsolidated Affiliate

We review the carrying values of our property, plant and equipment and other long-lived assets, including our finite-lived intangible assets, goodwill and our investment in an unconsolidated affiliate in accordance with U.S. GAAP in order to assess recoverability. Factors that we must estimate when performing impairment tests include production and sales volumes, selling prices, raw material costs, operating rates, operating expenses, inflation, discount rates, exchange rates, tax rates, capital spending and the impact that future market dynamics and geopolitical events could have on these factors. Judgment is involved in estimating each of these factors, which include inherent uncertainties. The factors we use are consistent with those used in our internal planning process. The recoverability of the values associated with our goodwill, long-lived assets and our investment in an unconsolidated affiliate is dependent upon future operating performance of the specific businesses to which they are attributed. Certain of the operating assumptions are particularly sensitive to the cyclical nature of the fertilizer business. Adverse changes in demand for our products, increases in supply and the availability and costs of key raw materials could significantly affect the results of our review.

The recoverability and impairment tests of long-lived assets are required only when conditions exist that indicate the carrying value may not be recoverable. For goodwill, impairment tests are required at least annually, or more frequently whenever events or circumstances indicate that the carrying value may not be recoverable. Our investment in an unconsolidated affiliate is reviewed for impairment whenever events or circumstances indicate that its carrying value may not be recoverable. When circumstances indicate that the fair value of our investment is less than its carrying value, and the reduction in value is other than temporary, the reduction in value would be recognized immediately in earnings.

We evaluate goodwill for impairment in the fourth quarter at the reporting unit level. Our evaluation generally begins with a qualitative assessment of the factors that could impact the significant inputs used to estimate fair value. If after performing the qualitative assessment, we determine that it is not more likely than not that the fair value of a reporting unit is less than its carrying amount, including goodwill, then no further analysis is necessary. However, if it is unclear based on the results of the qualitative test, we perform a quantitative test, which involves comparing the fair value of a reporting unit with its carrying amount, including goodwill. We use an income-based valuation method, determining the present value of future cash flows, to estimate the fair value of a reporting unit. If the fair value of a reporting unit exceeds its carrying amount, no further testing is necessary. If the fair value of the reporting unit is less than its carrying amount, goodwill impairment would be recognized equal to the amount of the carrying value in excess of the reporting unit's fair value, limited to the total amount of goodwill allocated to the reporting unit.

We review property, plant and equipment and other long-lived assets at the asset group level in order to assess recoverability based on expected future undiscounted cash flows. If the sum of the expected future net undiscounted cash flows is less than the carrying value, an impairment loss would be recognized. The impairment loss is measured as the amount by which the carrying value exceeds the fair value of the long-lived assets.

PLNL is our joint venture investment in Trinidad and operates an ammonia plant that relies on natural gas supplied, under a gas sales contract, by the National Gas Company of Trinidad and Tobago Limited (NGC). The joint venture is accounted for under the equity method. The joint venture experienced past curtailments in the supply of natural gas from NGC, which reduced the ammonia production at PLNL. The prior gas sales contract had an expiration date of September 2023. In the third quarter of 2023, PLNL entered into a new gas sales contract with NGC (the New NGC Contract), which is effective October 2023 through December 2025. Any NGC commitment to supply gas beyond December 2025 will be based on new agreements.

In the third quarter of 2023 and due to the terms of the New NGC Contract, we assessed our investment in PLNL for impairment and determined that the carrying value of our equity method investment in PLNL exceeded its fair value. As a result, we recorded an impairment of our equity method investment in PLNL of $43 million, which is reflected in equity in (loss) earnings of operating affiliate on our consolidated statement of operations for the year ended December 31, 2023. If NGC does not make sufficient quantities of natural gas available to PLNL at prices that permit profitable operations, PLNL may cease operating its facility and we would write off the remaining investment in PLNL. The carrying value of our equity method investment in PLNL at December 31, 2023 was $26 million.

Projected Benefit Obligations

The projected benefit obligations (PBOs) for our defined benefit pension plans are affected by plan design, actuarial estimates and discount rates. Key assumptions that affect our PBO are discount rates and, in addition for our United Kingdom plans, inflation rates, including an adjusted U.K. retail price index (RPI).

The December 31, 2023 PBO was computed based on a weighted-average discount rate of 4.8% for our North America plans and 4.6% for our United Kingdom plans, which were based on yields for high-quality (AA rated or better) fixed income debt securities that match the timing and amounts of expected benefit payments as of the measurement date of December 31, 2023. Declines in comparable bond yields would increase our PBO. For our United Kingdom plans, the 3.0% RPI used to calculate our PBO is developed using a U.K. government gilt prices only retail price inflation curve, which is based on the difference between yields on fixed interest government bonds and index-linked government bonds.

For North America qualified pension plans, our PBO was $292 million as of December 31, 2023, which was $21 million lower than pension plan assets. For our United Kingdom pension plans, our PBO was $367 million as of December 31, 2023, which was $7 million higher than pension plan assets. The table below estimates the impact of a 50 basis point increase or decrease in the key assumptions on our December 31, 2023 PBO:

	Increase/(Decrease) in December 31, 2023 PBO			
	North America Plans		United Kingdom Plans	
Assumption	**+50 bps**	**-50 bps**	**+50 bps**	**-50 bps**
	(in millions)			
Discount Rate	$ (16)	$ 18	$ (21)	$ 22
RPI	n/a	n/a	9	(9)

n/a—not applicable

See Note 13—Pension and Other Postretirement Benefits for further discussion of our pension plans.

Income Taxes

We are subject to the income tax laws of the many jurisdictions in which we operate, and we recognize expense, assets and liabilities based on estimates of amounts that ultimately will be determined to be taxable or deductible in tax returns filed in various jurisdictions. These tax laws are complex, and how they apply to our facts is sometimes open to interpretation. We recognize the effect of income tax positions only if sustaining those positions is more likely than not. Tax positions that meet the more likely than not recognition threshold but are not highly certain are measured based on the largest amount of benefit that is greater than 50% likely of being realized upon settlement with the taxing authority. Differences in interpretation of the tax laws and regulations, including negotiations with taxing authorities in various jurisdictions and resolution of disputes arising from federal, state and international tax audits, can result in differences in taxes paid, which may be higher or lower than our estimates. The judgments made at a point in time may change from previous conclusions based on the outcome of tax audits, as well as changes to, or further interpretations of, tax laws and regulations, and these changes could significantly impact the provision for income taxes, the amount of taxes payable and the deferred tax asset and liability balances. We adjust our income tax provision in the period in which these changes occur. As of December 31, 2023, we have recorded a reserve for unrecognized tax benefits, including penalties and interest, of $273 million.

We also engage in a significant amount of cross border transactions. The taxability of cross border transactions has received an increasing level of scrutiny among regulators across the globe, including the jurisdictions in which we operate. The tax rules and regulations of the various jurisdictions in which we operate are complex, and in many cases, there is not symmetry between the rules of the various jurisdictions. As a result, there are instances where regulators within the jurisdictions involved in a cross-border transaction may reach different conclusions regarding the taxability of the transaction in their respective jurisdictions based on the same set of facts and circumstances. We work closely with regulators to reach a common understanding and conclusion regarding the taxability of cross border transactions.

Recent Accounting Pronouncements

See Note 3—New Accounting Standards for a discussion of recent accounting pronouncements.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK.

We are exposed to the impact of changes in commodity prices, interest rates and foreign currency exchange rates.

Commodity Prices

Our gross margin, cash flows and estimates of future cash flows related to nitrogen-based products are sensitive not only to selling prices of our products, but also to changes in market prices of natural gas and other raw materials except to the extent the prices we pay for those inputs have been fixed or hedged. A $1.00 per MMBtu change in the price of natural gas would change the cost to produce a ton of ammonia, granular urea, UAN (32%) and AN by approximately $32, $22, $14 and $15, respectively.

Natural gas is the largest and most volatile component of the manufacturing cost for nitrogen-based products. At certain times, we have managed the risk of changes in natural gas prices through the use of derivative financial instruments. The derivative instruments that we may use for this purpose are primarily natural gas fixed price swaps, basis swaps and options. These derivatives settle using primarily a NYMEX futures price index, which represents the basis for fair value at any given time. The contracts represent anticipated natural gas needs for future periods and settlements are scheduled to coincide with anticipated natural gas purchases during those future periods. As of December 31, 2023, we had natural gas derivative contracts covering certain periods through March 2024.

As of December 31, 2023 and 2022, we had open natural gas derivative contracts for 49.0 million MMBtus and 66.3 million MMBtus, respectively. A $1.00 per MMBtu increase in the forward curve prices of natural gas at December 31, 2023 would result in a favorable change in the fair value of these derivative positions of approximately $48 million, and a $1.00 per MMBtu decrease in the forward curve prices of natural gas would change their fair value unfavorably by approximately $49 million.

From time to time, we may purchase nitrogen products on the open market to augment or replace production at our facilities.

Interest Rates

As of December 31, 2023, we had four series of senior notes totaling $3.00 billion of principal outstanding with maturity dates of December 1, 2026, March 15, 2034, June 1, 2043 and March 15, 2044. The senior notes have fixed interest rates. As of December 31, 2023, the carrying value and fair value of our senior notes was approximately $2.97 billion and $2.89 billion, respectively.

Borrowings under the Prior Revolving Credit Agreement bore current market rates of interest, and we were subject to interest rate risk on such borrowings. Borrowings under the New Revolving Credit Agreement bear current market rates of interest, and we are subject to interest rate risk on such borrowings. As of and during the years ended December 31, 2023 and 2022, there were no borrowings outstanding under either the Prior Revolving Credit Agreement or the New Revolving Credit Agreement.

Foreign Currency Exchange Rates

We are directly exposed to changes in the value of the Canadian dollar, the British pound and the euro. We generally do not maintain any exchange rate derivatives or hedges related to these currencies.

ITEM 8. FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA.

Report of Independent Registered Public Accounting Firm

To the Stockholders and Board of Directors
CF Industries Holdings, Inc.:

Opinion on the Consolidated Financial Statements

We have audited the accompanying consolidated balance sheets of CF Industries Holdings, Inc. and subsidiaries (the Company) as of December 31, 2023 and 2022, the related consolidated statements of operations, comprehensive income, equity, and cash flows for each of the years in the three-year period ended December 31, 2023, and the related notes (collectively, the consolidated financial statements). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company as of December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the years in the three-year period ended December 31, 2023, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission, and our report dated February 22, 2024 expressed an unqualified opinion on the effectiveness of the Company's internal control over financial reporting.

Basis for Opinion

These consolidated financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on these consolidated financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current period audit of the consolidated financial statements that were communicated or required to be communicated to the audit committee and that: (1) relate to accounts or disclosures that are material to the consolidated financial statements and (2) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matters below, providing separate opinions on the critical audit matters or on the accounts or disclosures to which they relate.

Measurements of projected benefit obligations

As discussed in Note 13 to the consolidated financial statements, the Company's projected benefit obligation (PBO) associated with its defined benefit pension plans established in North America and the United Kingdom was $292 million and $367 million, respectively, as of December 31, 2023. The Company's PBO represents an actuarially determined estimate of the present value of the vested benefits to which each eligible employee is currently entitled, based on the employee's expected date of separation or retirement. Determining the PBO requires the Company to make assumptions, including the selection of a discount rate for each of the North America and United Kingdom plans.

We identified the evaluation of the Company's measurements of the PBO as a critical audit matter. Specialized skills were needed to evaluate the discount rates utilized in the measurement of the PBO for each of the North America and United Kingdom plans. In addition, subjective auditor judgment was required to evaluate these discount rates, as minor changes to these discount rates could have a significant impact on the PBO.

The following are the primary procedures we performed to address this critical audit matter. We evaluated the design and tested the operating effectiveness of certain internal controls over the Company's PBO accounting process, including controls related to the determination of discount rates. We involved actuarial professionals with specialized skills and knowledge, who assisted in:

- developing an understanding and assessing the methods used by the Company's actuaries to develop the discount rates
- evaluating the relevance and reliability of information used by the Company's actuaries in the development of the discount rates
- evaluating the change in North America discount rates period over period using market trends based on published yield curves and indices
- recalculating the Company's discount rate for the North America plans using the PBO cash flows and the Company's actuaries' proprietary yield curve for the North America discount rates
- independently developing a single equivalent discount rate for the North America plans using the PBO cash flows and publicly available yield curves for pension plans in North America, and comparing that to the Company's selected discount rates for the North America plans
- evaluating discount rates using benchmark yield curves for the United Kingdom, adjusted for the assessment of the timing of payments expected to be made to beneficiaries under the Company's United Kingdom pension plans, and comparing those to the Company's selected discount rates for the United Kingdom plans.

Valuation of acquired customer relationships and assumed supply contract liability - Waggaman acquisition

As discussed in Note 6 to the consolidated financial statements, the Company completed the acquisition of an ammonia production facility located in Waggaman, Louisiana, from Dyno Nobel Louisiana Ammonia, LLC (DNLA), on December 1, 2023. In connection with the acquisition, the Company entered into a long-term ammonia offtake agreement (the Supply Contract). The terms of the Supply Contract were determined to be unfavorable compared to market as of the acquisition date. Accordingly, the assets acquired and liabilities assumed were recognized based on their acquisition date fair values, including customer relationships of $455 million and a Supply Contract liability of $757 million. The Company used valuation techniques under the income approach to determine the fair value of the identified customer relationships and Supply Contract liability.

We identified the evaluation of the acquisition date fair value of the customer relationships acquired and the Supply Contract liability assumed as a critical audit matter. Subjective auditor judgment and specialized skills and knowledge were necessary to evaluate the key assumptions used to estimate the fair value of the customer relationships and Supply Contract liability due to the degree of measurement uncertainty in the key assumptions, which could have a significant impact on the fair values of the customer relationships and the Supply Contract liability. The key assumptions included:

Customer relationships:

- Forecasted product selling prices
- Projected natural gas costs
- Discount rate

Supply Contract liability:

- Forecasted product selling prices

The following are the primary procedures we performed to address this critical audit matter. We evaluated the design and tested the operating effectiveness of certain internal controls over the Company's acquisition date valuation process. This included controls related to the determination of the key assumptions for customer relationships and the Supply Contract liability. We evaluated the reasonableness of forecasted product selling prices by comparing them to the Company's historical selling prices and external market data. We evaluated the reasonableness of projected natural gas costs by comparing them to external market data. We involved valuation professionals with specialized skills and knowledge, who assisted in independently developing a range of discount rates based on publicly available market data for comparable entities and comparing the range to the Company's discount rate.

(signed) KPMG LLP

We have served as the Company's auditor since 1983.

Chicago, Illinois
February 22, 2024

CF INDUSTRIES HOLDINGS, INC.

CONSOLIDATED STATEMENTS OF OPERATIONS

	Year ended December 31,		
	2023	2022	2021
	(in millions, except per share amounts)		
Net sales	$ 6,631	$ 11,186	$ 6,538
Cost of sales	4,086	5,325	4,151
Gross margin	2,545	5,861	2,387
Selling, general and administrative expenses	289	290	223
U.K. goodwill impairment	—	—	285
U.K. long-lived and intangible asset impairment	—	239	236
U.K. operations restructuring	10	19	—
Acquisition and integration costs	39	—	—
Other operating—net	(31)	10	(39)
Total other operating costs and expenses	307	558	705
Equity in (loss) earnings of operating affiliate	(8)	94	47
Operating earnings	2,230	5,397	1,729
Interest expense	150	344	184
Interest income	(158)	(65)	(1)
Loss on debt extinguishment	—	8	19
Other non-operating—net	(10)	15	(16)
Earnings before income taxes	2,248	5,095	1,543
Income tax provision	410	1,158	283
Net earnings	1,838	3,937	1,260
Less: Net earnings attributable to noncontrolling interest	313	591	343
Net earnings attributable to common stockholders	$ 1,525	$ 3,346	$ 917
Net earnings per share attributable to common stockholders:			
Basic	$ 7.89	$ 16.45	$ 4.27
Diluted	$ 7.87	$ 16.38	$ 4.24
Weighted-average common shares outstanding:			
Basic	193.3	203.3	215.0
Diluted	193.8	204.2	216.2

See Accompanying Notes to Consolidated Financial Statements.

CF INDUSTRIES HOLDINGS, INC.

CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Net earnings	$ 1,838	$ 3,937	$ 1,260
Other comprehensive income:			
Foreign currency translation adjustment—net of taxes	33	(38)	3
Derivatives—net of taxes	—	(1)	—
Defined benefit plans—net of taxes	(12)	66	60
	21	27	63
Comprehensive income	1,859	3,964	1,323
Less: Comprehensive income attributable to noncontrolling interest	313	591	343
Comprehensive income attributable to common stockholders	$ 1,546	$ 3,373	$ 980

See Accompanying Notes to Consolidated Financial Statements.

CF INDUSTRIES HOLDINGS, INC.

CONSOLIDATED BALANCE SHEETS

	December 31,	
	2023	2022
	(in millions, except share and per share amounts)	
Assets		
Current assets:		
Cash and cash equivalents	$ 2,032	$ 2,323
Accounts receivable—net	505	582
Inventories	299	474
Prepaid income taxes	167	215
Other current assets	47	79
Total current assets	3,050	3,673
Property, plant and equipment—net	7,141	6,437
Investment in affiliate	26	74
Goodwill	2,495	2,089
Intangible assets—net	538	15
Operating lease right-of-use assets	259	254
Other assets	867	771
Total assets	$ 14,376	$ 13,313
Liabilities and Equity		
Current liabilities:		
Accounts payable and accrued expenses	$ 520	$ 575
Income taxes payable	12	3
Customer advances	130	229
Current operating lease liabilities	96	93
Other current liabilities	42	95
Total current liabilities	800	995
Long-term debt	2,968	2,965
Deferred income taxes	999	958
Operating lease liabilities	168	167
Supply contract liability	754	—
Other liabilities	314	375
Equity:		
Stockholders' equity:		
Preferred stock—$0.01 par value, 50,000,000 shares authorized	—	—
Common stock—$0.01 par value, 500,000,000 shares authorized, 2023—188,188,401 shares issued and 2022—195,604,404 shares issued	2	2
Paid-in capital	1,389	1,412
Retained earnings	4,535	3,867
Accumulated other comprehensive loss	(209)	(230)
Total stockholders' equity	5,717	5,051
Noncontrolling interest	2,656	2,802
Total equity	8,373	7,853
Total liabilities and equity	$ 14,376	$ 13,313

See Accompanying Notes to Consolidated Financial Statements.

CF INDUSTRIES HOLDINGS, INC.

CONSOLIDATED STATEMENTS OF EQUITY

	Common Stockholders								
	$0.01 Par Value Common Stock	Treasury Stock	Paid-In Capital	Retained Earnings	Accumulated Other Comprehensive Loss	Total Stockholders' Equity	Noncontrolling Interest	Total Equity
				(in millions)				
Balance as of December 31, 2020	$ 2	$ (4)	$ 1,317	$ 1,927	$ (320)	$ 2,922	$ 2,681	$ 5,603
Net earnings	—	—	—	917	—	917	343	1,260
Other comprehensive income	—	—	—	—	63	63	—	63
Purchases of treasury stock	—	(540)	—	—	—	(540)	—	(540)
Retirement of treasury stock	—	554	(58)	(496)	—	—	—	—
Acquisition of treasury stock under employee stock plans	—	(13)	—	—	—	(13)	—	(13)
Issuance of $0.01 par value common stock under employee stock plans	—	1	65	—	—	66	—	66
Stock-based compensation expense	—	—	30	—	—	30	—	30
Cash dividends ($1.20 per share)	—	—	—	(260)	—	(260)	—	(260)
Deferred tax related to noncontrolling interest	—	—	21	—	—	21	—	21
Distributions declared to noncontrolling interest	—	—	—	—	—	—	(194)	(194)
Balance as of December 31, 2021	$ 2	$ (2)	$ 1,375	$ 2,088	$ (257)	$ 3,206	$ 2,830	$ 6,036
Net earnings	—	—	—	3,346	—	3,346	591	3,937
Other comprehensive income	—	—	—	—	27	27	—	27
Purchases of treasury stock	—	(1,346)	—	—	—	(1,346)	—	(1,346)
Retirement of treasury stock	—	1,370	(109)	(1,261)	—	—	—	—
Acquisition of treasury stock under employee stock plans	—	(23)	—	—	—	(23)	—	(23)
Issuance of $0.01 par value common stock under employee stock plans	—	1	105	—	—	106	—	106
Stock-based compensation expense	—	—	41	—	—	41	—	41
Cash dividends ($1.50 per share)	—	—	—	(306)	—	(306)	—	(306)
Distributions declared to noncontrolling interest	—	—	—	—	—	—	(619)	(619)
Balance as of December 31, 2022	$ 2	$ —	$ 1,412	$ 3,867	$ (230)	$ 5,051	$ 2,802	$ 7,853
Net earnings	—	—	—	1,525	—	1,525	313	1,838
Other comprehensive income	—	—	—	—	21	21	—	21
Purchases of treasury stock	—	(585)	—	—	—	(585)	—	(585)
Retirement of treasury stock	—	605	(59)	(546)	—	—	—	—
Acquisition of treasury stock under employee stock plans	—	(22)	—	—	—	(22)	—	(22)
Issuance of $0.01 par value common stock under employee stock plans	—	2	(1)	—	—	1	—	1
Stock-based compensation expense	—	—	37	—	—	37	—	37
Cash dividends ($1.60 per share)	—	—	—	(311)	—	(311)	—	(311)
Distributions declared to noncontrolling interest	—	—	—	—	—	—	(459)	(459)
Balance as of December 31, 2023	$ 2	$ —	$ 1,389	$ 4,535	$ (209)	$ 5,717	$ 2,656	$ 8,373

See Accompanying Notes to Consolidated Financial Statements.

CF INDUSTRIES HOLDINGS, INC.

CONSOLIDATED STATEMENTS OF CASH FLOWS

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Operating Activities:			
Net earnings	$ 1,838	$ 3,937	$ 1,260
Adjustments to reconcile net earnings to net cash provided by operating activities:			
Depreciation and amortization	869	850	888
Deferred income taxes	81	(107)	(196)
Stock-based compensation expense	37	41	30
Loss on debt extinguishment	—	8	19
Unrealized net (gain) loss on natural gas derivatives	(39)	41	25
Impairment of equity method investment in PLNL	43	—	—
(Gain) loss on embedded derivative	—	(14)	1
U.K. goodwill impairment	—	—	285
U.K. long-lived and intangible asset impairment	—	239	236
Pension settlement loss and curtailment gains	—	17	—
Gain on sale of emission credits	(39)	(6)	(49)
Loss on disposal of property, plant and equipment	4	2	3
Undistributed losses (earnings) of affiliate—net of taxes	3	(1)	(6)
Changes in assets and liabilities, net of acquisition:			
Accounts receivable—net	100	(110)	(235)
Inventories	152	(93)	(123)
Accrued and prepaid income taxes	(44)	(227)	94
Accounts payable and accrued expenses	(88)	1	142
Customer advances	(100)	(471)	570
Other—net	(60)	(252)	(71)
Net cash provided by operating activities	2,757	3,855	2,873
Investing Activities:			
Additions to property, plant and equipment	(499)	(453)	(514)
Proceeds from sale of property, plant and equipment	1	1	1
Purchase of Waggaman ammonia production facility	(1,223)	—	—
Distributions received from unconsolidated affiliate	—	6	—
Purchase of investments held in nonqualified employee benefit trust	(1)	(1)	(13)
Proceeds from sale of investments held in nonqualified employee benefit trust	1	1	12
Purchase of emission credits	(2)	(9)	(10)
Proceeds from sale of emission credits	39	15	58
Other—net	5	—	—
Net cash used in investing activities	(1,679)	(440)	(466)
Financing Activities:			
Payments of long-term borrowings	—	(507)	(518)
Payment to CHS related to credit provision	—	—	(5)
Financing fees	(2)	(4)	—
Dividends paid on common stock	(311)	(306)	(260)
Distributions to noncontrolling interest	(459)	(619)	(194)
Purchases of treasury stock	(580)	(1,347)	(539)
Proceeds from issuances of common stock under employee stock plans	2	106	64
Cash paid for shares withheld for taxes	(22)	(23)	(11)
Net cash used in financing activities	(1,372)	(2,700)	(1,463)
Effect of exchange rate changes on cash and cash equivalents	3	(20)	1
(Decrease) increase in cash and cash equivalents	(291)	695	945
Cash and cash equivalents at beginning of period	2,323	1,628	683
Cash and cash equivalents at end of period	$ 2,032	$ 2,323	$ 1,628

See Accompanying Notes to Consolidated Financial Statements.

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

1. Background and Basis of Presentation

Our mission is to provide clean energy to feed and fuel the world sustainably. With our employees focused on safe and reliable operations, environmental stewardship, and disciplined capital and corporate management, we are on a path to decarbonize our ammonia production network – the world's largest – to enable green and low-carbon hydrogen and nitrogen products for energy, fertilizer, emissions abatement and other industrial activities. Our nitrogen manufacturing complexes in the United States, Canada and the United Kingdom, an extensive storage, transportation and distribution network in North America, and logistics capabilities enabling a global reach underpin our strategy to leverage our unique capabilities to accelerate the world's transition to clean energy. Our principal customers are cooperatives, independent fertilizer distributors, traders, wholesalers and industrial users. Our core product is anhydrous ammonia (ammonia), which contains 82% nitrogen and 18% hydrogen. Our nitrogen products that are upgraded from ammonia are granular urea, urea ammonium nitrate solution (UAN) and ammonium nitrate (AN). Our other nitrogen products include diesel exhaust fluid (DEF), urea liquor, nitric acid and aqua ammonia, which are sold primarily to our industrial customers.

The accompanying consolidated financial statements have been prepared in conformity with accounting principles generally accepted in the United States (U.S. GAAP). Certain prior period amounts have been reclassified to conform with the current year presentation.

All references to "CF Holdings," "the Company," "we," "us" and "our" refer to CF Industries Holdings, Inc. and its subsidiaries, except where the context makes clear that the reference is to CF Industries Holdings, Inc. only and not its subsidiaries. All references to "CF Industries" refer to CF Industries, Inc., a 100% owned subsidiary of CF Industries Holdings, Inc.

Our principal assets as of December 31, 2023 include:

- six U.S. nitrogen manufacturing facilities, located in Donaldsonville, Louisiana; Sergeant Bluff, Iowa (our Port Neal complex); Yazoo City, Mississippi; Claremore, Oklahoma (our Verdigris complex); Woodward, Oklahoma; and Waggaman, Louisiana. The Waggaman facility is wholly owned by us, and the other five U.S. nitrogen manufacturing facilities are wholly owned directly or indirectly by CF Industries Nitrogen, LLC (CFN), of which we own approximately 89% and CHS Inc. (CHS) owns the remainder. See Note 19—Noncontrolling Interest for additional information on our strategic venture with CHS;
- two Canadian nitrogen manufacturing facilities, located in Medicine Hat, Alberta and Courtright, Ontario;
- a United Kingdom nitrogen manufacturing facility, located in Billingham;
- an extensive system of terminals and associated transportation equipment located primarily in the Midwestern United States; and
- a 50% interest in Point Lisas Nitrogen Limited (PLNL), an ammonia production joint venture located in the Republic of Trinidad and Tobago (Trinidad) that we account for under the equity method.

2. Summary of Significant Accounting Policies

Consolidation and Noncontrolling Interest

The consolidated financial statements of CF Holdings include the accounts of CF Industries and all majority-owned subsidiaries. All significant intercompany transactions and balances have been eliminated.

We own approximately 89% of the membership interests in CFN and consolidate CFN in our financial statements. CHS' minority equity interest in CFN is included in noncontrolling interest in our consolidated financial statements. See Note 19—Noncontrolling Interest for additional information.

Use of Estimates

The preparation of financial statements in conformity with U.S. GAAP requires us to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent liabilities at the date of the consolidated financial statements and the reported amounts of revenues and expenses for the periods presented. Such estimates and assumptions are used for, but are not limited to, net realizable value of inventories, environmental remediation liabilities, environmental and litigation contingencies, plant closure and asset retirement obligations, the cost of emission credits required

to meet environmental regulations, the cost of customer incentives, the fair values utilized in the allocation of purchase price in an acquisition, useful lives of property and identifiable intangible assets, the evaluation of potential impairments of property, investments, identifiable intangible assets and goodwill, income tax reserves and the assessment of the realizability of deferred tax assets, measurement of the fair values of investments for which markets are not active, the determination of the funded status and annual expense of defined benefit pension and other postretirement plans and the valuation of stock-based compensation awards granted to employees.

Revenue Recognition

We follow a five-step model for revenue recognition. The five steps are: (1) identification of the contract(s) with the customer, (2) identification of the performance obligation(s) in the contract(s), (3) determination of the transaction price, (4) allocation of the transaction price to the performance obligation(s), and (5) recognition of revenue when (or as) each performance obligation is satisfied. Control of our products transfers to our customers when the customer is able to direct the use of, and obtain substantially all of the benefits from, our products, which occurs at the later of when title or risk of loss transfers to the customer. Control generally transfers to the customer at a point in time upon loading of our product onto transportation equipment or delivery to a customer destination. Revenue from forward sales programs is recognized on the same basis as other sales regardless of when the customer advances are received.

In situations where we have agreed to arrange delivery of the product to the customer's intended destination and control of the product transfers upon loading of our product, we have elected to not identify delivery of the product as a performance obligation. We account for freight income associated with the delivery of these products as freight revenue, since this activity fulfills our obligation to transfer the product to the customer. Shipping and handling costs incurred by us are included in cost of sales.

We offer cash incentives to certain customers based on the volume of their purchases over a certain period. Customer incentives are reported as a reduction in net sales.

Cash and Cash Equivalents

Cash and cash equivalents include highly liquid investments that are readily convertible to known amounts of cash with original maturities of three months or less. The carrying value of cash and cash equivalents approximates fair value.

Investments

Short-term investments and noncurrent investments are accounted for primarily as available-for-sale securities reported at fair value. Changes in the fair value of available-for-sale debt securities are recognized in other comprehensive income. Changes in the fair value of available-for-sale equity securities are recognized through earnings. The carrying values of short-term investments, if any, approximate fair values because of the short maturities and the highly liquid nature of these investments.

Inventories

Inventories are reported at the lower of cost and net realizable value with cost determined on a first-in, first-out and average cost basis. Inventory includes the cost of materials, production labor and production overhead. Inventory at warehouses and terminals also includes distribution costs to move inventory to the distribution facilities. Net realizable value is reviewed at least quarterly. Fixed production costs related to idle capacity are not included in the cost of inventory but are charged directly to cost of sales in the period incurred.

Investment in Unconsolidated Affiliate

The equity method of accounting is used for our investment in an affiliate that we do not consolidate, but over which we have the ability to exercise significant influence. Our equity method investment for which the results are included in operating earnings consists of our 50% ownership interest in PLNL, which operates an ammonia production facility in Trinidad. Our share of the net earnings from this investment is reported as an element of earnings from operations because PLNL's operations provide additional production and are integrated with our supply chain and sales activities in the Ammonia segment.

Profits resulting from sales or purchases with equity method investees are eliminated until realized by the investee or investor, respectively. Investments in affiliates are reviewed for impairment whenever events or circumstances indicate that the carrying value may not be recoverable. If circumstances indicate that the fair value of an investment in an affiliate is less than its carrying value, and the reduction in value is other than temporary, the reduction in value would be recognized immediately in earnings.

See Note 10—Equity Method Investment for additional information.

Property, Plant and Equipment

Property, plant and equipment are stated at cost. Depreciation and amortization are computed using the straight-line method and are recorded over the estimated useful life of the property, plant and equipment. Useful lives are as follows:

	Years
Mobile and office equipment	3 to 10
Production facilities and related assets	2 to 30
Land improvements	10 to 30
Buildings	10 to 40

We periodically review the useful lives assigned to our property, plant and equipment and we change the estimates to reflect the results of those reviews.

Scheduled inspections, replacements and overhauls of plant machinery and equipment at our continuous process manufacturing facilities during a full plant shutdown are referred to as plant turnarounds. Plant turnarounds are accounted for under the deferral method, as opposed to the direct expense or built-in overhaul methods. Under the deferral method, expenditures related to turnarounds are capitalized in property, plant and equipment when incurred and amortized to production costs on a straight-line basis over the period benefited, which is until the next scheduled turnaround in up to five years. If the direct expense method were used, all turnaround costs would be expensed as incurred. Internal employee costs and overhead amounts are not considered turnaround costs and are not capitalized. Turnaround costs are classified as investing activities and included in capital expenditures in our consolidated statements of cash flows. See Note 8—Property, Plant and Equipment—Net for additional information.

Recoverability of Long-Lived Assets

We review property, plant and equipment and other long-lived assets at the asset group level in order to assess recoverability based on expected future undiscounted cash flows whenever events or circumstances indicate that the carrying value may not be recoverable. If the sum of the expected future net undiscounted cash flows is less than the carrying value, an impairment loss would be recognized. The impairment loss is measured as the amount by which the carrying value exceeds the fair value of the asset. For property, plant and equipment that is planned for abandonment, we first consider a market or income-based valuation method. In situations where a secondary market does not exist and the assets have been idled and planned for abandonment and therefore will not generate future cash flows from operations, we estimate a salvage value for those assets. See Note 7—United Kingdom Operations Restructuring and Impairment Charges and Note 8—Property, Plant and Equipment—Net for additional information.

Goodwill

Goodwill represents the excess of the purchase price of an acquired entity over the amounts assigned to the assets acquired and liabilities assumed. Goodwill is not amortized, but is reviewed for impairment annually or more frequently whenever events or circumstances indicate that the carrying value may not be recoverable. We perform our annual goodwill impairment review in the fourth quarter of each year at the reporting unit level. Our evaluation generally begins with a qualitative assessment of the factors that could impact the significant inputs used to estimate fair value. If after performing the qualitative assessment, we determine that it is not more likely than not that the fair value of a reporting unit is less than its carrying amount, including goodwill, then no further analysis is necessary. However, if the results of the qualitative test are unclear, we perform a quantitative test, which involves comparing the fair value of a reporting unit with its carrying amount, including goodwill. We use an income-based valuation method, determining the present value of future cash flows, to estimate the fair value of a reporting unit. If the fair value of a reporting unit exceeds its carrying amount, no further analysis is necessary. If the fair value of the reporting unit is less than its carrying amount, goodwill impairment would be recognized equal to the amount of the carrying value in excess of the reporting unit's fair value, limited to the total amount of goodwill allocated to the reporting unit. See Note 9—Goodwill and Other Intangible Assets for additional information regarding our goodwill.

Leases

Right-of-use (ROU) assets represent our right to use an underlying asset for the lease term and lease liabilities represent our obligation to make lease payments arising from the lease. ROU assets and lease liabilities are recognized at the lease commencement date based on the present value of lease payments over the lease term. The discount rate used to calculate the present value represents our secured incremental borrowing rate and is calculated based on the treasury yield curve commensurate with the term of each lease, and a spread representative of our secured borrowing costs. Our lease terms may include options to extend or terminate the lease when it is reasonably certain that we will exercise that option.

Leases may be classified as either operating leases or finance leases. We have made an accounting policy election to not include leases with an initial term of 12 months or less on the balance sheet. For finance leases, if any, ROU assets are amortized over the lease term on a straight-line basis and interest expense is recognized using the effective interest method and based on the lease liability at period end. For operating leases, rental payments, including rent holidays, leasehold incentives, and scheduled rent increases are expensed on a straight-line basis. Leasehold improvements are amortized over the shorter of the depreciable lives of the corresponding fixed assets or the lease term including any applicable renewals. For our rail car leases, barge tow charters, and terminal and warehouse storage agreements, we have made an accounting policy election to not separate lease and non-lease components, such as operating costs and maintenance, due to sufficient data not being available. As a result, the non-lease components are included in the ROU assets and lease liabilities on our consolidated balance sheet. See Note 25—Leases for additional information.

Income Taxes

Income taxes are accounted for under the asset and liability method. Deferred tax assets and liabilities are measured using enacted tax rates expected to apply to taxable income in the years in which those differences are projected to be recovered or settled. Realization of deferred tax assets is dependent on our ability to generate sufficient taxable income of an appropriate character in future periods. A valuation allowance is established if it is determined to be more likely than not that a deferred tax asset will not be realized. Significant judgment is applied in evaluating the need for and magnitude of appropriate valuation allowances against deferred tax assets.

We record our tax expense for Global Intangible Low-Taxed Income (GILTI) as an expense in the period in which incurred and as such do not record a deferred tax liability for taxes that may be due in future periods.

Interest and penalties related to unrecognized tax benefits are reported as interest expense and income tax expense, respectively.

See Note 12—Income Taxes for additional information.

Customer Advances

Customer advances represent cash received from customers following acceptance of orders under our forward sales programs. Under such advances, the customer prepays a portion of the value of the sales contract prior to obtaining control of the product, thereby reducing or eliminating accounts receivable from customers. Revenue is recognized when the customer obtains control of the product.

Derivative Financial Instruments

Natural gas is the principal raw material used to produce nitrogen-based products. We manage the risk of changes in natural gas prices primarily through the use of derivative financial instruments. The derivative instruments that we use are primarily natural gas fixed price swaps, basis swaps and options traded in the over-the-counter (OTC) markets. The derivatives reference primarily a NYMEX futures price index, which represent the basis for fair value at any given time. These derivatives are traded in months forward and settlements are scheduled to coincide with anticipated gas purchases during those future periods. We do not use derivatives for trading purposes and are not a party to any leveraged derivatives.

Derivative financial instruments are accounted for at fair value and recognized as current or noncurrent assets and liabilities on our consolidated balance sheets. We use natural gas derivatives as an economic hedge of natural gas price risk, but without the application of hedge accounting. As a result, changes in fair value of these contracts are recognized in earnings. The fair values of derivative instruments and any related cash collateral are reported on a gross basis rather than on a net basis. Cash flows related to natural gas derivatives are reported as operating activities.

See Note 17—Derivative Financial Instruments for additional information.

Debt Issuance Costs

Costs associated with the issuance of debt are recorded on the balance sheet as a direct deduction from the carrying amount of the related debt liability. Costs associated with entering into revolving credit facilities are recorded as an asset in noncurrent assets. All debt issuance costs are amortized over the term of the related debt using the effective interest rate method. Debt issuance discounts are netted against the related debt and are amortized over the term of the debt using the effective interest method. See Note 14—Financing Agreements for additional information.

Environmental

Environmental expenditures that relate to current operations are expensed or capitalized as appropriate. Expenditures that relate to an existing condition caused by past operations are expensed. Expenditures that increase the capacity or extend the useful life of an asset, improve the safety or efficiency of the operations, or mitigate or prevent future environmental contamination are capitalized. Liabilities are recorded when it is probable that an obligation has been incurred and the costs can be reasonably estimated. Environmental liabilities are not discounted.

Emission Credits

Emission credits may be generated by or granted to us through emissions trading systems or othe regulatory programs. From time to time, we may also purchase emission credits. We have elected to account for emission credits using the intangible asset model. Under this model, emission credits that are purchased are measured at their cost basis and tested for impairment annually. We do not recognize any internally generated emission credits under the intangible asset model until a monetary transaction occurs, such as a sale of the emission credits. If a facility exceeds regulatory emissions allowance levels and offsetting credits are not held by us, our obligation is recognized as an operating expense and a liability at the fair value of the emissions allowance deficit.

Stock-based Compensation

We grant stock-based compensation awards under our equity and incentive plans. The awards that have been granted to date are nonqualified stock options, restricted stock awards, restricted stock units and performance restricted stock units. The cost of employee services received in exchange for the awards is measured based on the fair value of the award on the grant date and is recognized as expense on a straight-line basis over the period during which the employee is required to provide the services. We have elected to recognize equity award forfeitures as they occur in determining the compensation cost to be recognized in each period. See Note 21—Stock-based Compensation for additional information.

Treasury Stock

We periodically retire treasury shares acquired through repurchases of our common stock and return those shares to the status of authorized but unissued. We account for treasury stock transactions under the cost method. For each reacquisition of common stock, the number of shares and the acquisition price for those shares is added to the treasury stock count and total value. When treasury shares are retired, we allocate the excess of the repurchase price over the par value of shares acquired to both retained earnings and paid-in capital. The portion allocated to paid-in capital is determined by applying the average paid-in capital per share, and the remaining portion is recorded to retained earnings. Payments of excise taxes associated with treasury stock repurchases are classified as a financing activity in our consolidated statements of cash flows.

Litigation

From time to time, we are subject to ordinary, routine legal proceedings related to the usual conduct of our business. We may also be involved in proceedings regarding public utility and transportation rates, environmental matters, taxes and permits relating to the operations of our various plants and facilities. Accruals for such contingencies are recorded to the extent management concludes their occurrence is probable and the financial impact of an adverse outcome is reasonably estimable. Legal fees are recognized as incurred and are not included in accruals for contingencies. Disclosure for specific legal contingencies is provided if the likelihood of occurrence is at least reasonably possible and the exposure is considered material to the consolidated financial statements.

In making determinations of likely outcomes of litigation matters, many factors are considered. These factors include, but are not limited to, history, scientific and other evidence, and the specifics and status of each matter. If the assessment of various factors changes, the estimates may change. Predicting the outcome of claims and litigation, and estimating related costs and exposure, involves substantial uncertainties that could cause actual costs to vary materially from estimates and accruals.

Foreign Currency Translation and Remeasurement

We translate the financial statements of our foreign subsidiaries with non-U.S. dollar functional currencies using period-end exchange rates for assets and liabilities and weighted-average exchange rates for each period for revenues and expenses. The resulting translation adjustments are recorded as a separate component of accumulated other comprehensive income (loss) within stockholders' equity.

Foreign currency-denominated assets and liabilities are remeasured into U.S. dollars at exchange rates existing at the respective balance sheet dates. Gains and losses resulting from these foreign currency transactions are included in other operating—net in our consolidated statements of operations. Gains and losses resulting from intercompany foreign currency transactions that are of a long-term investment nature, if any, are reported in other comprehensive income.

3. New Accounting Standards

In November 2023, the Financial Accounting Standards Board (FASB) issued Accounting Standards Update (ASU) No. 2023-07, Segment Reporting (Topic 280): Improvements to Reportable Segment Disclosures. This ASU is intended to improve reportable segment disclosures through enhanced disclosures about significant segment expenses. The guidance in this ASU is effective for fiscal years beginning after December 15, 2023, and interim periods within fiscal years beginning after December 15, 2024, with early adoption permitted. We are currently evaluating the impact that our adoption of this ASU will have on the disclosures in our consolidated financial statements.

In December 2023, the FASB issued ASU No. 2023-09, Income Taxes (Topic 740): Improvements to Income Tax Disclosures. This ASU adds new guidance that further enhances income tax disclosures, primarily through standardization and disaggregation of rate reconciliation categories and income taxes paid by jurisdiction. The amendments are effective for fiscal years beginning after December 15, 2024, with early adoption permitted. We are currently evaluating the impact that our adoption of this ASU will have on the disclosures in our consolidated financial statements.

4. Revenue Recognition

Our performance obligations under a customer contract correspond to each shipment of product that we make to our customer under the contract. As a result, each contract may have more than one performance obligation based on the number of products ordered, the quantity of product to be shipped and the mode of shipment requested by the customer. When we enter into a contract with a customer, we are obligated to provide the product in that contract during a mutually agreed upon time period. Depending on the terms of the contract, either we or the customer arranges delivery of the product to the customer's intended destination. When we arrange delivery of the product and control of the product transfers upon loading, we recognize freight revenue, which was $92 million for 2023, $91 million for 2022, and not material for 2021.

Certain of our contracts require us to supply products on a continuous basis to the customer. We recognize revenue on these contracts based on the quantity of products transferred to the customer during the period. For 2023, 2022 and 2021, the total amount of revenue for these contracts was $61 million, $65 million and $92 million, respectively.

From time to time, we will enter the marketplace to purchase product in order to satisfy the obligations of our customer contracts. When we purchase product for this purpose, we are the principal in the transaction and recognize revenue on a gross basis. As discussed in Note 10—Equity Method Investment, we have transactions in the normal course of business with PLNL, reflecting our obligation to purchase 50% of the ammonia produced by PLNLat current market prices. Other than products purchased from PLNL, products purchased in the marketplace in order to satisfy the obligations of our customers were not material during 2023 and 2022, and $68 million for 2021.

Transaction Price

We agree with our customers on the selling price of each transaction. This transaction price is generally based on the product, market conditions, including supply and demand balances, freight arrangements including where control transfers, and customer incentives. In our contracts with customers, we allocate the entire transaction price to the sale of product to the customer, which is the basis for the determination of the relative standalone selling price allocated to each performance obligation. Any sales tax, value added tax, and other tax we collect concurrently with our revenue-producing activities are excluded from revenue. Returns of our product by our customers are permitted only when the product is not to specification. Returns were not material during 2023, 2022 or 2021.

We offer cash incentives to certain customers generally based on the volume of their purchases over the fertilizer year ending June 30. Our cash incentives do not provide an option to the customer for additional product. Accrual of these incentives involves the use of estimates, including how much product the customer will purchase and whether the customer will achieve a

certain level of purchases within the incentive period. The balances of customer incentives accrued at December 31, 2023 and 2022 were not material.

Revenue Disaggregation

We track our revenue by product and by geography. See Note 22—Segment Disclosures for our revenue by reportable segment, which are Ammonia, Granular Urea, UAN, AN and Other. The following table summarizes our revenue by product and by geography (based on destination of our shipment) for 2023, 2022 and 2021:

	Ammonia	Granular Urea	UAN	AN	Other	Total
			(in millions)			
Year ended December 31, 2023						
North America	$ 1,387	$ 1,767	$ 1,646	$ 253	$ 486	$ 5,539
Europe and other	292	56	422	244	78	1,092
Total revenue	$ 1,679	$ 1,823	$ 2,068	$ 497	$ 564	$ 6,631
Year ended December 31, 2022						
North America	$ 2,659	$ 2,722	$ 2,930	$ 294	$ 605	$ 9,210
Europe and other	431	170	642	551	182	1,976
Total revenue	$ 3,090	$ 2,892	$ 3,572	$ 845	$ 787	$ 11,186
Year ended December 31, 2021						
North America	$ 1,575	$ 1,880	$ 1,667	$ 212	$ 400	$ 5,734
Europe and other	212	—	121	298	173	804
Total revenue	$ 1,787	$ 1,880	$ 1,788	$ 510	$ 573	$ 6,538

Accounts Receivable and Customer Advances

Our customers purchase our products through sales on credit or forward sales. Products sold to our customers on credit are recorded as accounts receivable when the customer obtains control of the product. Customers that purchase our products on credit are required to pay in accordance with our customary payment terms, which are generally less than 30 days. For 2023, 2022 and 2021, the amount of customer bad debt expense recognized was not material.

For forward sales, the customer prepays a portion of the value of the sales contract prior to obtaining control of the product. These prepayments, when received, are recorded as customer advances and are recognized as revenue when the customer obtains control of the product. Forward sales are customarily offered for periods of less than one year in advance of when the customer obtains control of the product. As of December 31, 2023 and 2022, we had $130 million and $229 million, respectively, in customer advances on our consolidated balance sheets. The decrease in the balance of customer advances was due primarily to lower average selling prices in 2023 compared to 2022. During 2023, all of our customer advances that were recorded as of December 31, 2022 were recognized as revenue.

We have certain customer contracts with performance obligations where if the customer does not take the required amount of product specified in the contract, then the customer is required to make a payment to us, the amount of which payment may vary based upon the terms and conditions of the applicable contract. As of December 31, 2023, excluding contracts with original durations of less than one year, and based on the minimum product tonnage to be sold and current market price estimates, our remaining performance obligations under these contracts were approximately $2.5 billion. We expect to recognize approximately 28% of these performance obligations as revenue in 2024, approximately 17% as revenue during 2025-2027, approximately 14% as revenue during 2028-2030, and the remainder as revenue thereafter. Subject to the terms and conditions of the applicable contracts, if these customers do not satisfy their purchase obligations under such contracts, the minimum amount that they would be required to pay to us under such contracts, in the aggregate, was approximately $1.3 billion as of December 31, 2023. Other than the performance obligations described above, any performance obligations with our customers that were unfulfilled or partially filled at December 31, 2023 will be satisfied in 2024.

All of our contracts require that the period between the payment for goods and the transfer of those goods to the customer occur within normal contractual terms that do not exceed one year; therefore, we have elected the practical expedient and not adjusted the transaction price of any of our contracts to recognize a significant financing component. We have also elected the practical expedient to not capitalize any incremental costs associated with obtaining a contract that has a duration of less than one year, and there were no costs capitalized during 2023, 2022 or 2021.

5. Net Earnings Per Share

Net earnings per share were computed as follows:

	Year ended December 31,		
	2023	2022	2021
	(in millions, except per share amounts)		
Net earnings attributable to common stockholders	$ 1,525	$ 3,346	$ 917
Basic earnings per common share:			
Weighted-average common shares outstanding	193.3	203.3	215.0
Net earnings attributable to common stockholders	$ 7.89	$ 16.45	$ 4.27
Diluted earnings per common share:			
Weighted-average common shares outstanding	193.3	203.3	215.0
Dilutive common shares—stock-based awards	0.5	0.9	1.2
Diluted weighted-average common shares outstanding	193.8	204.2	216.2
Net earnings attributable to common stockholders	$ 7.87	$ 16.38	$ 4.24

Diluted earnings per common share is calculated using weighted-average common shares outstanding, including the dilutive effect of stock-based awards as determined under the treasury stock method. In the computation of diluted earnings per common share, potentially dilutive stock-based awards are excluded if the effect of their inclusion is anti-dilutive. Shares for anti-dilutive stock-based awards not included in the computation of diluted earnings per common share were zero for the years ended December 31, 2023 and 2022 and0.9 million for the year ended December 31, 2021.

6. Acquisition of Waggaman Ammonia Production Facility

On December 1, 2023, we acquired an ammonia production facility located in Waggaman, Louisiana, from Dyno Nobel Louisiana Ammonia, LLC (DNLA), a U.S. subsidiary of Australia-based Incitec Pivot Limited (IPL), pursuant to an asset purchase agreement with DNLA and IPL. The facility has a nameplate capacity of 880,000 tons of ammonia annually. Our acquisition of the Waggaman facility is intended to expand our ammonia manufacturing and distribution capacity, including our ability to enable low-carbon ammonia production.

In connection with the acquisition, we entered into a long-term ammonia offtake agreement providing for us to supply up to200,000 tons of ammonia per year to IPL's Dyno Nobel, Inc. subsidiary (the Supply Contract). Under the terms of the asset purchase agreement, $425 million of the purchase price of $1.675 billion, subject to adjustment, was allocated by the parties to the ammonia offtake agreement. We funded the balance of the purchase price with $1.223 billion of cash on hand.

The consideration transferred reflects an estimated net working capital adjustment and other adjustments to the purchase price, which is subject to further adjustment pursuant to the terms of the asset purchase agreement. We expect any further purchase price adjustments required under the asset purchase agreement will be completed in 2024.

In connection with the Waggaman acquisition, we incurred $36 million of acquisition-related costs in 2023, which are included in acquisition and integration costs in our consolidated statement of operations.

We accounted for the acquisition as a business combination. The following table summarizes the preliminary allocation of the acquisition date fair value of the consideration transferred to the assets acquired and liabilities assumed. The estimated fair value of the assets acquired and liabilities assumed is based on the estimated net realizable value for inventory, a replacement cost approach for property, plant and equipment and the income approach for intangible assets and the Supply Contract liability. Final determination of the fair values may result in adjustments to the amounts presented below due primarily to customary post-closing purchase price adjustment procedures specified under the terms of the asset purchase agreement, as discussed above.

	Preliminary Valuation[1] (in millions)
Assets acquired and liabilities assumed	
Current assets	$ 37
Property, plant and equipment	1,022
Goodwill	406
Other assets	539
Total assets acquired	2,004
Current liabilities	18
Supply Contract liability	757
Other liabilities	6
Total liabilities assumed	781
Total net assets acquired	$ 1,223

[1] The purchase price related to the acquisition was initially allocated based on the information available at the acquisition date.

Current assets acquired included accounts receivable of $32 million and inventory of $5 million. The acquired property, plant and equipment will be depreciated over a period consistent with our existing fixed assets depreciation policy.

The acquisition resulted in the recognition of $406 million of goodwill, which is recognized in our ammonia segment. Goodwill arising from the acquisition is expected to be deductible for tax reporting purposes. Other assets acquired included intangible assets of $526 million, including $455 million of customer relationships and $71 million related to a favorable personal property tax agreement. The acquired intangible assets will be amortized over a period consistent with our existing intangible asset amortization policy. See Note 9—Goodwill and Other Intangible Assets, for additional information related to goodwill and the acquired intangible assets.

In addition, the terms of the Supply Contract were determined to be unfavorable compared to market as of the acquisition date. As a result, we recorded an intangible liability, reflected as Supply Contract liability in the table above, with an acquisition date fair value of $757 million, which will be amortized to net sales over the estimated life of the Supply Contract of25 years.

Financial results of the Waggaman facility are included in our consolidated statement of operations from the acquisition date of December 1, 2023. For the year ended December 31, 2023, the amount of net sales and net earnings of the Waggaman facility included in our consolidated statement of operations was $28 million and $7 million, respectively.

Estimates utilized in determining the fair value of acquired assets and assumed liabilities

The valuation of the assets acquired and liabilities assumed as a result of our acquisition of the Waggaman ammonia production facility on December 1, 2023, required us to make significant estimates and assumptions.

We estimated the fair value of the customer relationships of $455 million using the multi-period excess earnings method of the income approach, which incorporated the estimated future cash flows associated with the net earnings attributable to the acquired customer relationships. The estimated future cash flows were discounted to their present value using an appropriate risk-adjusted discount rate from the perspective of a market participant. Key assumptions used in estimating future cash flows include forecasted product selling prices, future production and sales volumes, probability of renewal, projected natural gas costs, operating rates, operating expenses, inflation, tax rates, capital spending, and contributory asset charges, among other factors. The discount rate utilized in estimating the fair value of the customer relationships was 9.0% and was derived using a capital asset pricing model and publicly available data for comparable companies to estimate the cost of equity financing. We used Moody's Baa corporate bonds as a benchmark to estimate the cost of debt. Additional assumptions utilized in the valuation of the customer relationships included the duration of the forecasted cash flows and the tax amortization benefit, among other factors.

The terms of the Supply Contract were determined to be unfavorable compared to market as of the acquisition date. Therefore, we estimated the fair value of the Supply Contract liability of $757 million using the with and without method of the income approach. This method included estimating the future cash flows by applying both a market price and the contract price in determining the respective forecasted sales. The difference in the estimated future cash flows using the resulting forecasted

sales was then discounted to present value at the date of acquisition using an appropriate risk-adjusted discount rate from the perspective of a market participant. Key assumptions used in estimating future cash flows include forecasted product selling price, specifically the market price in excess of the contract price, future production and sales volumes, probability of renewal, and inflation, among other factors. The discount rate utilized in estimating the fair value of the Supply Contract liability was 6.2% and was derived based on the historical cost of our issued debt and treasury and corporate bond yields as of the acquisition date. Additional assumptions utilized in the valuation of the Supply Contract liability included the duration of the forecasted cash flows, among other factors.

Due to the inherent uncertainties involved in making estimates and assumptions, actual results may differ from those assumed in our forecasts.

7. United Kingdom Operations Restructuring and Impairment Charges

2021 Impairment

During the third quarter of 2021, the United Kingdom began experiencing an energy crisis that included a substantial increase in the price of natural gas. In the first half of 2021, natural gas prices had increased to levels that were considered high compared to historical prices, and prices then more than doubled within the third quarter of 2021. On September 15, 2021, we announced the halt of operations at both our Ince and Billingham manufacturing facilities in the United Kingdom due to negative profitability driven by the high cost of natural gas. Shortly thereafter, our Billingham facility resumed operations.

The U.K. energy crisis necessitated an evaluation of the long-lived assets, including definite-lived intangible assets, and goodwill of our U.K. operations to determine if their fair value had declined to below their carrying value. We performed the impairment evaluations on the U.K. Ammonia, U.K. AN and U.K. Other asset groups' long-lived assets, including definite-lived intangible assets, and the U.K. Ammonia, U.K. AN and U.K. Other reporting units' goodwill as of September 30, 2021. Our assets groups are the same as our reporting units. Based on these analyses, we concluded that a decline in fair value below carrying value had occurred, and we recognized impairment charges of $495 million in the third quarter of 2021, consisting of long-lived and intangible asset impairment charges of $236 million and a goodwill impairment charge of $259 million.

In the fourth quarter of 2021, natural gas prices in the United Kingdom continued to rise, which triggered an additional impairment test of long-lived assets and goodwill and resulted in an additional goodwill impairment charge of $26 million. The results of our long-lived asset impairment test indicated thatno additional long-lived asset impairment existed, as the undiscounted cash flows were in excess of the carrying values for each of the U.K. asset groups.

For the full year ended December 31, 2021, these evaluations resulted in total impairment charges of $521 million, consisting of goodwill impairment of $285 million and long-lived and intangible asset impairment of $236 million, of which $50 million related to definite-lived intangible assets. As of December 31, 2021, no goodwill related to our U.K. reporting units remained.

2022 Impairment and Restructuring

During the first quarter of 2022, we concluded that the continued impacts of the U.K. energy crisis, including further increases and volatility in natural gas prices due in part to geopolitical events as a result of Russia's invasion of Ukraine in February 2022, triggered an additional long-lived asset impairment test. The results of this test indicated that no additional long-lived asset impairment existed, as the undiscounted estimated future cash flows were in excess of the carrying values for each of the U.K. asset groups.

In the second quarter of 2022, we approved and announced our proposed plan to restructure our U.K. operations, including the planned permanent closure of the Ince facility, which had been idled since September 2021, and optimization of the remaining manufacturing operations at our Billingham facility. Pursuant to our proposed plan to restructure our U.K. operations and dispose of the Ince facility assets before we originally intended, we concluded that an evaluation of our long-lived assets and an additional impairment test was required. Our assessment then identified the U.K. asset groups as U.K. Ammonia, U.K. AN and U.K. Other, comprising our ongoing U.K. operations, and Ince, U.K. In response to this impairment indicator, we compared the undiscounted cash flows expected to result from the use and eventual disposition of the Ince, U.K. asset group to its carrying amount and concluded the carrying amount was not recoverable and should be adjusted to its fair value. As a result, in the second quarter of 2022, we recorded total charges of $162 million related to the Ince facility as follows:

- asset impairment charges of $152 million consisting of the following:

- an impairment charge of $135 million related to property, plant and equipment that is planned for abandonment at the Ince facility, including a liability of approximately $9 million for the costs of certain asset retirement activities related to the Ince site;
- an intangible asset impairment charge of $8 million related to trade names; and
- an impairment charge of $9 million related to the write-down of spare parts and certain raw materials at the Ince facility;

and

- a charge for post-employment benefits totaling $10 million, which is included in the U.K. operations restructuring line item in our consolidated statements of operations, related to contractual and statutory obligations due to employees whose employment would be terminated in the proposed plan.

There was no additional asset impairment indicated for the three asset groups that comprise the continuing U.K. operations as the undiscounted estimated future cash flows were in excess of the carrying values for each of these asset groups.

In the third quarter of 2022, the United Kingdom continued to experience extremely high and volatile natural gas prices. Russian natural gas flows to Europe via the Nord Stream 1 pipeline ceased, causing the United Kingdom to experience unprecedented natural gas prices. In addition, the European Union announced a desire to cap the price that Europe would pay Russia for natural gas deliveries, further contributing to the uncertainty in European energy markets. Given these factors and the lack of a corresponding increase in global nitrogen product market prices, in September 2022, we temporarily idled ammonia production at our Billingham complex. As a result, we concluded that an additional impairment test was triggered for the asset groups that comprise the continuing U.K. operations. The results of our impairment test indicated that the carrying values for our U.K. Ammonia and U.K. AN asset groups exceeded the undiscounted estimated future cash flows. As a result, we recognized asset impairment charges of $87 million, primarily related to property, plant and equipment and definite-lived intangible assets.

In August 2022, the final restructuring plan for our U.K. operations was approved, and decommissioning activities were initiated. As a result, in the third quarter of 2022, we incurred additional charges related to our U.K. restructuring of $8 million, primarily related to one-time termination benefits. In the fourth quarter of 2022, we incurred additional charges related to our U.K. restructuring of $1 million, primarily related to one-time termination benefits.

For the full year ended December 31, 2022, as a result of the above, we recognized total impairment and restructuring charges of $258 million, consisting of long-lived and intangible asset impairment charges of $239 million and restructuring charges of $19 million.

As of December 31, 2022, amounts accrued related to the final restructuring plan for our U.K. operations consisted of $3 million for employee contractual and one-time termination benefits and $6 million for asset retirement obligations.

See Note 8—Property, Plant and Equipment—Net, Note 9—Goodwill and Other Intangible Assets and Note 24—Asset Retirement Obligations for additional information.

Assumptions in the impairment evaluations

The valuation of our asset groups and reporting units requires significant judgment in evaluating recent indicators of market activity and estimating future cash flows, discount rates, and other factors. The expected cash flows used in the long-lived asset and goodwill impairment tests reflected assumptions about product selling prices and natural gas costs, as well as estimates of future production and sales volumes, operating rates, operating expenses, inflation, discount rates, tax rates and capital spending. The valuations also incorporate assumptions regarding the time it could take for the U.K. energy crisis to be resolved. In addition, assumptions were used to estimate the fair value of the long-lived assets in our asset groups, which included replacement cost and, for the Ince, U.K. asset group that is planned for abandonment, salvage value.

For purposes of our goodwill impairment analyses in 2021, we estimated the fair value of the reporting units using the income approach, which incorporated the estimated future cash flows and a terminal value discounted to their present value using an appropriate risk-adjusted discount rate from the perspective of a market participant. The estimated future cash flows were based on our internal forecasts, updated for recent events at that time. These estimated future cash flows went beyond the specific operating plans, using a terminal value calculation, which incorporated historical and forecasted trends and an estimate of long-term future growth rates. The future growth rates were based on our view of the long-term outlook for each reporting unit.

The discount rates utilized in the income approach, for our goodwill impairment tests, and to discount the cash flows in calculating long-lived asset impairment, were derived using a capital asset pricing model and analyzing published rates for industries relevant to our reporting units to estimate the cost of equity financing. The discount rates were commensurate with the risks and uncertainties inherent in the business and in the United Kingdom and our cash flow forecasts, updated for recent events at that time.

Additional assumptions utilized in the long-lived asset impairment analyses were royalty rates and attrition rates in estimating the fair value of our definite-lived intangible assets, consisting of trade names and customer relationships, for which we used the relief from royalty method of the income approach and the multi-period excess earnings method, respectively.

For the asset groups that comprise the continuing U.K. operations, the fair value of our property, plant and equipment utilized in the long-lived asset impairment analyses was estimated using the indirect method of the cost approach by determining the reproduction cost new, or replacement cost, of the assets and applying appropriate adjustments for depreciation including an inutility adjustment based on the cash flows expected to be generated by those asset groups. For property, plant and equipment within the Ince, U.K. asset group, an asset group planned for abandonment, we first considered use of a market or income-based valuation method. However, given that a secondary market did not exist and the assets had been idled with a planned abandonment and therefore would not generate future cash flows from operations, we estimated the fair value of the asset group by determining the replacement cost of the underlying assets and then adjusting each of the asset categories to an estimated salvage value utilizing industry recognized price publications.

Due to the inherent uncertainties involved in making estimates and assumptions, actual results may differ from those assumed in our forecasts.

2023 Restructuring

In July 2023, we approved and announced our proposed plan to permanently close the ammonia plant at our Billingham complex, and, in September 2023, the final plan was approved. As a result, in 2023, we recognized total charges of $10 million consisting primarily of the recognition of an asset retirement obligation and post-employment benefits related to contractual and statutory obligations due to employees whose employment would be terminated.

8. Property, Plant and Equipment—Net

Property, plant and equipment—net consists of the following:

	December 31,	
	2023	2022
	(in millions)	
Land	$ 114	$ 113
Machinery and equipment	13,716	12,633
Buildings and improvements	1,020	914
Construction in progress	394	203
Property, plant and equipment[1]	15,244	13,863
Less: Accumulated depreciation and amortization	8,103	7,426
Property, plant and equipment—net	$ 7,141	$ 6,437

(1) As of December 31, 2023 and 2022, we had property, plant and equipment that was accrued but unpaid of approximately $68 million and $53 million, respectively.

Depreciation and amortization related to property, plant and equipment was $861 million, $838 million and $871 million in 2023, 2022 and 2021, respectively.

In June 2022, we approved and announced our proposed plan to restructure our U.K. operations, including the planned permanent closure of our Ince facility and optimization of the remaining manufacturing operations at our Billingham facility. As a result, in the second quarter of 2022, we recorded an asset impairment charge of $135 million to write down the property, plant and equipment at the Ince facility to its estimated salvage value. The asset impairment consisted of $128 million related to machinery and equipment, $2 million relating to buildings and improvements, and $5 million related to construction in progress.

In the third quarter of 2022, the United Kingdom continued to experience extremely high and volatile natural gas prices. Given the increase in the price of natural gas in the United Kingdom and the lack of a corresponding increase in global nitrogen product market prices, in September 2022, we temporarily idled ammonia production at our Billingham complex. As a result, we concluded that an additional impairment test was triggered for the asset groups that comprise the continuing U.K. operations. The results of our impairment test indicated that the carrying values for our U.K. Ammonia and U.K. AN asset groups exceeded the undiscounted estimated future cash flows. As a result, we recognized asset impairment charges of $87 million, of which $69 million related to property, plant and equipment, consisting of $57 million related to machinery and equipment and $12 million related to construction in progress. See Note 7—United Kingdom Operations Restructuring and Impairment Charges for additional information.

Plant turnarounds—Scheduled inspections, replacements and overhauls of plant machinery and equipment at our continuous process manufacturing facilities during a full plant shutdown are referred to as plant turnarounds. The expenditures related to turnarounds are capitalized in property, plant and equipment when incurred.

Scheduled replacements and overhauls of plant machinery and equipment during a plant turnaround include the dismantling, repair or replacement and installation of various components including piping, valves, motors, turbines, pumps, compressors and heat exchangers and the replacement of catalysts when a full plant shutdown occurs. Scheduled inspections, including required safety inspections which entail the disassembly of various components such as steam boilers, pressure vessels and other equipment requiring safety certifications, are also conducted during full plant shutdowns. Internal employee costs and overhead amounts are not considered turnaround costs and are not capitalized.

The following is a summary of capitalized plant turnaround costs:

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Net capitalized turnaround costs as of January 1	$ 312	$ 355	$ 226
Additions	165	118	250
Depreciation	(138)	(134)	(121)
Impairment related to U.K. operations	—	(21)	—
Acquisition of Waggaman ammonia production facility	16	—	—
Effect of exchange rate changes and other	(3)	(6)	—
Net capitalized turnaround costs as of December 31	$ 352	$ 312	$ 355

9. Goodwill and Other Intangible Assets

Goodwill

The following table shows the carrying amount of goodwill by reportable segment as of December 31, 2023 and 2022:

	Ammonia	Granular Urea	UAN	AN	Other	Total
	(in millions)					
ınce as of December 31, 2022	$ 57[illegible]7	828	$ 5[illegible]76	$ [illegible]69	$ [illegible]9	2,089
uisition[(1)]	406	—	—	—	—	406
ınce as of December 31, 2023	$ 9[illegible]83	828	$ 5[illegible]76	$ [illegible]69	$ [illegible]9	2,495

(1) See Note 6—Acquisition of Waggaman Ammonia Production Facility for additional information.

Goodwill is not amortized, but is reviewed for impairment annually in the fourth quarter or more frequently whenever events or circumstances indicate that the carrying value may not be recoverable. During the third quarter of 2021, in light of the unprecedented increase in natural gas prices in the United Kingdom and its estimated impact on our U.K. operations, we identified a triggering event indicating possible impairment of goodwill within our U.K. Ammonia, U.K. AN and U.K. Other reporting units. Due to the triggering event, we performed an interim quantitative goodwill impairment analysis as of September 30, 2021 for our U.K. Ammonia, U.K. AN and U.K. Other reporting units. We estimated the fair value of the reporting units using the income approach described in Note 7—United Kingdom Operations Restructuring and Impairment Charges. Based on the evaluation performed, we determined that the carrying value of all three reporting units exceeded their

fair value, which resulted in a goodwill impairment charge totaling $259 million in the third quarter of 2021. The goodwill impairment was calculated as the amount that the carrying value of the reporting unit, including any goodwill, exceeded its fair value, limited to the total amount of goodwill allocated to the reporting unit.

In the fourth quarter of 2021, the continued impacts of the U.K. energy crisis triggered an additional impairment test of goodwill, which resulted in an additional goodwill impairment charge of $26 million. As a result, we had no remaining goodwill related to our U.K. operations on our consolidated balance sheet as of December 31, 2021. For the year ended December 31, 2021, goodwill impairment totaled $285 million, of which $9 million related to our Ammonia segment, $241 million related to our AN segment and $35 million related to our Other segment. See Note 7—United Kingdom Operations Restructuring and Impairment Charges for additional information.

Other Intangible Assets

All of our identifiable intangible assets have definite lives and are presented on our consolidated balance sheets at gross carrying amount, net of accumulated amortization, as follows:

	December 31, 2023			December 31, 2022		
	Gross Carrying Amount	Accumulated Amortization	Net	Gross Carrying Amount	Accumulated Amortization	Net
	(in millions)					
Customer relationships	$ 505	$ (40)	$ 465	$ 50	$ (35)	$ 15
Personal property tax agreement	71	—	71	—	—	—
Carbon credits	2	—	2	—	—	—
Total intangible assets	$ 578	$ (40)	$ 538	$ 50	$ (35)	$ 15

In connection with our acquisition of the Waggaman ammonia production facility, we recorded $526 million of intangible assets, including $455 million of customer relationships and $71 million related to a favorable personal property tax agreement. See Note 6—Acquisition of Waggaman Ammonia Production Facility for additional information.

Our customer relationships are being amortized over a weighted-average life of approximately20 years. For the years ended December 31, 2023, 2022 and 2021, amortization expense of our customer relationships was $5 million, $3 million and $8 million, respectively. Estimated amortization expense related to customer relationships for each of the fiscal years from 2024 to 2028 is approximately $26 million.

The intangible asset related to the favorable personal property tax agreement is being amortized over the agreement's remaining term of approximately12 years. Estimated amortization expense related to this intangible asset for each of the fiscal years from 2024 to 2028 is approximately $6 million.

In June 2022, we approved and announced our proposed plan to restructure our U.K. operations, including the planned permanent closure of our Ince facility and optimization of the remaining manufacturing operations at our Billingham facility. As a result, in the second quarter of 2022, we recorded an intangible asset impairment charge of $8 million related to trade names.

In the third quarter of 2022, the United Kingdom continued to experience extremely high and volatile natural gas prices. Given the increase in the price of natural gas in the United Kingdom and the lack of a corresponding increase in global nitrogen product market prices, in September 2022, we temporarily idled ammonia production at our Billingham complex. As a result, we concluded that an additional impairment test was triggered for the asset groups that comprise the continuing U.K. operations, which resulted in asset impairment charges of $87 million in our U.K. Ammonia and U.K. AN asset groups, of which $15 million related to intangible assets, consisting of $6 million related to customer relationships and $9 million related to trade names. As a result of these impairment charges, intangible assets related to our U.K. operations were fully written off. See Note 7—United Kingdom Operations Restructuring and Impairment Charges for additional information.

During the fourth quarter of 2021, as we had estimated that we had sufficient emission credits for our 2021 obligations, we sold excess U.K. emission credits, including those purchased in the third quarter of 2021, for approximately $46 million and recognized a corresponding gain of $27 million, which is included in other operating—net in our consolidated statement of operations for the year ended December 31, 2021.

10. Equity Method Investment

We have a 50% ownership interest in PLNL, which operates an ammonia production facility in Trinidad. We include our share of the net earnings from this equity method investment as an element of earnings from operations because PLNL provides additional production to our operations and is integrated with our other supply chain and sales activities in the Ammonia segment.

PLNL's ammonia production facility relies on natural gas supplied, under a gas sales contract, by The National Gas Company of Trinidad and Tobago Limited (NGC). The prior gas sales contract had an expiration date of September 2023. In the third quarter of 2023, PLNL entered into a new gas sales contract with NGC (the New NGC Contract), which is effective October 2023 through December 2025.

In the third quarter of 2023 and due to the terms of the New NGC Contract, we assessed our investment in PLNL for impairment and determined that the carrying value of our equity method investment in PLNL exceeded its fair value. As a result, we recorded an impairment of our equity method investment in PLNL of $43 million, which is reflected in equity in (loss) earnings of operating affiliate on our consolidated statement of operations for the year ended December 31, 2023. As of December 31, 2023, the total carrying value of our equity method investment in PLNL was $26 million.

We have transactions in the normal course of business with PLNL reflecting our obligation to purchase50% of the ammonia produced by PLNLat current market prices. Our ammonia purchases from PLNL totaled $142 million, $259 million and $150 million in 2023, 2022 and 2021, respectively.

11. Fair Value Measurements

Our cash and cash equivalents and other investments consist of the following:

	December 31, 2023			
	Cost Basis	Unrealized Gains	Unrealized Losses	Fair Value
	(in millions)			
Cash	$ 208	$ —	$ —	$ 208
Cash equivalents:				
U.S. and Canadian government obligations	1,488	—	—	1,488
Other debt securities	336	—	—	336
Total cash and cash equivalents	$ 2,032	$ —	$ —	$ 2,032
Nonqualified employee benefit trusts	16	1	—	17

	December 31, 2022			
	Cost Basis	Unrealized Gains	Unrealized Losses	Fair Value
	(in millions)			
Cash	$ 153	$ —	$ —	$ 153
Cash equivalents:				
U.S. and Canadian government obligations	1,902	—	—	1,902
Other debt securities	268	—	—	268
Total cash and cash equivalents	$ 2,323	$ —	$ —	$ 2,323
Nonqualified employee benefit trusts	16	—	—	16

Under our short-term investment policy, we may invest our cash balances, either directly or through mutual funds, in several types of investment-grade securities, including notes and bonds issued by governmental entities or corporations. Securities issued by governmental entities include those issued directly by the U.S. and Canadian federal governments; those issued by state, local or other governmental entities; and those guaranteed by entities affiliated with governmental entities.

Assets and Liabilities Measured at Fair Value on a Recurring Basis

The following tables present assets and liabilities included in our consolidated balance sheets as of December 31, 2023 and 2022 that are recognized at fair value on a recurring basis, and indicate the fair value hierarchy utilized to determine such fair value:

	December 31, 2023			
	Total Fair Value	Quoted Prices in Active Markets (Level 1)	Significant Other Observable Inputs (Level 2)	Significant Unobservable Inputs (Level 3)
	(in millions)			
Cash equivalents	$ 1,824	$ 1,824	$ —	$ —
Nonqualified employee benefit trusts	17	17	—	—
Derivative assets	1	—	1	—
Derivative liabilities	(35)	—	(35)	—

	December 31, 2022			
	Total Fair Value	Quoted Prices in Active Markets (Level 1)	Significant Other Observable Inputs (Level 2)	Significant Unobservable Inputs (Level 3)
	(in millions)			
Cash equivalents	$ 2,170	$ 2,170	$ —	$ —
Nonqualified employee benefit trusts	16	16	—	—
Derivative assets	12	—	12	—
Derivative liabilities	(85)	—	(85)	—

Cash Equivalents

Cash equivalents include highly liquid investments that are readily convertible to known amounts of cash and have original maturities of three months or less. As of December 31, 2023 and 2022, our cash equivalents consisted primarily of U.S. and Canadian government obligations and money market mutual funds that invest in U.S. government obligations and other investment-grade securities.

Nonqualified Employee Benefit Trusts

We maintain trusts associated with certain nonqualified supplemental pension plans. The fair values of the trust assets are based on daily quoted prices in an active market, which represent the net asset values of the shares held in the trusts, and are included on our consolidated balance sheets in other assets. Debt securities are accounted for as available-for-sale securities, and changes in fair value are reported in other comprehensive income. Changes in the fair value of available-for-sale equity securities in the trust assets are recognized through earnings.

Derivative Instruments

The derivative instruments that we use are primarily natural gas fixed price swaps, basis swaps and options traded in the OTC markets with multi-national commercial banks, other major financial institutions or large energy companies. The natural gas derivative contracts represent anticipated natural gas needs for future periods, and settlements are scheduled to coincide with anticipated natural gas purchases during those future periods. The natural gas derivative contracts settle using primarily a NYMEX futures price index. To determine the fair value of these instruments, we use quoted market prices from NYMEX and standard pricing models with inputs derived from or corroborated by observable market data such as forward curves supplied by an industry-recognized independent third party. See Note 17—Derivative Financial Instruments for additional information.

Financial Instruments

The carrying amounts and estimated fair values of our financial instruments are as follows:

	December 31, 2023		December 31, 2022	
	Carrying Amount	Fair Value	Carrying Amount	Fair Value
	(in millions)			
Long-term debt	$ 2,968	$ 2,894	$ 2,965	$ 2,764

The fair value of our long-term debt was based on quoted prices for identical or similar liabilities in markets that are not active or valuation models in which all significant inputs and value drivers are observable and, as a result, they are classified as Level 2 inputs.

The carrying amounts of cash and cash equivalents, as well as any instruments included in other current assets and other current liabilities that meet the definition of financial instruments, approximate fair values because of their short-term maturities.

Assets and Liabilities Measured at Fair Value on a Nonrecurring Basis

We also have assets and liabilities that may be measured at fair value on a nonrecurring basis; that is, the assets and liabilities are not measured at fair value on an ongoing basis, but are subject to fair value adjustments in certain circumstances, such as when there is evidence of impairment, when there is allocation of purchase price in an acquisition or when a new liability is being established that requires fair value measurement. These include long-lived assets, goodwill and other intangible assets and investments in unconsolidated subsidiaries, such as equity method investments, which may be written down to fair value as a result of impairment. The fair value measurements related to assets and liabilities measured at fair value on a nonrecurring basis rely primarily on Company-specific inputs. Since certain of the Company's assumptions would involve inputs that are not observable, these fair values would reside within Level 3 of the fair value hierarchy.

On December 1, 2023, we acquired the Waggaman ammonia production facility. See Note 6—Acquisition of Waggaman Ammonia Production Facility for information on the inputs utilized in the allocation of purchase price to the fair value of assets acquired and liabilities assumed.

In the third quarter of 2023, we determined the carrying value of our equity method investment in PLNL exceeded its fair value and recorded an impairment of our equity method investment in PLNL of $43 million. See Note 10—Equity Method Investment for additional information.

12. Income Taxes

The components of earnings before income taxes and the components of our income tax provision are as follows:

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Domestic	$ 2,248	$ 4,699	$ 1,979
Non-U.S.	—	396	(436)
Earnings before income taxes	$ 2,248	$ 5,095	$ 1,543
Current			
Federal	$ 271	$ 702	$ 394
Foreign	(26)	395	30
State	84	168	55
	329	1,265	479
Deferred			
Federal	108	(102)	(137)
Foreign	(5)	(18)	(50)
State	(22)	13	(9)
	81	(107)	(196)
Income tax provision	$ 410	$ 1,158	$ 283

Differences in the expected income tax provision based on statutory rates applied to earnings before income taxes and the income tax provision reflected in the consolidated statements of operations are summarized below.

	Year ended December 31,		
	2023	2022	2021
	(in millions, except percentages)		
Earnings before income taxes	$ 2,248	$ 5,095	$ 1,543
Expected tax provision at U.S. statutory rate of 21%	$ 472	$ 1,070	$ 324
State income taxes, net of federal	44	143	34
Net earnings attributable to noncontrolling interest	(66)	(124)	(72)
Foreign tax rate differential	(1)	(9)	(1)
U.S. tax on foreign earnings	4	3	—
Non-deductible goodwill impairment	—	—	60
Transfer pricing arbitration	—	69	—
Federal income tax return audits	—	—	(38)
Other	(43)	6	(24)
Income tax provision	$ 410	$ 1,158	$ 283
Effective tax rate	18.3 %	22.7 %	18.3 %

Our effective tax rate is impacted by earnings attributable to the noncontrolling interest in CFN, as our consolidated income tax provision does not include a tax provision on the earnings attributable to the noncontrolling interest. As a result, earnings attributable to the noncontrolling interest of $313 million, $591 million and $343 million in 2023, 2022 and 2021, respectively, which are included in earnings before income taxes, impacted the effective tax rate in all three years. See Note 19—Noncontrolling Interest for additional information.

The foreign tax rate differential is impacted by the inclusion of equity earnings from our equity method investment in PLNL, a foreign operating affiliate, which are included in pre-tax earnings on an after-tax basis. In 2021, the foreign tax rate differential includes $12 million of tax expense for the revaluing of deferred taxes due to an enacted rate change in the jurisdiction of a foreign subsidiary.

U.S. tax on foreign earnings is inclusive of the current year tax on global intangible low-tax income (GILTI), benefit from the GILTI Section 250 deduction and foreign tax credits, as well as adjustments to prior year amounts for these items.

Non-deductible goodwill impairment in the table above relates to the goodwill impairment recognized in 2021 as described in Note 7—United Kingdom Operations Restructuring and Impairment Charges. We did not record an income tax benefit for the goodwill impairment as it is nondeductible for income tax purposes.

In 2021, we reached agreement on certain issues related to U.S. federal income tax audits for the tax years 2012 through 2016 and reversed accruals for unrecognized tax benefits of $13 million related to those tax years. This resulted in a $38 million federal income tax benefit, which included the reduction in our unrecognized tax benefits. The federal income tax benefit was offset by $12 million of state income tax liability resulting from adjustments to U.S. federal taxable income, which is included in the line "State income tax, net of federal" in the table above.

Canada Revenue Agency Competent Authority Matter

In 2016, the Canada Revenue Agency (CRA) and Alberta Tax and Revenue Administration (Alberta TRA) issued Notices of Reassessment for tax years 2006 through 2009 to one of our Canadian subsidiaries asserting a disallowance of certain patronage deductions. We filed Notices of Objection with respect to the Notices of Reassessment with the CRA and Alberta TRA and posted letters of credit in lieu of paying the additional tax liability assessed. The letters of credit served as security until the matter was resolved, as discussed below. In 2018, the matter, including the related transfer pricing topic regarding the allocation of profits between Canada and the United States, was accepted for consideration under the bilateral settlement provisions of the U.S.-Canada tax treaty (the Treaty) by the United States and Canadian competent authorities, and included tax years 2006 through 2011. In the second quarter of 2021, the Company submitted the transfer pricing aspect of the matter into the arbitration process under the terms of the Treaty.

In February 2022, we were informed that a decision was reached by the arbitration panel for tax years 2006 through 2011. In March 2022, we received further details of the results of the arbitration proceedings and the settlement provisions between the United States and Canadian competent authorities, and we accepted the decision of the arbitration panel. Under the terms of the arbitration decision, additional income for tax years 2006 through 2011 was subject to tax in Canada, resulting in our having additional Canadian tax liability for those tax years of approximately $129 million.

As a result of the impact of these events on our Canadian and U.S. federal and state income taxes, we recognized an income tax provision of $38 million, reflecting the net impact of $129 million of accrued income taxes payable to Canada for tax years 2006 to 2011, partially offset by net income tax receivables of approximately $1 million in the United States, and we accrued net interest of $102 million, primarily reflecting the estimated interest payable to Canada. The $69 million in the effective tax rate table above excludes the state income tax liability of $9 million, which is included in the line "State income tax, net of federal."

In the second half of 2022, this tax liability and the related interest was assessed and paid, resulting in total payments of $224 million, which also reflect the impact of changes in foreign currency exchange rates. As a result, the letters of credit we had posted in lieu of paying the additional tax liability assessed by the Notices of Reassessment were cancelled. Due primarily to the availability of additional foreign tax credits to offset in part the increased Canadian tax referenced above, we filed amended tax returns in the United States to request a refund of taxes paid.

Deferred Taxes

Deferred tax assets and deferred tax liabilities are as follows:

	December 31,	
	2023	2022
	(in millions)	
Deferred tax assets:		
Net operating loss and capital loss carryforwards, state	$ 68	$ 34
Net operating loss and capital loss carryforwards, foreign	116	114
Retirement and other employee benefits	14	25
Foreign tax credits	55	44
State tax credits	16	7
Operating lease liabilities	63	64
Other	38	35
	370	323
Valuation allowance	(194)	(190)
	176	133
Deferred tax liabilities:		
Depreciation and amortization	(300)	(139)
Investments in partnerships	(780)	(858)
Operating lease right-of-use assets	(63)	(63)
Foreign earnings	(9)	(12)
Other	(23)	(19)
	(1,175)	(1,091)
Net deferred tax liability	$ (999)	$ (958)

As of December 31, 2023, we recorded a deferred tax liability of $9 million on the undistributed earnings of our Canadian subsidiaries for which the Company does not have an indefinite reinvestment assertion. We have not provided for deferred taxes on the remainder of undistributed earnings from our foreign subsidiaries because such earnings would not give rise to additional tax liabilities upon repatriation or such earnings are considered to be indefinitely reinvested.

As of December 31, 2023, our net operating loss and capital loss carryforwards consist primarily of state net operating loss carryforwards of $66 million, of which $32 million will expire at various dates between 2031 and 2043 and the remaining $34 million can be carried forward indefinitely and foreign capital loss carryforwards of $116 million, which can be carried forward indefinitely. Our foreign subsidiaries have operations that do not normally generate capital gains and have no practical plans to do so in the future. As a result, we have recorded a full valuation allowance against all foreign capital loss carryforwards.

As of December 31, 2023, we have state tax credit carryforwards resulting in a deferred tax asset of $16 million. The state tax credits have expiration dates generally ranging from 2038 to 2043.

In 2023, the net increase in the valuation allowance is primarily attributable to excess foreign tax credits associated with certain U.S. taxed foreign branch income and the impact of changes in foreign currency exchange rates, partially offset by the utilization of deferred tax assets of one of our foreign subsidiaries in the United Kingdom. The excess foreign tax credits carried forward, subject to U.S. foreign tax credit limitation rules, are not expected to be utilized prior to expiration and have a full valuation allowance of $55 million reflecting an increase of $11 million in 2023. Based on recent income generated in the United Kingdom, a portion of the deferred tax assets for which a full valuation allowance had been recorded in 2022 was utilized in 2023, resulting in a $7 million decrease to the deferred tax assets and the valuation allowance of one of our foreign subsidiaries in the United Kingdom.

In 2022, the net increase in the valuation allowance is primarily attributable to excess foreign tax credits associated with certain U.S. taxed foreign branch income and the reversal of future deductible temporary differences of one of our foreign subsidiaries in the United Kingdom, partially offset by the impact of changes in foreign currency exchange rates. The excess foreign tax credits carried forward, subject to U.S. foreign tax credit limitation rules, are not expected to be utilized prior to expiration and have a full valuation allowance of $44 million reflecting an increase of $26 million in 2022. Based on recent losses generated in the United Kingdom, and projections for future income over the period for which the deferred tax assets will

reverse, we believe it is more likely than not that the foreign subsidiary in the United Kingdom will not realize the deferred tax assets and therefore have recorded a full valuation allowance of $24 million. See Note 7—United Kingdom Operations Restructuring and Impairment Charges for additional detail.

In 2021, the valuation allowance activity was primarily attributable to state tax credit carryforwards and excess foreign tax credits associated with certain U.S. taxed foreign branch income. Due to the expiration of statute of limitations on state tax credits and increases in taxable income, we no longer have valuation allowances on the remaining state tax credit carryforwards resulting in a decrease of $27 million. The excess foreign tax credits carried forward, subject to U.S. foreign tax credit limitation rules, are not expected to be utilized prior to expiration and have a full valuation allowance of approximately $18 million.

Unrecognized Tax Benefits

A reconciliation of the beginning and ending amount of unrecognized tax benefits is as follows:

	December 31,	
	2023	2022
	(in millions)	
Unrecognized tax benefits:		
Balance as of January 1	$ 181	$ 27
Additions for tax positions taken during the current year	5	—
Additions for tax positions taken during prior years	69	154
Reductions related to lapsed statutes of limitations	—	—
Reductions related to settlements with tax jurisdictions	(33)	—
Balance as of December 31	$ 222	$ 181

In 2023, we increased the amount of our unrecognized tax benefits by $74 million, which primarily relates to refunds claimed on the U.S. amended returns filed during the year, as discussed above under Canada Revenue Agency Competent Authority Matter. In addition, we reduced the amount of unrecognized tax benefits by $33 million, reflecting primarily the settlement of issues raised on state and Canadian income tax audits for various open tax years.

As of December 31, 2023, we had $222 million of unrecognized tax benefits. The majority of our unrecognized tax benefits relate to transfer pricing positions, which have a corollary receivable for the other jurisdiction impacted by the transfer pricing relationship. Recognizing these unrecognized tax benefits would result in additional tax expense of $29 million in the future. These receivables are included in other assets on our consolidated balance sheet.

In 2022, we increased the amount of our unrecognized tax benefits by $154 million, which relates primarily to the Canada Revenue Agency Competent Authority Matter discussed above. As a result of the outcome of the arbitration decision, we evaluated our transfer pricing positions between Canada and the United States for open years 2012 and after. In order to mitigate the assessment of future Canadian interest on these Canadian transfer pricing positions, in the fourth quarter of 2022, we made payments to the Canadian taxing authorities of CAD $363 million (approximately $267 million), which were recorded as noncurrent income tax receivables and included in other assets on our consolidated balance sheet. For the amounts ultimately owed and paid to the Canadian tax authorities upon resolution of these tax years, we would seek refunds of related taxes paid in the United States.

In 2021, we increased the amount of our unrecognized tax benefits by $5 million related to an addition for state investment tax credits. In addition, we reduced the amount of unrecognized tax benefits in 2021 by $59 million primarily related to the effective settlement of the U.S. federal income tax audits for the tax years 2012 through 2016, as described above.

We file federal, provincial, state and local income tax returns principally in the United States, Canada and the United Kingdom, as well as in certain other foreign jurisdictions. In general, filed tax returns remain subject to examination by United States tax jurisdictions for years 2017 and thereafter, by Canadian tax jurisdictions for years 2012 and thereafter, and by the United Kingdom for years 2021 and thereafter. As a result of uncertainties regarding tax audits and their possible outcomes, an estimate of the range of possible impacts to unrecognized tax benefits in the next twelve months cannot be made at this time.

Interest expense and penalties related to our unrecognized tax benefits of $(4) million and $66 million were recorded for the years ended December 31, 2023 and 2022, respectively. Interest expense and penalties recorded for the year ended December 31, 2021 were not material. Amounts recognized in our consolidated balance sheets for accrued interest and penalties

related to our unrecognized tax benefits of $51 million and $62 million as of December 31, 2023 and 2022, respectively, are included in other liabilities.

13. Pension and Other Postretirement Benefits

We maintain five funded pension plans, consisting of three in North America (one U.S. plan and two Canadian plans) and two in the United Kingdom. Both U.K. pension plans are closed to new employees and future accruals. All of our North American plans are closed to new employees. As a result of plan amendments in the fourth quarter of 2022, as further described below, one of the Canadian plans ceased future accruals effective December 31, 2023, and the cash balance portion of the U.S. plan that provided benefits based on years of service and interest credits was closed to new employees effective December 31, 2022. We also provide group medical insurance benefits, which vary by group and location, to certain retirees in North America.

On July 15, 2022, we entered into an agreement with an insurance company to purchase a non-participating group annuity contract and transfer approximately $75 million of our primary U.S. defined benefit pension plan's projected benefit obligation. The transaction closed on July 22, 2022 and was funded with plan assets. Under the transaction, the insurance company assumed responsibility for pension benefits and annuity administration for approximately 4,000 retirees or their beneficiaries. As a result of this transaction, in the third quarter of 2022, we remeasured the plan's projected benefit obligation and plan assets and recognized a non-cash pre-tax pension settlement loss of $24 million, reflecting the unamortized net unrecognized postretirement benefit costs related to the settled obligations, with a corresponding offset to accumulated other comprehensive loss. In the fourth quarter of 2022, the final settlement of the non-participating group annuity contract resulted in a refund of $4 million, which decreased the settlement loss by $3 million to $21 million.

In the fourth quarter of 2022, we remeasured certain of our defined benefit pension plans due to plan amendments resulting from a revision to our North American retirement plan strategy, which, among other things, closed the cash balance portion of the U.S. plan that was previously open to new employees and established effective dates for each of the three North America plans to freeze future benefit accruals through the end of 2025. The plan curtailments resulted in a reduction in our benefit obligations of $20 million and curtailment gains of $4 million, which are reflected in other non-operating—net in our consolidated statement of operations.

Our plan assets, benefit obligations, funded status and amounts recognized on our consolidated balance sheets for our North America and United Kingdom plans as of the December 31 measurement date are as follows:

CF INDUSTRIES HOLDINGS, INC.

	Pension Plans				Retiree Medical Plans	
	North America		United Kingdom		North America	
	December 31,		December 31,		December 31,	
	2023	2022	2023	2022	2023	2022
	(in millions)					
Change in plan assets						
Fair value of plan assets as of January 1	$ 273	$ 830	$ 320	$ 505	$ —	$ —
Return on plan assets	29	(142)	19	(136)	—	—
Employer contributions	19	2	25	25	2	1
Plan participant contributions	—	—	—	—	—	1
Pension retiree annuity purchase	—	(372)	—	—	—	—
Benefit payments	(11)	(35)	(22)	(23)	(2)	(2)
Foreign currency translation	3	(10)	18	(51)	—	—
Fair value of plan assets as of December 31	313	273	360	320	—	—
Change in benefit obligation						
Benefit obligation as of January 1	(274)	(841)	(347)	(590)	(23)	(32)
Service cost	(5)	(16)	—	—	—	—
Interest cost	(13)	(19)	(16)	(10)	(1)	(1)
Benefit payments	11	35	22	23	2	2
Foreign currency translation	(3)	9	(19)	58	—	—
Pension retiree annuity purchase	—	372	—	—	—	—
Plan curtailments	—	20	—	—	—	—
Change in assumptions and other	(8)	166	(7)	172	2	8
Benefit obligation as of December 31	(292)	(274)	(367)	(347)	(20)	(23)
Funded status as of December 31	$ 21	$ (1)	$ (7)	$ (27)	$ (20)	$ (23)

The line titled “change in assumptions and other” for our North America pension plans primarily reflects the impact of losses due to the decrease in discount rates for 2023 and gains due to the increase in discount rates for 2022.

The line titled “change in assumptions and other” for our United Kingdom pension plans primarily reflects losses due to the decrease in discount rates and increase in the inflation rate assumptions, partially offset by the change in mortality assumptions for 2023 and gains due to the increase in discount rates for 2022.

Amounts recognized on the consolidated balance sheets consist of the following:

	Pension Plans				Retiree Medical Plans	
	North America		United Kingdom		North America	
	December 31,		December 31,		December 31,	
	2023	2022	2023	2022	2023	2022
	(in millions)					
Other assets	$ 23	$ 23	$ —	$ —	$ —	$ —
Accrued expenses	—	—	—	—	(2)	(2)
Other liabilities	(2)	(24)	(7)	(27)	(18)	(21)
	$ 21	$ (1)	$ (7)	$ (27)	$ (20)	$ (23)

Pre-tax amounts recognized in accumulated other comprehensive loss consist of the following:

	Pension Plans				Retiree Medical Plans	
	North America		United Kingdom		North America	
	December 31,		December 31,		December 31,	
	2023	2022	2023	2022	2023	2022
	(in millions)					
Prior service cost	$ —	$ —	$ 1	$ 1	$ —	$ —
Net actuarial loss (gain)	1	6	73	56	(6)	(4)
	$ 1	$ 6	$ 74	$ 57	$ (6)	$ (4)

Net periodic benefit cost (income) and other amounts recognized in other comprehensive (income) loss for the years ended December 31 included the following:

	Pension Plans						Retiree Medical Plans		
	North America			United Kingdom			North America		
	2023	2022	2021	2023	2022	2021	2023	2022	2021
	(in millions)								
Service cost	$ 5	$ 16	$ 20	$ —	$ —	$ —	$ —	$ —	$ —
Interest cost	13	19	21	16	10	9	1	1	1
Expected return on plan assets	(15)	(22)	(24)	(25)	(14)	(14)	—	—	—
Settlement loss	—	21	—	—	—	—	—	—	—
Curtailment gains	—	(4)	—	—	—	—	—	—	—
Amortization of prior service cost	—	1	1	—	—	—	—	—	—
Amortization of actuarial loss (gain)	—	—	5	—	2	4	(1)	—	—
Net periodic benefit cost (income)	3	31	23	(9)	(2)	(1)	—	1	1
Net actuarial (gain) loss	(6)	(2)	(31)	14	(22)	(36)	(2)	(8)	—
Settlement loss	—	(21)	—	—	—	—	—	—	—
Curtailment effects	—	(20)	—	—	—	—	—	—	—
Curtailment gains	—	4	—	—	—	—	—	—	—
Amortization of prior service (cost) benefit	—	(1)	(1)	—	—	—	—	—	—
Amortization of actuarial (loss) gain	—	—	(5)	—	(2)	(4)	1	—	—
Total recognized in other comprehensive (income) loss	(6)	(40)	(37)	14	(24)	(40)	(1)	(8)	—
Total recognized in net periodic benefit cost (income) and other comprehensive (income) loss	$ (3)	$ (9)	$ (14)	$ 5	$ (26)	$ (41)	$ (1)	$ (7)	$ 1

Service cost is recognized in cost of sales and selling, general and administrative expenses, and the other components of net periodic benefit cost are recognized in other non-operating—net in our consolidated statements of operations.

The accumulated benefit obligation (ABO) in aggregate for the defined benefit pension plans in North America was approximately $290 million and $269 million as of December 31, 2023 and 2022, respectively. The ABO in aggregate for the defined benefit pension plans in the United Kingdom was approximately $367 million and $347 million as of December 31, 2023 and 2022, respectively.

The following table presents aggregated information for those individual defined benefit pension plans that have an ABO in excess of plan assets as of December 31, which, for 2023, excludes all three of the North America defined benefit pension plans and, for 2022, excludes two of the North America defined benefit pension plans, as each has plan assets in excess of its ABO:

	North America		United Kingdom	
	2023	2022	2023	2022
	(in millions)			
Accumulated benefit obligation	$ —	$ (163)	$ (367)	$ (347)
Fair value of plan assets	—	143	360	320

The following table presents aggregated information for those individual defined benefit pension plans that have a projected benefit obligation (PBO) in excess of plan assets as of December 31, which excludes two North America defined benefit pension plans each of which has plan assets in excess of its PBO:

	North America		United Kingdom	
	2023	2022	2023	2022
	(in millions)			
Projected benefit obligation	$ (176)	$ (167)	$ (367)	$ (347)
Fair value of plan assets	174	143	360	320

Our pension funding policy in North America is to contribute amounts sufficient to meet minimum legal funding requirements plus discretionary amounts that we may deem to be appropriate. Actual contributions may vary from estimated amounts depending on changes in assumptions, actual returns on plan assets, changes in regulatory requirements and funding decisions.

In accordance with United Kingdom pension legislation, our United Kingdom pension funding policy is to contribute amounts sufficient to meet the funding level target agreed between the employer and the trustees of the United Kingdom plans. Actual contributions are usually agreed with the plan trustees in connection with each triennial valuation and may vary following each such review depending on changes in assumptions, actual returns on plan assets, changes in regulatory requirements and funding decisions.

We currently estimate that our consolidated pension funding contributions for 2024 will be approximately $27 million, which consists of contributions to our United Kingdom plans as agreed with the plans' trustees.

The expected future benefit payments for our pension and retiree medical plans are as follows:

	Pension Plans		Retiree Medical Plans
	North America	United Kingdom	North America
		(in millions)	
2024	$ 14	$ 23	$ 2
2025	15	24	2
2026	16	24	2
2027	17	25	2
2028	17	25	2
2029-2033	91	138	6

The following assumptions were used in determining the benefit obligations and expense:

	Pension Plans						Retiree Medical Plans		
	North America			United Kingdom			North America		
	2023	2022	2021	2023	2022	2021	2023	2022	2021
Weighted-average discount rate—obligation	4.8 %	5.1 %	2.8 %	4.6 %	4.8 %	2.0 %	4.8 %	5.0 %	2.7 %
Weighted-average discount rate—expense	5.1 %	3.6 %	2.4 %	4.8 %	2.0 %	1.5 %	5.0 %	2.7 %	2.2 %
Weighted-average cash balance interest crediting rate—obligation	3.9 %	3.9 %	3.0 %	n/a	n/a	n/a	n/a	n/a	n/a
Weighted-average cash balance interest crediting rate—expense	3.9 %	3.0 %	3.0 %	n/a	n/a	n/a	n/a	n/a	n/a
Weighted-average rate of increase in future compensation	3.3 %	3.8 %	4.2 %	n/a	n/a	n/a	n/a	n/a	n/a
Weighted-average expected long-term rate of return on assets—expense	4.8 %	3.9 %	3.2 %	6.1 %	3.4 %	3.3 %	n/a	n/a	n/a
Weighted-average retail price index—obligation	n/a	n/a	n/a	3.0 %	3.2 %	3.3 %	n/a	n/a	n/a
Weighted-average retail price index—expense	n/a	n/a	n/a	3.2 %	3.3 %	3.0 %	n/a	n/a	n/a

n/a—not applicable

The discount rates for all plans are developed by plan using spot rates derived from a hypothetical yield curve of high quality (AA rated or better) fixed income debt securities as of the year-end measurement date to calculate discounted cash flows (the projected benefit obligation) and solving for a single equivalent discount rate that produces the same projected benefit obligation. In determining our benefit obligation, we use the actuarial present value of the vested benefits to which each eligible employee is currently entitled, based on the employee's expected date of separation or retirement.

The cash balance interest crediting rate for the U.S. plan is based on the greater of 10-year Treasuries or 3.0%.

For our North America plans, the expected long-term rate of return on assets is based on analysis of historical rates of return achieved by equity and non-equity investments and current market characteristics, adjusted for estimated plan expenses and weighted by target asset allocation percentages. As of January 1, 2024, our weighted-average expected long-term rate of return on assets for our North America plans is 5.0%, which will be used in determining net periodic benefit cost for our North America plans for 2024.

For our United Kingdom plans, the expected long-term rate of return on assets is based on the expected long-term performance of the underlying investments, adjusted for investment managers' fees and estimated plan expenses. As of January 1, 2024, our weighted-average expected long-term rate of return on assets for our United Kingdom plans is 6.5%, which will be used in determining net periodic benefit cost for our United Kingdom plans for 2024.

The retail price index for our United Kingdom plans is developed using a U.K. Government Gilt Prices Only retail price inflation curve, which is based on the difference between yields on fixed interest government bonds and index-linked government bonds.

For the measurement of the benefit obligation at December 31, 2023 for our primary (U.S.) retiree medical benefit plans, the assumed health care cost trend rates, for pre-age 65 retirees, start with an 8.0% increase in 2024, followed by a gradual decline in increases to 4.5% for 2033 and thereafter. For post-age 65 retirees, the assumed health care cost trend rates start with an 8.4% increase in 2024, followed by a gradual decline in increases to 4.5% for 2033 and thereafter. For the measurement of the benefit obligation at December 31, 2022 for our primary (U.S.) retiree medical benefit plans, the assumed health care cost trend rates, for pre-age 65 retirees, started with a 7.0% increase in 2023, followed by a gradual decline in increases to 4.5% for 2031 and thereafter. For post-age 65 retirees, the assumed health care cost trend rates started with a 7.5% increase in 2023, followed by a gradual decline in increases to 4.5% for 2031 and thereafter.

The objectives of the investment policies governing the pension plans are to administer the assets of the plans for the benefit of the participants in compliance with all laws and regulations, and to establish an asset mix that provides for diversification and considers the risk of various different asset classes with the purpose of generating favorable investment returns. The investment policies consider circumstances such as participant demographics, time horizon to retirement and liquidity needs, and provide guidelines for asset allocation, planning horizon, general portfolio issues and investment manager evaluation criteria. The investment strategies for the plans, including target asset allocations and investment vehicles, are subject to change within the guidelines of the policies.

The target asset allocation for our U.S. pension plan is 80% non-equity and 20% equity, which has been determined based on analysis of actual historical rates of return and plan needs and circumstances. The equity investments are tailored to exceed the growth of the benefit obligation and are a combination of U.S. and non-U.S. total stock market index mutual funds. The non-equity investments consist primarily of investments in debt securities and money market instruments that are selected based on investment quality and duration to mitigate volatility of the funded status and annual required contributions. The non-equity investments have a duration profile that is similar to the benefit obligation in order to mitigate the impact of interest rate changes on the funded status. This investment strategy is achieved through the use of mutual funds and individual securities.

The target asset allocation for one of the Canadian plans is 80% non-equity and 20% equity and for the other Canadian plan is 100% non-equity. This investment strategy is achieved through the use of a mutual fund for equity investments and individual securities for non-equity investments. The equity investment is a passively managed portfolio that diversifies assets across multiple securities, economic sectors and countries. The non-equity investments consist primarily of investments in debt securities that are selected based on investment quality and duration to mitigate volatility of the funded status and annual required contributions. The non-equity investments have a duration profile that is similar to the benefit obligation in order to mitigate the impact of interest rate changes on the funded status.

The pension assets in the United Kingdom plans are each administered by a Board of Trustees consisting of employer-nominated trustees, member-nominated trustees and an independent trustee, with a requirement that member-nominated trustees represent at least one-third of each Board of Trustees. It is the responsibility of the trustees to ensure prudent management and investment of the assets in the plans. The trustees meet on a quarterly basis to review and discuss fund performance and other administrative matters.

The trustees' investment objectives are to hold assets that generate returns sufficient to cover prudently each plan's liability without exposing the plans to unacceptable risk. This is accomplished through the asset allocation strategy of each plan. For both plans, if the asset allocation moves more than plus or minus 5% from the target allocation, the plans' appointed investment manager would amend the asset allocation. At a minimum, the trustees review the investment strategy at every triennial actuarial valuation to ensure that the strategy remains consistent with its funding principles. The trustees may review the strategy more frequently if opportunities arise to reduce risk within the investments without jeopardizing the funding position.

Assets of the United Kingdom plans are invested in externally managed pooled funds. The assets are allocated between a growth portfolio and a matching portfolio. The growth portfolio seeks a return premium on investments across multiple asset classes. Growth portfolio funds may include, among others, traditional equities and bonds, growth fixed income, hedged funds, and may use derivatives. The matching portfolio seeks to align asset changes with changes in liabilities due to interest rates and inflation expectations. Matching portfolio funds are composed of corporate bonds, U.K. gilts and liability-driven investment funds and generally invest in fixed income debt securities including government bonds, gilts, gilt repurchase agreements, swaps and investment grade corporate bonds and may use derivatives. The target asset allocation for one of the United Kingdom plans is 46% in the growth portfolio and 54% in the matching portfolio and for the other United Kingdom plan is 57% in the growth portfolio (including a legacy holding in an actively managed property fund) and 43% in the matching portfolio.

The fair values of our pension plan assets as of December 31, 2023 and 2022, by major asset class, are as follows:

	North America			
	December 31, 2023			
	Total Fair Value	Quoted Prices in Active Markets (Level 1)	Significant Other Observable Inputs (Level 2)	Significant Unobservable Inputs (Level 3)
	(in millions)			
Cash and cash equivalents[1]	$ 1	$ —	$ 1	$ —
Short-term investments[2]	3	1	2	—
Equity mutual funds				
Index equity[3]	35	35	—	—
Pooled equity[4]	16	—	16	—
Fixed income				
U.S. Treasury bonds and notes[5]	17	17	—	—
Fixed income mutual funds[6]	45	11	34	—
Corporate bonds and notes[7]	101	—	101	—
Government and agency securities[8]	88	—	88	—
Other[9]	8	—	8	—
Total assets at fair value by fair value levels	$ 314	$ 64	$ 250	$ —
Accruals and payables—net	(1)			
Total assets	$ 313			

	United Kingdom			
	December 31, 2023			
	Total Fair Value	Quoted Prices in Active Markets (Level 1)	Significant Other Observable Inputs (Level 2)	Significant Unobservable Inputs (Level 3)
	(in millions)			
Cash and cash funds[10]	$ 13	$ 13	$ —	$ —
Pooled equity funds[11]	44	—	44	—
Pooled diversified funds[12]	54	—	54	—
Debt funds				
Pooled U.K. government fixed and index-linked securities funds[13]	54	—	54	—
Pooled global debt funds[14]	74	—	74	—
Pooled liability-driven investment funds[15]	68	—	68	—
Total assets at fair value by fair value levels	$ 307	$ 13	$ 294	$ —
Funds measured at NAV as a practical expedient[16]	47			
Total assets at fair value	$ 354			
Receivable from redemption	6			
Total assets	$ 360			

	North America			
	December 31, 2022			
	Total Fair Value	Quoted Prices in Active Markets (Level 1)	Significant Other Observable Inputs (Level 2)	Significant Unobservable Inputs (Level 3)
	(in millions)			
Cash and cash equivalents[1]	$ 3	$ 2	$ 1	$ —
Equity mutual funds				
Index equity[3]	28	28	—	—
Pooled equity[4]	16	—	16	—
Fixed income				
U.S. Treasury bonds and notes[5]	14	14	—	—
Fixed income mutual funds[6]	16	—	16	—
Corporate bonds and notes[7]	109	—	109	—
Government and agency securities[8]	81	—	81	—
Other[9]	6	—	6	—
Total assets at fair value by fair value levels	$ 273	$ 44	$ 229	$ —
Accruals and payables—net	—			
Total assets	$ 273			

	United Kingdom			
	December 31, 2022			
	Total Fair Value	Quoted Prices in Active Markets (Level 1)	Significant Other Observable Inputs (Level 2)	Significant Unobservable Inputs (Level 3)
	(in millions)			
Cash and cash funds[10]	$ 26	$ 9	$ 17	$ —
Pooled equity funds[11]	41	—	41	—
Pooled diversified funds[12]	44	—	44	—
Debt funds				
Pooled U.K. government fixed and index-linked securities funds[13]	67	—	67	—
Pooled global debt funds[14]	47	—	47	—
Pooled liability-driven investment funds[15]	30	—	30	—
Total assets at fair value by fair value levels	$ 255	$ 9	$ 246	$ —
Funds measured at NAV as a practical expedient[16]	61			
Total assets at fair value	$ 316			
Receivable from redemption	4			
Total assets	$ 320			

(1) Cash and cash equivalents are primarily short-term money market funds.
(2) Short-term investments are primarily treasury bills with original maturities longer than three months but less than a year.
(3) The index equity funds are mutual funds that utilize a passively managed investment approach designed to track specific equity indices. They are valued at quoted market prices in an active market, which represent the net asset values of the shares held by the plan.
(4) The equity pooled mutual funds consist of pooled funds that invest in common stock and other equity securities that are traded on U.S., Canadian, and foreign markets.
(5) U.S. Treasury bonds and notes are valued based on quoted market prices in an active market.
(6) The fixed income mutual funds invest primarily in high-quality longer duration fixed income securities which include bonds, debt securities and other similar instruments. The funds are priced based on a daily published net asset value.

(7) Corporate bonds and notes, including private placement securities, are valued by institutional bond pricing services, which gather information from market sources and integrate credit information, observed market movements and sector news into their pricing applications and models.

(8) Government and agency securities consist of U.S. municipal bonds and Canadian provincial bonds that are valued by institutional bond pricing services, which gather information on current trading activity, market movements, trends, and specific data on specialty issues.

(9) Other includes primarily mortgage-backed, asset-backed securities and U.S. Treasury strips. Mortgage-backed and asset-backed securities are valued by institutional pricing services, which gather information from market sources and integrate credit information, observed market movements and sector news into their pricing applications and models. U.S. Treasury strips are valued using stripped interest and stripped principal yield curves based on data obtained from various dealer contacts and live data sources.

(10) Cash and cash funds as of December 31, 2022 include a cash fund that holds primarily short-dated term money market securities.

(11) Pooled equity funds invest in a broad array of global equity, equity-related securities, a range of diversifiers and may use derivatives for efficient portfolio management. The funds are valued at net asset value (NAV) as determined by the fund managers based on the value of the underlying net assets of the fund.

(12) Pooled diversified funds invest in a broad array of asset classes and a range of diversifiers including the use of derivatives. The funds are valued at NAV as determined by the fund managers based on the value of the underlying net assets of the fund.

(13) Pooled U.K. government fixed and index-linked securities funds invest primarily in Sterling denominated fixed income and inflation-linked fixed income securities issued or guaranteed by the U.K. government and may use derivatives for efficient portfolio management. The funds are valued at NAV as determined by the fund managers based on the value of the underlying net assets of the fund.

(14) Pooled global debt funds invest in a broad array of debt securities from corporate and government bonds to emerging markets and high-yield fixed and floating rate securities of varying maturities and may use derivatives for efficient portfolio management. The funds are valued at NAV as determined by the fund managers based on the value of the underlying net assets of the fund.

(15) Pooled liability-driven investment funds primarily invest, either through a sub-fund or directly, in gilt repurchase agreements, physical U.K. government gilts, other inflation linked fixed income securities, and derivatives to provide exposure to interest rates and inflation, thus hedging these elements of risk associated with pension liabilities. The funds are valued at NAV as determined by the fund managers based on the value of the underlying net assets of the fund.

(16) Funds measured at NAV as a practical expedient include funds of funds with return strategies with exposure to varying asset classes and credit strategies, as well as alternative investment strategies not precluding multi-asset credit strategies, global macro strategies, commodities, fixed income, equities and currency, and funds that invest primarily in freehold and leasehold property in the United Kingdom. The funds are valued using NAV as determined by the fund managers based on the value of the underlying assets of the fund.

We have defined contribution plans covering substantially all employees in North America and the United Kingdom. Depending on the specific provisions of each plan, qualified employees receive company contributions based on a percentage of base salary or base salary and incentive pay, matching of employee contributions up to specified limits, or a combination of both. In 2023, 2022 and 2021, we recognized expense related to our contributions to the defined contribution plans of $34 million, $19 million and $25 million, respectively.

In addition to our qualified defined benefit pension plans, we also maintain certain nonqualified supplemental pension plans for highly compensated employees as defined under federal law. The amounts recognized in accrued expenses and other liabilities in our consolidated balance sheets for these plans were $1 million and $10 million, respectively, as of December 31, 2023, and $1 million and $10 million, respectively, as of December 31, 2022. We recognized expense for these plans of $1 million, $1 million and $2 million in 2023, 2022 and 2021, respectively.

14. Financing Agreements

Revolving Credit Agreement

On October 26, 2023, we entered into a new senior unsecured revolving credit agreement (the New Revolving Credit Agreement), which replaced our prior senior unsecured revolving credit agreement (the Prior Revolving Credit Agreement) that was scheduled to mature on December 5, 2024.

The New Revolving Credit Agreement provides for a revolving credit facility of up to $750 million with a maturity of October 26, 2028 and includes a letter of credit sub-limit of $125 million. Borrowings under the New Revolving Credit Agreement may be used for working capital, capital expenditures, acquisitions, share repurchases and other general corporate purposes. CF Industries is the lead borrower, and CF Holdings is the sole guarantor, under the New Revolving Credit Agreement.

Borrowings under the New Revolving Credit Agreement can be denominated in U.S. dollars, Canadian dollars, euros and British pounds. Borrowings under the New Revolving Credit Agreement in U.S. dollars bear interest at a per annum rate equal to, at our option, an applicable adjusted term Secured Overnight Financing Rate or base rate plus, in either case, a specified margin. We are required to pay an undrawn commitment fee on the undrawn portion of the commitments under the New Revolving Credit Agreement and customary letter of credit fees. The specified margin and the amount of the commitment fee depend on CF Holdings' credit rating at the time.

As of December 31, 2023, we had unused borrowing capacity under the New Revolving Credit Agreement of $750 million and no outstanding letters of credit. As of December 31, 2023 and 2022, and during the years ended, there were no borrowings outstanding under either the Prior Revolving Credit Agreement or the New Revolving Credit Agreement.

The New Revolving Credit Agreement contains representations and warranties and affirmative and negative covenants, including a financial covenant. As of December 31, 2023, we were in compliance with all covenants under the New Revolving Credit Agreement.

Letters of Credit Under Bilateral Agreement

We are party to a bilateral agreement providing for the issuance of up to $425 million (reflecting an increase of $75 million in December 2023) of letters of credit. As of December 31, 2023, approximately $291 million of letters of credit were outstanding under this agreement.

Senior Notes

Long-term debt presented on our consolidated balance sheets as of December 31, 2023 and 2022 consisted of the following debt securities issued by CF Industries:

		December 31, 2023		December 31, 2022	
	Effective Interest Rate	Principal	Carrying Amount[1]	Principal	Carrying Amount[1]
		(in millions)			
Public Senior Notes:					
5.150% due March 2034	5.293%	$ 750	$ 741	$ 750	$ 741
4.950% due June 2043	5.040%	750	742	750	742
5.375% due March 2044	5.478%	750	741	750	740
Senior Secured Notes:					
4.500% due December 2026[2]	4.783%	750	744	750	742
Total long-term debt		$ 3,000	$ 2,968	$ 3,000	$ 2,965

(1) Carrying amount is net of unamortized debt discount and deferred debt issuance costs. Total unamortized debt discount was $ 7 million as of both December 31, 2023 and 2022, and total deferred debt issuance costs were $25 million and $28 million as of December 31, 2023 and 2022, respectively.

(2) Effective August 23, 2021, these notes are no longer secured, in accordance with the terms of the applicable indenture.

Under the indentures (including the applicable supplemental indentures) governing the senior notes due 2034, 2043 and 2044 identified in the table above (the Public Senior Notes), each series of Public Senior Notes is guaranteed by CF Holdings.

Under the terms of the indenture governing the 4.500% senior secured notes due December 2026 (the 2026 Notes) identified in the table above, the 2026 Notes are guaranteed by CF Holdings.

Interest on the Public Senior Notes and the 2026 Notes is payable semiannually, and the Public Senior Notes and the 2026 Notes are redeemable at our option, in whole at any time or in part from time to time, at specified make-whole redemption prices.

On April 21, 2022, we redeemed in full all of the $500 million outstanding principal amount of the 3.450% senior notes due June 2023 (the 2023 Notes) in accordance with the optional redemption provisions in the indenture governing the 2023 Notes. The total aggregate redemption price paid in connection with the April 2022 redemption of the 2023 Notes, which was funded with cash on hand, was $513 million, including accrued interest. As a result, we recognized a loss on debt extinguishment of $8 million, consisting primarily of the premium paid on the redemption of the $500 million principal amount of the 2023 Notes prior to their scheduled maturity.

On September 10, 2021, we redeemed $250 million principal amount, representing one-third of the $750 million principal amount outstanding immediately prior to such redemption, of the 2023 Notes, in accordance with the optional redemption provisions in the indenture governing the 2023 Notes. The total aggregate redemption price paid in connection with the redemption of the $250 million principal amount of the 2023 Notes was approximately $265 million, including accrued interest. As a result, we recognized a loss on debt extinguishment of $13 million in 2021, consisting primarily of a premium paid on the redemption of the $250 million principal amount of the 2023 Notes prior to their scheduled maturity.

On March 20, 2021, we redeemed in full all of the $250 million outstanding principal amount of the 3.400% senior secured notes due December 2021 (the 2021 Notes), in accordance with the optional redemption provisions in the indenture governing the 2021 Notes. The total aggregate redemption price paid in connection with the redemption of the 2021 Notes, which was funded with cash on hand, was $258 million, including accrued interest. As a result, we recognized a loss on debt extinguishment of $8 million, consisting primarily of the premium paid on the redemption of the $250 million principal amount of the 2021 Notes prior to their scheduled maturity.

15. Interest Expense

Details of interest expense are as follows:

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Interest on borrowings[1]	$ 150	$ 155	$ 175
Fees on financing agreements[1]	8	8	9
Interest on tax liabilities[2]	(2)	184	1
Interest capitalized	(6)	(3)	(1)
Interest expense	$ 150	$ 344	$ 184

(1) See Note 14—Financing Agreements for additional information.

(2) Interest on tax liabilities for the year ended December 31, 2022 consists primarily of interest accrued on reserves for unrecognized tax benefits related to Canadian transfer pricing. See Note 12—Income Taxes for additional information.

16. Other Operating—Net

Details of other operating—net are as follows:

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Loss on disposal of property, plant and equipment	$ 4	$ 2	$ 3
Gain on sale of emission credits	(39)	(6)	(29)
Loss on foreign currency transactions[1]	—	28	6
(Gain) loss on embedded derivative[2]	—	(14)	1
Other[3]	4	—	(20)
Other operating—net	$ (31)	$ 10	$ (39)

(1) Loss on foreign currency transactions consists of foreign currency exchange rate impacts on foreign currency denominated transactions, including the impact of changes in foreign currency exchange rates on intercompany loans that were not permanently invested.

(2) (Gain) loss on embedded derivative consists of unrealized and realized net (gains) and losses related to a provision of our strategic venture with CHS.

(3) Other includes the front-end engineering and design study costs related to our clean energy initiatives, the recovery of certain precious metals used in the manufacturing process, litigation expenses, and, in 2021, the amount received under the terms of an agreement with the U.K. government associated with the restart of our Billingham facility.

17. Derivative Financial Instruments

We use derivative financial instruments to reduce our exposure to changes in prices for natural gas that will be purchased in the future. Natural gas is the largest and most volatile component of our manufacturing cost for nitrogen-based products. From time to time, we may also use derivative financial instruments to reduce our exposure to changes in foreign currency exchange rates. The derivatives that we use to reduce our exposure to changes in prices for natural gas are primarily natural gas fixed price swaps, basis swaps and options traded in the OTC markets. These natural gas derivatives settle using primarily a NYMEX futures price index, which represents the basis for fair value at any given time. We enter into natural gas derivative contracts with respect to natural gas to be consumed by us in the future, and settlements of those derivative contracts are scheduled to coincide with our anticipated purchases of natural gas used to manufacture nitrogen products during those future periods. We use natural gas derivatives as an economic hedge of natural gas price risk, but without the application of hedge accounting. As a result, changes in fair value of these contracts are recognized in earnings. As of December 31, 2023, we had natural gas derivative contracts covering certain periods through March 2024.

As of December 31, 2023, our open natural gas derivative contracts consisted of natural gas fixed price swaps, basis swaps and options for 49.0 million MMBtus of natural gas. As of December 31, 2022, we had open natural gas derivative contracts consisting of natural gas fixed price swaps, basis swaps and options for 66.3 million MMBtus of natural gas. For the year ended December 31, 2023, we used derivatives to cover approximately 27% of our natural gas consumption.

The effect of derivatives in our consolidated statements of operations is shown in the table below.

	Gain (loss) recognized in income			
		Year ended December 31,		
	Location	2023	2022	2021
		(in millions)		
Natural gas derivatives				
Unrealized net gains (losses)	Cost of sales	$ 39	$ (41)	$ (25)
Realized net (loss) gains	Cost of sales	(139)	10	1
Gain on net settlement of natural gas derivatives due to Winter Storm Uri	Cost of sales	—	—	112
Net derivative (losses) gains		$ (100)	$ (31)	$ 88

Gain on net settlement of natural gas derivatives due to Winter Storm Uri

We also enter into supply agreements to facilitate the availability of natural gas to operate our plants. When we purchase natural gas under these agreements, we intend to take physical delivery for use in our plants. Certain of these supply agreements allow us to fix the price of the deliveries for the following month using an agreed upon first of month price. We utilize the Normal Purchase Normal Sales (NPNS) derivative scope exception for these fixed price contracts and, therefore, we do not account for them as derivatives.

In the first quarter of 2021, the central portion of the United States experienced extreme and unprecedented cold weather due to the impact of Winter Storm Uri. Certain natural gas suppliers and natural gas pipelines declared force majeure events due to frozen equipment. This occurred at the same time as large increases in natural gas demand were occurring due to the extreme cold temperatures. Due to these unprecedented factors, several states declared a state of emergency and natural gas was redirected for residential usage. We net settled certain natural gas contracts with our suppliers and received prevailing market prices, which were in excess of our cost. We no longer qualified for the NPNS derivative scope exception for the natural gas that was net settled with our suppliers due to the impact of Winter Storm Uri. As a result, we recognized a gain of $112 million from the net settlement of these natural gas contracts, which is reflected in cost of sales in our consolidated statement of operations for the year ended December 31, 2021.

The fair values of derivatives on our consolidated balance sheets are shown below. As of December 31, 2023 and 2022, none of our derivative instruments were designated as hedging instruments. See Note 11—Fair Value Measurements for additional information on derivative fair values.

	Asset Derivatives			Liability Derivatives		
	Balance Sheet Location	December 31,		Balance Sheet Location	December 31,	
		2023	2022		2023	2022
		(in millions)			(in millions)	
Natural gas derivatives	Other current assets	$ 1	$ 12	Other current liabilities	$ (35)	$ (85)

The counterparties to our derivative contracts are multinational commercial banks, major financial institutions and large energy companies. Our derivative contracts are executed with several counterparties under International Swaps and Derivatives Association (ISDA) agreements. The ISDA agreements are master netting arrangements commonly used for OTC derivatives that mitigate exposure to counterparty credit risk, in part, by creating contractual rights of netting and setoff, the specifics of which vary from agreement to agreement. These rights are described further below:

- Settlement netting generally allows us and our counterparties to net, into a single net payable or receivable, ordinary settlement obligations arising between us and our counterparties under the ISDA agreement on the same day, in the same currency, for the same types of derivative instruments, and through the same pairing of offices.
- Close-out netting rights are provided in the event of a default or other termination event (as defined in the ISDA agreements), including bankruptcy. Depending on the cause of early termination, the non-defaulting party may elect to terminate all or some transactions outstanding under the ISDA agreement. The values of all terminated transactions and certain other payments under the ISDA agreement are netted, resulting in a single net close-out amount payable to or by the non-defaulting party.
- Setoff rights are provided by certain of our ISDA agreements and generally allow a non-defaulting party to elect to set off, against the final net close-out payment, other matured and contingent amounts payable between us and our counterparties under the ISDA agreement or otherwise. Typically, these setoff rights arise upon the early termination of all transactions outstanding under an ISDA agreement following a default or specified termination event.

Most of our ISDA agreements contain credit-risk-related contingent features such as cross default provisions. In the event of certain defaults or termination events, our counterparties may request early termination and net settlement of certain derivative trades or, under certain ISDA agreements, may require us to collateralize derivatives in a net liability position. As of December 31, 2023 and 2022, the aggregate fair value of the derivative instruments with credit-risk-related contingent features in net liability positions was $34 million and $73 million, respectively, which also approximates the fair value of the assets that may be needed to settle the obligations if the credit-risk-related contingent features were triggered at the reporting dates. The credit support documents executed in connection with certain of our ISDA agreements generally provide us and our counterparties the right to set off collateral against amounts owing under the ISDA agreements upon the occurrence of a default or a specified termination event. As of December 31, 2023 and 2022, we had no cash collateral on deposit with counterparties for derivative contracts.

The following table presents amounts relevant to offsetting of our derivative assets and liabilities as of December 31, 2023 and 2022:

	Amounts presented in consolidated balance sheets[1]	Gross amounts not offset in consolidated balance sheets		Net amount
		Financial instruments	Cash collateral received (pledged)	
	(in millions)			
December 31, 2023				
Total derivative assets	$ 1	$ —	$ —	$ 1
Total derivative liabilities	(35)	—	—	(35)
Net derivative liabilities	$ (34)	$ —	$ —	$ (34)
December 31, 2022				
Total derivative assets	$ 12	$ —	$ —	$ 12
Total derivative liabilities	(85)	—	—	(85)
Net derivative liabilities	$ (73)	$ —	$ —	$ (73)

[1] We report the fair values of our derivative assets and liabilities on a gross basis on our consolidated balance sheets. As a result, the gross amounts recognized and net amounts presented are the same.

We do not believe the contractually allowed netting, close-out netting or setoff of amounts owed to, or due from, the counterparties to our ISDA agreements would have a material effect on our financial position.

18. Supplemental Balance Sheet Data

Accounts Receivable—Net

Accounts receivable—net consist of the following:

	December 31,	
	2023	2022
	(in millions)	
Trade	$ 459	$ 542
Other	46	40
Accounts receivable—net	$ 505	$ 582

Inventories

Inventories consist of the following:

	December 31,	
	2023	2022
	(in millions)	
Finished goods	$ 256	$ 437
Raw materials, spare parts and supplies	43	37
Total inventories	$ 299	$ 474

Other Assets

Other assets consist of the following:

	December 31,	
	2023	2022
	(in millions)	
Spare parts	$ 200	$ 180
Nonqualified employee benefit trusts	17	16
Tax-related assets	610	545
Other	40	30
Total other assets	$ 867	$ 771

Tax-related assets include long-term receivables related to U.S. and Canadian transfer pricing and the related interest, and certain payments to Canadian taxing authorities. See Note 12—Income Taxes for additional information.

Accounts Payable and Accrued Expenses

Accounts payable and accrued expenses consist of the following:

	December 31,	
	2023	2022
	(in millions)	
Accounts payable[(1)]	$ 114	$ 63
Accrued capital expenditures[(1)]	48	39
Accrued natural gas costs	85	200
Payroll and employee-related costs	81	82
Accrued interest	30	30
Other	162	161
Total accounts payable and accrued expenses	$ 520	$ 575

(1) As of December 31, 2023 and 2022, accrued capital expenditures totaled $68 million and $53 million, respectively, of which $20 million and $14 million, respectively, are included within accounts payable in the table above.

Payroll and employee-related costs include accrued salaries and wages, vacation, benefits, incentive plans and payroll taxes.

Accrued interest includes interest payable on our outstanding senior notes. See Note 14—Financing Agreements and Note 15—Interest Expense for additional information.

Other includes accrued utilities, property and other taxes, sales incentives and other credits, accrued litigation settlement costs, and accrued maintenance and professional services.

Other Current Liabilities

As of December 31, 2023, other current liabilities of $42 million consists primarily of $35 million of unrealized loss on natural gas derivatives and $6 million for asset retirement obligations related to our Ince and Billingham complexes.

As of December 31, 2022, other current liabilities of $95 million consists primarily of $85 million of unrealized loss on natural gas derivatives and $6 million for asset retirement obligations related to our Ince complex.

See Note 7—United Kingdom Operations Restructuring and Impairment Charges, Note 17—Derivative Financial Instruments and Note 24—Asset Retirement Obligations for additional information.

Other Liabilities

Other liabilities consist of the following:

	December 31,	
	2023	2022
	(in millions)	
Benefit plans and deferred compensation	$ 50	$ 92
Tax-related liabilities	247	267
Unrealized loss on embedded derivative	1	1
Other	16	15
Other liabilities	$ 314	$ 375

Benefit plans and deferred compensation include liabilities for pensions, retiree medical benefits, and the noncurrent portion of incentive plans. See Note 13—Pension and Other Postretirement Benefits for additional information.

Tax-related liabilities include reserves for unrecognized tax benefits and the related interest. See Note 12—Income Taxes for additional information.

19. Noncontrolling Interest

A reconciliation of the beginning and ending balances of noncontrolling interest and distributions payable to the noncontrolling interest on our consolidated balance sheets is provided below.

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Noncontrolling interest:			
Balance as of January 1	$ 2,802	$ 2,830	$ 2,681
Earnings attributable to noncontrolling interest	313	591	343
Declaration of distributions payable	(459)	(619)	(194)
Balance as of December 31	$ 2,656	$ 2,802	$ 2,830
Distributions payable to noncontrolling interest:			
Balance as of January 1	$ —	$ —	$ —
Declaration of distributions payable	459	619	194
Distributions to noncontrolling interest	(459)	(619)	(194)
Balance as of December 31	$ —	$ —	$ —

We have a strategic venture with CHS under which CHS owns an equity interest in CFN, a subsidiary of CF Holdings, which represents approximately 1% of the membership interests of CFN. We own the remaining membership interests. Under the terms of CFN's limited liability company agreement, each member's interest will reflect, over time, the impact of the profitability of CFN, any member contributions made to CFN and withdrawals and distributions received from CFN. For financial reporting purposes, the assets, liabilities and earnings of the strategic venture are consolidated into our financial statements. CHS' interest in the strategic venture is recorded in noncontrolling interest in our consolidated financial statements. CHS also receives deliveries pursuant to a supply agreement under which CHS has the right to purchase annually from CFN up to approximately 1.1 million tons of granular urea and 580,000 tons of UAN at market prices. As a result of its equity interest in CFN, CHS is entitled to semi-annual cash distributions from CFN. We are also entitled to semi-annual cash distributions from CFN. The amounts of distributions from CFN to us and CHS are based generally on the profitability of CFN and determined based on the volume of granular urea and UAN sold by CFN to us and CHS pursuant to supply agreements, less a formula driven amount based primarily on the cost of natural gas used to produce the granular urea and UAN, and adjusted for the allocation of items such as operational efficiencies and overhead amounts.

On January 31, 2024, the CFN Board of Managers approved semi-annual distribution payments for the distribution period ended December 31, 2023, in accordance with CFN's limited liability company agreement. On January 31, 2024, CFN distributed $144 million to CHS for the distribution period ended December 31, 2023.

20. Stockholders' Equity

Common Stock

Our Board of Directors (the Board) has authorized certain programs to repurchase shares of our common stock. These programs have generally permitted repurchases to be made from time to time in the open market, through privately-negotiated transactions, through block transactions, through accelerated share repurchase programs, or otherwise. Our management has determined the manner, timing and amount of repurchases under these programs based on the evaluation of market conditions, stock price and other factors.

On November 3, 2021, the Board authorized the repurchase of up to $1.5 billion of CF Holdings common stock through December 31, 2024 (the 2021 Share Repurchase Program). On November 2, 2022, the Board authorized the repurchase of up to $3 billion of CF Holdings common stock commencing upon completion of the 2021 Share Repurchase Program and effective through December 31, 2025 (the 2022 Share Repurchase Program).

The following table summarizes the share repurchases under the 2022 Share Repurchase Program and the 2021 Share Repurchase Program.

	2022 Share Repurchase Program		2021 Share Repurchase Program	
	Shares	Amounts[1]	Shares	Amounts[1]
		(in millions)		
Shares repurchased in 2022:				
First quarter	—	$ —	1.3	$ 100
Second quarter	—	—	5.3	490
Third quarter	—	—	6.1	532
Fourth quarter	—	—	2.2	223
Total shares repurchased in 2022	—	—	14.9	1,345
Shares repurchased as of December 31, 2022	—	$ —	14.9	$ 1,345
Shares repurchased in 2023:				
First quarter	—	$ —	1.1	$ 75
Second quarter	0.8	50	1.2	80
Third quarter	1.9	150	—	—
Fourth quarter	2.9	225	—	—
Total shares repurchased in 2023	5.6	425	2.3	155
Shares repurchased as of December 31, 2023	5.6	$ 425	17.2	$ 1,500

[1] As defined in the share repurchase programs, amounts reflect the price paid for the shares of common stock repurchased, excluding commissions paid to brokers and excise taxes.

In the first half of 2023, we completed the 2021 Share Repurchase Program with the repurchase of approximately2.3 million shares for $155 million. In 2023, we repurchased 5.6 million shares under the 2022 Share Repurchase Program for $425 million.

The shares we repurchase are held as treasury stock. If the Board authorizes us to retire the shares, they are returned to the status of authorized but unissued shares. As part of the retirements, we reduce our treasury stock, paid-in capital and retained earnings balances. In 2022, we retired 15.2 million shares of repurchased stock. In 2023, we retired8.1 million shares of repurchased stock, including shares repurchased under the 2021 Share Repurchase Program and the 2022 Share Repurchase Program. As of December 31, 2023 and 2022, we held no shares of treasury stock.

Changes in common shares outstanding are as follows:

	Year ended December 31,		
	2023	2022	2021
Beginning balance	195,604,404	207,575,978	213,954,858
Exercise of stock options	39,106	2,475,550	1,806,940
Issuance of restricted stock[1]	664,200	740,025	643,882
Purchase of treasury shares[2]	(8,119,309)	(15,187,149)	(8,829,702)
Ending balance	188,188,401	195,604,404	207,575,978

(1) Includes shares issued from treasury.
(2) Consists of shares repurchased under share repurchase programs and shares withheld to pay employee tax obligations upon the vesting of restricted stock or the exercise of stock options.

Preferred Stock

CF Holdings is authorized to issue 50 million shares of $0.01 par value preferred stock. Our Third Amended and Restated Certificate of Incorporation authorizes the Board, without any further stockholder action or approval, to issue these shares in one or more classes or series, and (except in the case of our Series A Junior Participating Preferred Stock, 500,000 shares of which are authorized and the terms of which were specified in the original certificate of incorporation of CF Holdings) to fix the rights, preferences and privileges of the shares of each wholly unissued class or series and any of its qualifications, limitations or restrictions. The Series A Junior Participating Preferred Stock had been established in CF Holdings' original certificate of incorporation in connection with our former stockholder rights plan that expired in 2015. No shares of preferred stock have been issued.

Accumulated Other Comprehensive Loss

Changes to accumulated other comprehensive loss and the impact on other comprehensive income (loss) are as follows:

	Foreign Currency Translation Adjustment	Unrealized Gain (Loss) on Derivatives	Defined Benefit Plans	Accumulated Other Comprehensive Loss
	(in millions)			
Balance as of December 31, 2020	$ (144)	$ 4	$ (180)	$ (320)
Gain arising during the period	—	—	67	67
Reclassification to earnings[1]	—	—	12	12
Effect of exchange rate changes and deferred taxes	3	—	(19)	(16)
Balance as of December 31, 2021	$ (141)	$ 4	$ (120)	$ (257)
Gain arising during the period	—	—	55	55
Reclassification to earnings[1]:				
Settlement loss	—	—	21	21
Curtailment gains	—	—	(4)	(4)
Other	—	(1)	4	3
Effect of exchange rate changes and deferred taxes	(38)	—	(10)	(48)
Balance as of December 31, 2022	$ (179)	$ 3	$ (54)	$ (230)
Loss arising during the period	—	—	(6)	(6)
Reclassification to earnings[1]	—	—	(1)	(1)
Effect of exchange rate changes and deferred taxes	33	—	(5)	28
Balance as of December 31, 2023	$ (146)	$ 3	$ (66)	$ (209)

(1) Reclassifications out of accumulated other comprehensive loss to the consolidated statements of operations were not material.

21. Stock-based Compensation

2022 Equity and Incentive Plan

In May 2022, our shareholders approved the CF Industries Holdings, Inc. 2022 Equity and Incentive Plan (the 2022 Equity and Incentive Plan), including 2.5 million new shares of the Company's common stock available for grant thereunder as part of our pay-for-performance compensation program, which we use to provide incentives that are aligned with the interests of our shareholders. The 2022 Equity and Incentive Plan replaced the CF Industries Holdings, Inc. 2014 Equity and Incentive Plan (the 2014 Equity and Incentive Plan) and permits grants of stock options, stock appreciation rights, restricted stock, restricted stock units and other stock-based awards, which in each case may be conditioned on performance criteria, to employees and certain consultants of the Company and its subsidiaries and non-employee directors of the Company.

Share Reserve and Individual Award Limits

The maximum number of shares reserved for the grant of awards under the 2022 Equity and Incentive Plan is the sum of (i) 2.5 million shares, plus (ii) the number of shares that remain available for new grants under the 2014 Equity and Incentive Plan when the 2022 Equity and Incentive Plan was approved by shareholders, plus (iii) the number of shares subject to stock options granted under the 2014 Equity and Incentive Plan or the CF Industries Holdings, Inc. 2009 Equity and Incentive Plan that were outstanding when the 2022 Equity and Incentive Plan was approved by shareholders, but only to the extent such awards terminate or expire without the delivery of shares, plus (iv) 1.61 times the number of shares subject to restricted stock or restricted stock unit awards (including performance restricted stock unit awards) granted under the 2014 Equity and Incentive Plan that were outstanding when the 2022 Equity and Incentive Plan was approved by shareholders, but only to the extent such awards terminate or expire without the delivery of shares. In no event will the number of shares available for issuance under the 2022 Equity and Incentive Plan exceed 10,615,515 shares. Shares issued with respect to all awards granted under the 2022 Equity and Incentive Plan are counted against the share reserve on a one-for-one basis. The shares subject to any outstanding award under the 2022 Equity and Incentive Plan will be available for subsequent award and issuance under the 2022 Equity and Incentive Plan to the extent those awards subsequently expire, are forfeited or cancelled, or terminate for any reason prior to issuance of the shares subject to those awards. In addition, shares tendered or withheld in payment of the exercise price of an award and shares withheld by the Company to satisfy tax withholding obligations related to an award will be available for subsequent award under the 2022 Equity and Incentive Plan. As of December 31, 2023, we had approximately 6.7 million shares available for future awards under the 2022 Equity and Incentive Plan. The 2022 Equity and Incentive Plan provides that no more than 5.0 million shares may be issued pursuant to the exercise of incentive stock options, subject to adjustment upon certain capitalization events.

Restricted Stock Awards, Restricted Stock Units and Performance Restricted Stock Units

The fair value of a restricted stock award (RSA) or a restricted stock unit (RSU) is equal to the number of shares subject to the award multiplied by the closing market price of our common stock on the date of grant. We estimated the fair value of each performance restricted stock unit (PSU) on the date of grant using a Monte Carlo simulation. Generally, RSUs vest in three equal annual installments following the date of grant. PSUs are granted to key employees and generally vest three years from the date of grant subject to the attainment of applicable performance goals during the performance period. The RSAs awarded to non-management members of the Board vest the earlier of one year from the date of the grant or the date of the next annual stockholder meeting. During the vesting period, the holders of the RSAs are entitled to dividends and voting rights. During the vesting period, the holders of the RSUs are paid dividend equivalents in cash to the extent we pay cash dividends. PSUs accrue dividend equivalents to the extent we pay cash dividends on our common stock during the performance and vesting periods. Upon vesting of the PSUs, holders are paid the cash equivalent of the dividends paid during the performance and vesting periods based on the shares of common stock, if any, delivered in settlement of PSUs. Holders of RSUs and PSUs are not entitled to voting rights unless and until the awards have vested.

A summary of restricted stock activity during the year ended December 31, 2023 is presented below.

	Restricted Stock Awards		Restricted Stock Units		Performance Restricted Stock Units	
	Shares	Weighted-Average Grant-Date Fair Value	Shares	Weighted-Average Grant-Date Fair Value	Shares	Weighted-Average Grant-Date Fair Value
Outstanding as of December 31, 2022	16,736	$ 95.59	546,680	$ 53.21	411,685	$ 63.95
Granted	24,873	74.79	182,122	81.44	181,160	93.61
Restrictions lapsed (vested)[1]	(16,736)	95.59	(281,791)	49.58	(184,400)	59.69
Forfeited	—	—	(14,372)	64.86	—	—
Outstanding as of December 31, 2023	24,873	74.79	432,639	67.07	408,445	79.03

(1) For performance restricted stock units, the shares represent the performance restricted stock units granted in 2020, for which the three-year performance period ended December 31, 2022.

The 2023, 2022 and 2021 weighted-average grant-date fair value for RSAs was $74.79, $95.59 and $49.28, for RSUs was $81.44, $71.68 and $38.69, and for PSUs was $93.61, $81.38 and $48.25, respectively.

The actual tax benefit realized from restricted stock vested in each of the years ended December 31, 2023, 2022 and 2021 was $3 million, $14 million and $7 million, respectively. The fair value of restricted stock vested was $55 million, $60 million and $29 million for the years ended December 31, 2023, 2022 and 2021, respectively

Stock Options

Under the 2014 Equity and Incentive Plan and our other predecessor plans, we granted to plan participants nonqualified stock options to purchase shares of our common stock. The exercise price of these options was equal to the market price of our common stock on the date of grant. The contractual life of each option was ten years and generally one-third of the options vested on each of the first three anniversaries of the date of grant. No stock option awards were granted under the 2014 Equity and Incentive Plan or our other predecessor plans after 2017, and no stock option awards have been granted under the 2022 Equity and Incentive Plan.

A summary of stock option activity during the year ended December 31, 2023 is presented below:

	Shares	Weighted-Average Exercise Price
Outstanding as of December 31, 2022	162,036	$ 37.72
Exercised	(39,106)	41.99
Outstanding as of December 31, 2023	122,930	36.36
Exercisable as of December 31, 2023	122,930	36.36

	Weighted-Average Remaining Contractual Term (years)	Aggregate Intrinsic Value[1] (in millions)
Outstanding as of December 31, 2023	2.2	$ 5
Exercisable as of December 31, 2023	2.2	5

(1) The aggregate intrinsic value represents the total pre-tax intrinsic value, based on our closing stock price of $79.50 as of December 31, 2023, which would have been received by the option holders had all option holders exercised their options as of that date.

Selected amounts pertaining to stock option exercises are as follows:

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Cash received from stock option exercises	$ 2	$ 106	$ 64
Actual tax benefit realized from stock option exercises	—	23	9
Pre-tax intrinsic value of stock options exercised	1	100	39

Compensation Cost

Compensation cost is recorded primarily in selling, general and administrative expenses.The following table summarizes stock-based compensation costs and related income tax benefits:

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Stock-based compensation expense	$ 37	$ 41	$ 30
Income tax benefit	(9)	(9)	(7)
Stock-based compensation expense, net of income taxes	$ 28	$ 32	$ 23

As of December 31, 2023, pre-tax unrecognized compensation cost was $16 million for RSAs and RSUs, which will be recognized over a weighted-average period of1.6 years, and $4 million for PSUs, which will be recognized over a weighted-average period of 1.3 years.

Excess tax benefits realized from the vesting of restricted stock or stock option exercises are recognized as an income tax benefit in our consolidated statements of operations and are required to be reported as an operating cash inflow rather than a reduction of taxes paid. The excess tax benefits realized in 2023, 2022 and 2021 were $19 million, $96 million and $22 million, respectively.

22. Segment Disclosures

Our reportable segments consist of Ammonia, Granular Urea, UAN, AN and Other. These segments are differentiated by products. Our management uses gross margin to evaluate segment performance and allocate resources. Total other operating costs and expenses (consisting primarily of selling, general and administrative expenses and other operating—net) and non-operating expenses (consisting primarily of interest and income taxes) are centrally managed and are not included in the measurement of segment profitability reviewed by management.

Our assets, with the exception of goodwill, are not monitored by or reported to our chief operating decision maker by segment; therefore, we do not present total assets by segment. Goodwill by segment is presented in Note 9—Goodwill and Other Intangible Assets. Segment data for sales, cost of sales and gross margin for 2023, 2022 and 2021 are presented in the table below.

	Ammonia[1]	Granular Urea[2]	UAN[2]	AN[2]	Other[2]	Consolidated
			(in millions)			
Year ended December 31, 2023						
Net sales	$ 1,679	$ 1,823	$ 2,068	$ 497	$ 564	$ 6,631
Cost of sales	1,138	1,010	1,251	359	328	4,086
Gross margin	$ 541	$ 813	$ 817	$ 138	$ 236	2,545
Total other operating costs and expenses						307
Equity in loss of operating affiliate[3]						(8)
Operating earnings						$ 2,230
Year ended December 31, 2022						
Net sales	$ 3,090	$ 2,892	$ 3,572	$ 845	$ 787	$ 11,186
Cost of sales	1,491	1,328	1,489	597	420	5,325
Gross margin	$ 1,599	$ 1,564	$ 2,083	$ 248	$ 367	5,861
Total other operating costs and expenses[4]						558
Equity in earnings of operating affiliate						94
Operating earnings						$ 5,397
Year ended December 31, 2021						
Net sales	$ 1,787	$ 1,880	$ 1,788	$ 510	$ 573	$ 6,538
Cost of sales	1,162	992	1,119	475	403	4,151
Gross margin	$ 625	$ 888	$ 669	$ 35	$ 170	2,387
Total other operating costs and expenses[4]						705
Equity in earnings of operating affiliate						47
Operating earnings						$ 1,729

(1) Cost of sales and gross margin for the Ammonia segment for the year ended December 31, 2021 include a $112 million gain on the net settlement of certain natural gas contracts with our suppliers. See Note 17—Derivative Financial Instruments for additional information.

(2) The cost of the products that are upgraded into other products is transferred at cost into the upgraded product results.

(3) Equity in loss of operating affiliate for the year ended December 31, 2023 includes an impairment of our equity method investment in PLNL of $43 million. See Note 10—Equity Method Investment for additional information.

(4) Total other operating costs and expenses for the year ended December 31, 2022 include $258 million of asset impairment and restructuring charges related to our U.K. operations. Total other operating costs and expenses for the year ended December 31, 2021 include $521 million of asset impairment charges related to our U.K. operations. See Note 7—United Kingdom Operations Restructuring and Impairment Charges for additional information.

Depreciation and amortization by segment for 2023, 2022 and 2021 is as follows:

	Ammonia	Granular Urea	UAN	AN	Other	Corporate	Consolidated
				(in millions)			
Depreciation and amortization							
Year ended December 31, 2023	$ 171	$ 285	$ 288	$ 48	$ 64	$ 13	$ 869
Year ended December 31, 2022	166	272	269	61	67	15	850
Year ended December 31, 2021	209	235	259	77	87	21	888

Enterprise-wide data by geographic region is as follows:

	Year ended December 31,		
	2023	2022	2021
		(in millions)	
Sales by geographic region (based on destination of shipments):			
United States	$ 4,856	$ 8,212	$ 5,086
Foreign:			
Canada	607	849	568
North America, excluding U.S. and Canada	75	149	79
United Kingdom	346	642	464
Other foreign	747	1,334	341
Total foreign	1,775	2,974	1,452
Consolidated	$ 6,631	$ 11,186	$ 6,538

	December 31,		
	2023	2022	2021
		(in millions)	
Property, plant and equipment—net by geographic region:			
United States	$ 6,538	$ 5,812	$ 6,211
Foreign:			
Canada	466	506	485
United Kingdom	137	119	385
Total foreign	603	625	870
Consolidated	$ 7,141	$ 6,437	$ 7,081

Our principal customers are cooperatives, independent fertilizer distributors, traders, wholesalers and industrial users. In 2023, 2022 and 2021, CHS accounted for approximately13%, 13% and 14% of our consolidated net sales, respectively. See Note 19—Noncontrolling Interest for additional information.

23. Supplemental Cash Flow Information

The following provides additional information relating to cash flow activities:

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Cash paid during the year for			
Interest—net of interest capitalized	$ 145	$ 257	$ 176
Income taxes—net of refunds	373	1,776	430
Supplemental disclosure of noncash investing and financing activities:			
Change in capitalized expenditures in accounts payable and accrued expenses	$ 15	$ 18	$ (8)
Change in accrued share repurchases, including accrued excise taxes	5	(1)	(1)

Interest—net of interest capitalized for the year ended December 31, 2022 includes interest paid to Canadian taxing authorities of approximately $300 million related to tax years 2006 through 2011. See Note 12—Income Taxes—"Canada Revenue Agency Competent Authority Matter" for additional information.

Income taxes—net of refunds for the year ended December 31, 2022 includes certain payments of CAD $363 million (approximately $267 million) to Canadian taxing authorities, which are reflected in the line "Other—net" in our consolidated statement of cash flows. These payments were made in order to mitigate the assessment of future Canadian interest on transfer pricing positions. See Note 12—Income Taxes—"Unrecognized Tax Benefits" for additional information.

24. Asset Retirement Obligations

Asset retirement obligations (AROs) are legal obligations associated with the retirement of long-lived assets that result from the acquisition, construction, development or normal operation of such assets. AROs are initially recognized as incurred when sufficient information exists to estimate fair value. We have AROs at our nitrogen manufacturing complexes and at our distribution and storage facilities that are conditional upon cessation of operations. These AROs include certain decommissioning activities as well as the removal and disposal of certain chemicals, waste materials, structures, equipment, vessels, piping and storage tanks. Also included are reclamation of land and the closure of certain effluent ponds and/or waste storage areas. The most recent estimate of the aggregate cost of AROs for our complexes and facilities, expressed in 2023 dollars, is approximately $126 million, which excludes the recorded AROs related to our U.K. operations discussed below.

We have not recorded a liability for these conditional AROs as of December 31, 2023, because we do not believe there is currently a reasonable basis for estimating a date or range of dates of cessation of operations at our nitrogen manufacturing facilities or our distribution and storage facilities, which is necessary in order to estimate fair value. In reaching this conclusion, we considered the historical performance of each complex or facility and considered factors such as planned maintenance, asset replacements and upgrades of plant and equipment, which if conducted as in the past, can extend the physical lives of our nitrogen manufacturing facilities and our distribution and storage facilities indefinitely. We also considered the possibility of changes in technology, risk of obsolescence, and availability of raw materials in arriving at our conclusion.

In July 2023, we approved and announced our proposed plan to permanently close the ammonia plant at our Billingham complex, and, in September 2023, the final plan was approved. As a result, in the third quarter of 2023, we recognized charges of $3 million for the costs of certain asset retirement activities related to the Billingham site. As of December 31, 2023, asset retirement obligations related to our U.K. operations, including the closed Ince complex, recorded in other current liabilities in our consolidated balance sheet was approximately $6 million. See Note 7—United Kingdom Operations Restructuring and Impairment Charges for additional information.

25. Leases

We have operating leases for certain property and equipment under various noncancelable agreements, the most significant of which are rail car leases and barge tow charters for the distribution of our products. The rail car leases currently have minimum terms ranging from one to eleven years and the barge tow charter commitments range fromone to six years. Our rail car leases and barge tow charters commonly contain provisions for automatic renewal that can extend the lease term unless cancelled by either party. We also have operating leases for terminal and warehouse storage for our distribution system, some of which contain minimum throughput requirements. The storage agreements contain minimum terms generally ranging from one to five years and commonly contain provisions for automatic renewal thereafter unless cancelled by either party. The renewal provisions for our rail car leases, barge tow charters and terminal and warehouse storage agreements are not reasonably certain to be exercised.

The components of lease costs were as follows:

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Operating lease cost	$ 113	$ 103	$ 102
Short-term lease cost	30	48	25
Variable lease cost	4	6	7
Total lease cost	$ 147	$ 157	$ 134

Supplemental cash flow information related to leases was as follows:

	Year ended December 31,		
	2023	2022	2021
	(in millions)		
Operating cash flows - cash paid for amounts included in the measurement of operating lease liabilities	$ 107	$ 100	$ 97
Right-of-use (ROU) assets obtained in exchange for operating lease obligations	103	106	80

Supplemental balance sheet information related to leases was as follows:

	December 31,	
	2023	2022
	(in millions)	
Operating lease ROU assets	$ 259	$ 254
Current operating lease liabilities	$ 96	$ 93
Operating lease liabilities	168	167
Total operating lease liabilities	$ 264	$ 260

	December 31,	
	2023	2022
Operating leases		
Weighted-average remaining lease term	5 years	4 years
Weighted-average discount rate	4.7 %	3.9 %

As of December 31, 2023, we have entered into two additional leases that have not yet commenced, with future minimum lease payments totaling $26 million and each with a lease term of seven years.

The following table reconciles the undiscounted cash flows for our operating leases to the operating lease liabilities recorded on our consolidated balance sheet as of December 31, 2023.

	Operating lease payments
	(in millions)
2024	$ 93
2025	65
2026	47
2027	34
2028	22
Thereafter	47
Total lease payments	308
Less: imputed interest	(44)
Present value of lease liabilities	264
Less: Current operating lease liabilities	(96)
Operating lease liabilities	$ 168

26. Subsequent Event

On January 31, 2024, the Board declared a quarterly dividend of $0.50 per common share, representing an increase from the quarterly dividend of $0.40 per common share that was declared and paid in the fourth quarter of 2023. The dividend will be paid on February 29, 2024 to stockholders of record as of February 15, 2024.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE.

None.

ITEM 9A. CONTROLS AND PROCEDURES.

(a) *Disclosure Controls and Procedures.* The Company's management, with the participation of the Company's principal executive officer and principal financial officer, has evaluated the effectiveness of the Company's disclosure controls and procedures (as such term is defined in Rules 13a-15(e) and 15d-15(e) under the Securities Exchange Act of 1934, as amended (the Exchange Act)) as of the end of the period covered by this report. Based on such evaluation, the Company's principal executive officer and principal financial officer have concluded that, as of the end of such period, the Company's disclosure controls and procedures are effective in (i) ensuring that information required to be disclosed by the Company in the reports that it files or submits under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms and (ii) ensuring that information required to be disclosed by the Company in the reports that it files or submits under the Exchange Act is accumulated and communicated to the Company's management, including the Company's principal executive officer and principal financial officer, as appropriate to allow timely decisions regarding required disclosure.

(b) *Management's Report on Internal Control over Financial Reporting.*

Management's Report on Internal Control over Financial Reporting

The Company's management is responsible for establishing and maintaining adequate internal control over financial reporting, as such term is defined in Rules 13a-15(f) and 15d-15(f) under the Exchange Act, for the Company. Under the supervision and with the participation of our senior management, including our principal executive officer and principal financial officer, we assessed the effectiveness of our internal control over financial reporting as of December 31, 2023, using the criteria set forth in the *Internal Control—Integrated Framework* issued by the Committee of Sponsoring Organizations of the Treadway Commission in 2013. Based on this assessment, management has concluded that our internal control over financial reporting is effective as of December 31, 2023. KPMG LLP, the independent registered public accounting firm that audited the Company's consolidated financial statements, has issued an attestation report on the Company's internal control over financial reporting as of December 31, 2023, which appears on the following page.

The Company acquired an ammonia production facility located in Waggaman, Louisiana, on December 1, 2023. As contemplated by SEC staff guidance with respect to newly acquired businesses, the Company's management excluded this acquired business from its assessment of the effectiveness of its internal control over financial reporting as of December 31, 2023. The financial results of the acquired business are included in the Company's consolidated statement of operations since December 1, 2023, and net sales of the acquired business represented 0.4% of the Company's net sales for the year ended December 31, 2023. The assets of the acquired business accounted for 14.0% of the Company's total assets as of December 31, 2023.

(c) *Changes in Internal Control over Financial Reporting.* There have not been any changes in the Company's internal control over financial reporting (as such term is defined in Rules 13a-15(f) and 15d-15(f) under the Exchange Act) during the quarter ended December 31, 2023 that have materially affected, or are reasonably likely to materially affect, the Company's internal control over financial reporting.

While there was no impact on the Company's internal control over financial reporting during the quarter ended December 31, 2023, in the first quarter of 2024, the Company is upgrading its financial reporting and business consolidation system. As a result, related changes in its internal control over financial reporting are expected due to the implementation.

CF INDUSTRIES HOLDINGS, INC.

Report of Independent Registered Public Accounting Firm

To the Stockholders and Board of Directors
CF Industries Holdings, Inc.:

Opinion on Internal Control Over Financial Reporting

We have audited CF Industries Holdings, Inc. and subsidiaries' (the Company) internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission. In our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of the Company as of December 31, 2023 and 2022, the related consolidated statements of operations, comprehensive income, equity, and cash flows for each of the years in the three-year period ended December 31, 2023, and the related notes (collectively, the consolidated financial statements), and our report dated February 22, 2024 expressed an unqualified opinion on those consolidated financial statements.

The Company acquired an ammonia production facility located in Waggaman, Louisiana, on December 1, 2023, and management excluded from its assessment of the effectiveness of the Company's internal control over financial reporting as of December 31, 2023, Waggaman's internal control over financial reporting associated with net sales of 0.4% and total assets of 14.0%, included in the consolidated financial statements of the Company as of and for the year ended December 31, 2023. Our audit of internal control over financial reporting of the Company also excluded an evaluation of the internal control over financial reporting of Waggaman.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Report on Internal Control over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audit also included performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

(signed) KPMG LLP

Chicago, Illinois
February 22, 2024

ITEM 9B. OTHER INFORMATION.

During the quarter ended December 31, 2023, there were no Rule 10b5-1 trading arrangements (as defined in Item 408(a) of Regulation S-K) or non-Rule 10b5-1 trading arrangements (as defined in Item 408(c) of Regulation S-K) adopted or terminated by any director or officer (as defined in Rule 16a-1(f) under the Exchange Act) of CF Industries Holdings, Inc.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS.

Not applicable.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE.

Information appearing in the Proxy Statement under the headings "Proposal 1: Election of Directors—Director Nominees"; "Proposal 1: Election of Directors—Director Nominee Biographies"; "Executive Officers"; "Corporate Governance—Committees of the Board—Audit Committee"; and, if required, "Delinquent Section 16(a) Reports" is incorporated herein by reference.

We have adopted a Code of Corporate Conduct that applies to our employees, directors and officers, including our principal executive officer, principal financial officer and principal accounting officer. The Code of Corporate Conduct is posted on our Internet website, *www.cfindustries.com*. We will provide an electronic or paper copy of this document free of charge upon request. We intend to disclose on our Internet website any amendment to any provision of the Code of Corporate Conduct that relates to any element of the definition of "code of ethics" enumerated in Item 406(b) of Regulation S-K under the Exchange Act and any waiver from any such provision granted to our principal executive officer, principal financial officer, principal accounting officer or controller or persons performing similar functions.

ITEM 11. EXECUTIVE COMPENSATION.

During the last completed fiscal year, John W. Eaves, Javed Ahmed, Stephen J. Hagge, Anne P. Noonan and Michael J. Toelle served as the members of the Compensation and Management Development Committee of the Board.

Information appearing under the following headings of the Proxy Statement is incorporated herein by reference: "Compensation Discussion and Analysis," "Compensation Discussion and Analysis—Compensation Discussion and Analysis: In Detail—Other Compensation Governance Practices and Considerations—Compensation and Benefits Risk Analysis," "Compensation Committee Report," "Executive Compensation" and "Corporate Governance—Director Compensation."

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS.

Information appearing under the following headings of the Proxy Statement is incorporated herein by reference: "Common Stock Ownership—Common Stock Ownership of Certain Beneficial Owners" and "Common Stock Ownership—Common Stock Ownership of Directors and Management."

We currently issue stock-based compensation under the CF Industries Holdings, Inc. 2022 Equity and Incentive Plan (the 2022 Equity and Incentive Plan) which permits grants of stock options, stock appreciation rights, restricted stock, restricted stock units and other stock-based awards, which in each case may be conditioned on performance criteria, to employees and certain consultants of the Company and its subsidiaries and non-employee directors of the Company.

Equity Compensation Plan Information as of December 31, 2023

Plan category	Number of securities to be issued upon exercise of outstanding options, warrants and rights[1]	Weighted-average exercise price of outstanding options, warrants and rights[2]	Number of securities remaining available for future issuance under equity compensation plans (excluding securities reflected in the first column)[3]
Equity compensation plans approved by security holders	1,859,962	$ 36.36	6,713,324
Equity compensation plans not approved by security holders	—	—	—
Total	1,859,962	$ 36.36	6,713,324

(1) Includes 122,930 shares issuable pursuant to outstanding nonqualified stock options, 432,639 shares issuable pursuant to restricted stock units (RSUs) and 1,304,393 shares issuable pursuant to performance restricted stock units (PSUs) under the 2022 Equity and Incentive Plan, the CF Industries Holdings, Inc. 2014 Equity and Incentive Plan (the 2014 Equity and Incentive Plan) and the CF Industries Holdings, Inc. 2009 Equity Incentive Plan. PSUs are subject to attainment of the applicable performance goals during the three-year performance period and are reflected at their maximum potential payout. The PSUs included in this table reflect the full amount awarded to plan participants in 2021, 2022 and 2023. The three-year performance periods for the PSUs awarded in 2021, 2022 and 2023 are in each case composed of three one-year periods with performance goals set annually. Because accounting rules require performance goals to be set before a PSU is determined for accounting purposes to have been granted, the number of PSUs reported as outstanding as of December 31, 2023 in Note 21—Stock-based Compensation reflects all of the PSUs awarded in 2021, but only two-thirds of the PSUs awarded in 2022 and one-third of the PSUs awarded in 2023.

(2) RSUs and PSUs are not reflected in the weighted-average exercise price as these awards do not have an exercise price.

(3) Under the 2022 Equity and Incentive Plan, upon the grant of an award, the number of shares available for issuance is reduced by one share for each share subject to or issued in respect of such awards. Under the 2022 Equity and Incentive Plan, shares withheld for taxes on awards are added to the number of shares available for issuance. If any restricted stock units (including any performance restricted stock units) granted under the 2014 Equity and Incentive Plan terminates or expires without delivery of shares, the number of shares available for issuance under the 2022 Equity and Incentive Plan is increased by 1.61 shares for each share that had been subject to such restricted stock unit at the time of such termination or expiration.

See Note 21—Stock-based Compensation for additional information on the 2022 Equity and Incentive Plan.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE.

Information appearing in the Proxy Statement under the headings "Corporate Governance—Director Independence" and "Policy Regarding Related Person Transactions" is incorporated herein by reference.

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES.

Information appearing in the Proxy Statement under the headings "Proposal 3: Ratification of Selection of Independent Registered Public Accounting Firm for 2024—Audit and Non-Audit Fees" and "Proposal 3: Ratification of Selection of Independent Registered Public Accounting Firm for 2024—Pre-Approval of Audit and Non-Audit Services" is incorporated herein by reference.

PART IV

ITEM 15. EXHIBITS AND FINANCIAL STATEMENT SCHEDULES.

(a) Documents filed as part of this report:

(1) All financial statements:

The following financial statements are included in Part II, Item 8. Financial Statements and Supplementary Data:

Report of Independent Registered Public Accounting Firm (KPMG LLP, Chicago, IL, PCAOB ID: 185)	62
Consolidated Statements of Operations	65
Consolidated Statements of Comprehensive Income	66
Consolidated Balance Sheets	67
Consolidated Statements of Equity	68
Consolidated Statements of Cash Flows	69
Notes to Consolidated Financial Statements	70

Financial statement schedules are omitted because they are not applicable or the required information is included in the consolidated financial statements or notes thereto.

(2) Exhibits

A list of exhibits filed with this Annual Report on Form 10-K (or incorporated by reference to exhibits previously filed or furnished) is provided in the Exhibit Index on page 123 of this report.

ITEM 16. FORM 10-K SUMMARY.

None.

CF INDUSTRIES HOLDINGS, INC.

EXHIBIT INDEX

EXHIBIT NO.	DESCRIPTION
2.1	Agreement and Plan of Merger, dated as of July 21, 2005, by and among CF Industries Holdings, Inc., CF Merger Corp. and CF Industries, Inc. (incorporated by reference to Exhibit 2.1 to Amendment No. 3 to CF Industries Holdings, Inc.'s Registration Statement on Form S-1 filed with the SEC on July 26, 2005, File No. 333-124949)
2.2	Agreement and Plan of Merger, dated as of March 12, 2010, by and among CF Industries Holdings, Inc., Composite Merger Corporation and Terra Industries Inc. (incorporated by reference to Exhibit 2.1 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on March 12, 2010)
2.3	Asset Purchase Agreement, dated October 28, 2013, among CF Industries Holdings, Inc., CF Industries, Inc. and The Mosaic Company (incorporated by reference to Exhibit 2.1 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on November 1, 2013)*
2.4	Second Amended and Restated Limited Liability Company Agreement of CF Industries Nitrogen, LLC, dated as of December 18, 2015, by and between CF Industries Sales, LLC and CHS Inc. (incorporated by reference to Exhibit 2.4 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 24, 2022) *,**
2.5	First Amendment to the Second Amended and Restated Limited Liability Company Agreement of CF Industries Nitrogen, LLC, dated as of March 30, 2018, by and among CF Industries Nitrogen, LLC, CF Industries Sales, LLC, CF USA Holdings, LLC and CHS Inc. (incorporated by reference to Exhibit 2.1 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 3, 2018)*
2.6	Asset Purchase Agreement, dated as of March 20, 2023, by and among Dyno Nobel Louisiana Ammonia, LLC, CF Industries East Point, LLC, and, solely for the purposes of Article X, Incitec Pivot Limited and CF Industries Holdings, Inc. (incorporated by reference to Exhibit 2.1 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on March 20, 2023)*
3.1	Third Amended and Restated Certificate of Incorporation (incorporated by reference to Exhibit 3.1 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on May 5, 2023)
3.2	Fifth Amended and Restated Bylaws of CF Industries Holdings, Inc., effective December 13, 2022 (incorporated by reference to Exhibit 3.1 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on December 15, 2022)
4.1	Specimen common stock certificate (incorporated by reference to Exhibit 4.3 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on July 25, 2017)
4.2	Description of common stock of CF Industries Holdings, Inc.
4.3	Indenture, dated as of May 23, 2013, among CF Industries, Inc., CF Industries Holdings, Inc. andComputershare Trust Company, N.A., as successor trustee (incorporated by reference to Exhibit 4.1 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on May 23, 2013)
4.4	Second Supplemental Indenture, dated as of May 23, 2013, among CF Industries, Inc., CF Industries Holdings, Inc. andComputershare Trust Company, N.A., as successor trustee, relating to CF Industries, Inc.'s 4.950% Senior Notes due 2043 (includes form of note) (the "2043 Notes Supplement") (incorporated by reference to Exhibit 4.3 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on May 23, 2013)
4.5	First Supplement, dated as of November 21, 2016, relating to the 2043 Notes Supplement (incorporated by reference to Exhibit 4.12 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 23, 2017)
4.6	Second Supplement, dated as of March 29, 2018, relating to the 2043 Notes Supplement (incorporated by reference to Exhibit 4.3 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 3, 2018)
4.7	Third Supplement, dated as of March 22, 2019, relating to the 2043 Notes Supplement (incorporated by reference to Exhibit 4.3 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 2, 2019)
4.8	Fourth Supplement, dated as of January 28, 2022, relating to the 2043 Notes Supplement (incorporated by reference to Exhibit 4.4 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on February 1, 2022)

CF INDUSTRIES HOLDINGS, INC.

EXHIBIT NO.	DESCRIPTION
4.9	Third Supplemental Indenture, dated as of March 11, 2014, among CF Industries, Inc., CF Industries Holdings, Inc. and Computershare Trust Company, N.A., as successor trustee, relating to CF Industries, Inc.'s 5.150% Senior Notes due 2034 (includes form of note) (the "2034 Notes Supplement") (incorporated by reference to Exhibit 4.2 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on March 11, 2014)
4.10	First Supplement, dated as of November 21, 2016, relating to the 2034 Notes Supplement (incorporated by reference to Exhibit 4.14 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 23, 2017)
4.11	Second Supplement, dated as of March 29, 2018, relating to the 2034 Notes Supplement (incorporated by reference to Exhibit 4.4 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 3, 2018)
4.12	Third Supplement, dated as of March 22, 2019, relating to the 2034 Notes Supplement (incorporated by reference to Exhibit 4.4 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 2, 2019)
4.13	Fourth Supplement, dated as of January 28, 2022, relating to the 2034 Notes Supplement (incorporated by reference to Exhibit 4.3 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on February 1, 2022)
4.14	Fourth Supplemental Indenture, dated as of March 11, 2014, among CF Industries, Inc., CF Industries Holdings, Inc. and Computershare Trust Company, N.A., as successor trustee, relating to CF Industries, Inc.'s 5.375% Senior Notes due 2044 (includes form of note) (the "2044 Notes Supplement") (incorporated by reference to Exhibit 4.3 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on March 11, 2014)
4.15	First Supplement, dated as of November 21, 2016, relating to the 2044 Notes Supplement (incorporated by reference to Exhibit 4.16 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 23, 2017)
4.16	Second Supplement, dated as of March 29, 2018, relating to the 2044 Notes Supplement (incorporated by reference to Exhibit 4.5 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 3, 2018)
4.17	Third Supplement, dated as of March 22, 2019, relating to the 2044 Notes Supplement (incorporated by reference to Exhibit 4.5 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 2, 2019)
4.18	Fourth Supplement, dated as of January 31, 2022, relating to the 2044 Notes Supplement (incorporated by reference to Exhibit 4.5 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on February 1, 2022)
4.19	Indenture, dated as of November 21, 2016, among CF Industries Holdings, Inc., CF Industries, Inc., the Subsidiary Guarantors (as defined therein) party thereto and Computershare Trust Company, N.A., as successor trustee, relating to CF Industries, Inc.'s 4.500% Senior Secured Notes due 2026 (includes form of note) (the "2026 Notes Indenture") (incorporated by reference to Exhibit 4.2 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on November 22, 2016)
4.20	First Supplemental Indenture, dated as of March 29, 2018, relating to the 2026 Notes Indenture (incorporated by reference to Exhibit 4.7 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 3, 2018)
4.21	Second Supplemental Indenture, dated as of March 22, 2019, relating to the 2026 Notes Indenture (incorporated by reference to Exhibit 4.7 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 2, 2019)
4.22	Third Supplemental Indenture, dated as of January 28, 2022, relating to the 2026 Notes Indenture (incorporated by reference to Exhibit 4.2 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on February 1, 2022)
10.1	Change in Control Severance Agreement, effective as of April 29, 2005, and amended and restated as of July 24, 2007, by and among CF Industries, Inc., CF Industries Holdings, Inc. and Douglas C. Barnard (incorporated by reference to Exhibit 10.3 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on November 5, 2007)***

CF INDUSTRIES HOLDINGS, INC.

EXHIBIT NO.	DESCRIPTION
10.2	Change in Control Severance Agreement, effective as of September 1, 2009, amended as of October 20, 2010, and amended further and restated as of February 17, 2014, by and between CF Industries Holdings, Inc. and Christopher D. Bohn (incorporated by reference to Exhibit 10.3 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 27, 2014)***
10.3	Change in Control Severance Agreement, effective as of November 21, 2008, by and between CF Industries Holdings, Inc. and Bert A. Frost (incorporated by reference to Exhibit 10.11 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 26, 2009)***
10.4	Change in Control Severance Agreement, effective as of November 19, 2007 and amended and restated as of March 6, 2009, by and between CF Industries Holdings, Inc. and Richard A. Hoker (incorporated by reference to Exhibit (e)(9) to CF Industries Holdings, Inc.'s Solicitation/Recommendation Statement on Schedule 14D-9 filed with the SEC on March 23, 2009)***
10.5	Change in Control Severance Agreement, effective as of October 9, 2017, by and between CF Industries Holdings, Inc. and Susan L. Menzel (incorporated by reference to Exhibit 10.1 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on November 2, 2017)***
10.6	Change in Control Severance Agreement, effective as of April 24, 2007, amended as of July 24, 2007, and amended further and restated as of February 17, 2014, by and between CF Industries Holdings, Inc. and W. Anthony Will (incorporated by reference to Exhibit 99.1 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on February 20, 2014)***
10.7	Change in Control Severance Agreement, effective as of February 2, 2012, and amended and restated as of September 1, 2019, by and between CF Industries Holdings, Inc. and Ashraf K. Malik (incorporated by reference to Exhibit 10.1 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on October 31, 2019)***
10.8	Change in Control Severance Agreement, effective as of February 27, 2020, by and between CF Industries Holdings, Inc. and Linda M. Dempsey (incorporated by reference to Exhibit 10.1 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 7, 2020)***
10.9	Change in Control Severance Agreement, effective as of October 17, 2023, by and between CF Industries Holdings, Inc. and Michael P. McGrane (incorporated by reference to Exhibit 10.2 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on November 2, 2023)***
10.10	Form of Amendment to Change in Control Severance Agreement (incorporated by reference to Exhibit 10.3 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on December 24, 2015)***
10.11	Form of Indemnification Agreement with Officers and Directors (incorporated by reference to Exhibit 10.10 to Amendment No. 2 to CF Industries Holdings, Inc.'s Registration Statement on Form S-1 filed with the SEC on July 20, 2005, File No. 333-124949)***
10.12	CF Industries Holdings, Inc. 2009 Equity and Incentive Plan (incorporated by reference to Appendix A to CF Industries Holdings, Inc.'s Definitive Proxy Statement on Schedule 14A filed with the SEC on March 16, 2009)***
10.13	Amendment, dated as of July 21, 2016, to the CF Industries Holdings, Inc. 2009 Equity and Incentive Plan (incorporated by reference to Exhibit 10.3 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on August 4, 2016)***
10.14	CF Industries Holdings, Inc. 2014 Equity and Incentive Plan (incorporated by reference to Appendix C to CF Industries Holdings, Inc.'s Definitive Proxy Statement on Schedule 14A filed with the SEC on April 3, 2014)***
10.15	Amendment, dated as of July 21, 2016, to the CF Industries Holdings, Inc. 2014 Equity and Incentive Plan (incorporated by reference to Exhibit 10.4 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on August 4, 2016)***
10.16	CF Industries Holdings, Inc. 2022 Equity and Incentive Plan (incorporated by reference to Appendix B to CF Industries Holdings, Inc.'s definitive proxy statement on Schedule 14A filed with the SEC on March 30, 2022)***
10.17	CF Industries Holdings, Inc. Supplemental Benefit and Deferral Plan(incorporated by reference to Exhibit 10.16 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 23, 2023)***

EXHIBIT NO.	DESCRIPTION
10.18	First Amendment of CF Industries Holdings, Inc. Supplemental Benefit and Deferral Plan(incorporated by reference to Exhibit 10.17 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 23, 2023)***
10.19	Second Amendment of CF Industries Holdings, Inc. Supplemental Benefit and Deferral Plan(incorporated by reference to Exhibit 10.18 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 23, 2023)***
10.20	Form of Non-Qualified Stock Option Award Agreement (incorporated by reference to Exhibit 10.6 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on August 3, 2009)***
10.21	Form of Non-Qualified Stock Option Award Agreement (incorporated by reference to Exhibit 10.17 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 27, 2014)***
10.22	Form of Non-Qualified Stock Option Award Agreement (incorporated by reference to Exhibit 10.2 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on November 6, 2014)***
10.23	Form of Non-Qualified Stock Option Award Agreement (incorporated by reference to Exhibit 10.2 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 7, 2015)***
10.24	Form of Amendment to Non-Qualified Stock Option Award Agreements (incorporated by reference to Exhibit 10.5 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on May 7, 2015)***
10.25	Form of Non-Qualified Stock Option Award Agreement (incorporated by reference to Exhibit 10.23 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 25, 2016)***
10.26	Form of Non-Qualified Stock Option Award Agreement (incorporated by reference to Exhibit 10.6 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on August 4, 2016)***
10.27	Form of Non-Qualified Stock Option Award Amendment Letter Agreement, dated as of July 19, 2018 (incorporated by reference to Exhibit 10.2 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on August 2, 2018)***
10.28	Form of Restricted Stock Unit Award Agreement (incorporated by reference to Exhibit 10.32 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 23, 2017)***
10.29	Form of Restricted Stock Unit Award Agreement under CF Industries Holdings, Inc. 2022 Equity and Incentive Plan (incorporated by reference to Exhibit 10.3 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on May 13, 2022)***
10.30	Form of Performance Restricted Stock Unit Award Agreement (incorporated by reference to Exhibit 10.43 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 22, 2019)***
10.31	Form of Performance Restricted Stock Unit Award Agreement (incorporated by reference to Exhibit 10.2 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on December 15, 2022)***
10.32	Form of Non-Employee Director Restricted Stock Award Agreement under CF Industries Holdings, Inc. 2022 Equity and Incentive Plan (incorporated by reference to Exhibit 10.2 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on May 13, 2022)***
10.33	Form of Equity Award Amendment Letter Agreement, dated as of July 21, 2016 (incorporated by reference to Exhibit 10.5 to CF Industries Holdings, Inc.'s Quarterly Report on Form 10-Q filed with the SEC on August 4, 2016)***
10.34	CF Industries Holdings, Inc. Annual Incentive Plan effective as of January 1, 2019 (incorporated by reference to Exhibit 10.1 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on December 14, 2018)***
10.35	CF Industries Holdings, Inc. Annual Incentive Plan, as amended and restated effective as of January 1, 2023 (incorporated by reference to Exhibit 10.1 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on December 15, 2022)***
10.36	Amended and Restated Nitrogen Fertilizer Purchase Agreement, dated December 18, 2015, between CF Industries Nitrogen, LLC and CHS Inc. (incorporated by reference to Exhibit 10.37 to CF Industries Holdings, Inc.'s Annual Report on Form 10-K filed with the SEC on February 24, 2022)****

CF INDUSTRIES HOLDINGS, INC.

EXHIBIT NO.	DESCRIPTION
10.37	Revolving Credit Agreement, dated as of October 26, 2023, by and among CF Industries Holdings, Inc., CF Industries, Inc., Citibank, N.A., as administrative agent, the issuing banks from time to time party thereto, and the lenders from time to time party thereto (incorporated by reference to Exhibit 10.1 to CF Industries Holdings, Inc.'s Current Report on Form 8-K filed with the SEC on October 27, 2023)
21	Subsidiaries of the registrant
23	Consent of KPMG LLP, independent registered public accounting firm
31.1	Certification of Principal Executive Officer pursuant to Section 302 of the Sarbanes-Oxley Act of 2002
31.2	Certification of Principal Financial Officer pursuant to Section 302 of the Sarbanes-Oxley Act of 2002
32.1	Certification of Principal Executive Officer pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002
32.2	Certification of Principal Financial Officer pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002
97	Policy related to recovery of erroneously awarded compensation
101	The following financial information from CF Industries Holdings, Inc.'s Annual Report on Form 10-K for the fiscal year ended December 31, 2023, formatted in Inline XBRL (eXtensible Business Reporting Language): (1) Consolidated Statements of Operations, (2) Consolidated Statements of Comprehensive Income, (3) Consolidated Balance Sheets, (4) Consolidated Statements of Equity, (5) Consolidated Statements of Cash Flows and (6) Notes to Consolidated Financial Statements
104	Cover Page Interactive Data File (included in the Exhibit 101 Inline XBRL Document Set)

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* Schedules (or similar attachments) have been omitted pursuant to Item 601(a)(5) of Regulation S-K.

** Portions omitted pursuant to Item 601(b)(2)(ii) of Regulation S-K.

*** Management contract or compensatory plan or arrangement required to be filed (and/or incorporated by reference) as an exhibit to this Annual Report on Form 10-K pursuant to Item 15(a)(3) of Form 10-K.

**** Portions omitted to pursuant to Item 601(b)(10)(iv) of Regulation S-K.

CF INDUSTRIES HOLDINGS, INC.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

CF INDUSTRIES HOLDINGS, INC.

Date: February 22, 2024

By: /s/ W. ANTHONY WILL
W. Anthony Will
President and Chief Executive Officer

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

Signature	Title(s)	Date
/s/ W. ANTHONY WILL W. Anthony Will	President and Chief Executive Officer, Director (Principal Executive Officer)	February 22, 2024
/s/ CHRISTOPHER D. BOHN Christopher D. Bohn	Executive Vice President and Chief Operating Officer, Director (Principal Financial Officer)	February 22, 2024
/s/ RICHARD A. HOKER Richard A. Hoker	Vice President and Corporate Controller (Principal Accounting Officer)	February 22, 2024
/s/ STEPHEN J. HAGGE Stephen J. Hagge	Chairman of the Board of Directors	February 22, 2024
/s/ JAVED AHMED Javed Ahmed	Director	February 22, 2024
/s/ ROBERT C. ARZBAECHER Robert C. Arzbaecher	Director	February 22, 2024
/s/ DEBORAH L. DEHAAS Deborah L. DeHaas	Director	February 22, 2024
/s/ JOHN W. EAVES John W. Eaves	Director	February 22, 2024
/s/ SUSAN A. ELLERBUSCH Susan A. Ellerbusch	Director	February 22, 2024
/s/ JESUS MADRAZO YRIS Jesus Madrazo Yris	Director	February 22, 2024
/s/ ANNE P. NOONAN Anne P. Noonan	Director	February 22, 2024
/s/ MICHAEL J. TOELLE Michael J. Toelle	Director	February 22, 2024
/s/ THERESA E. WAGLER Theresa E. Wagler	Director	February 22, 2024
/s/ CELSO L. WHITE Celso L. White	Director	February 22, 2024